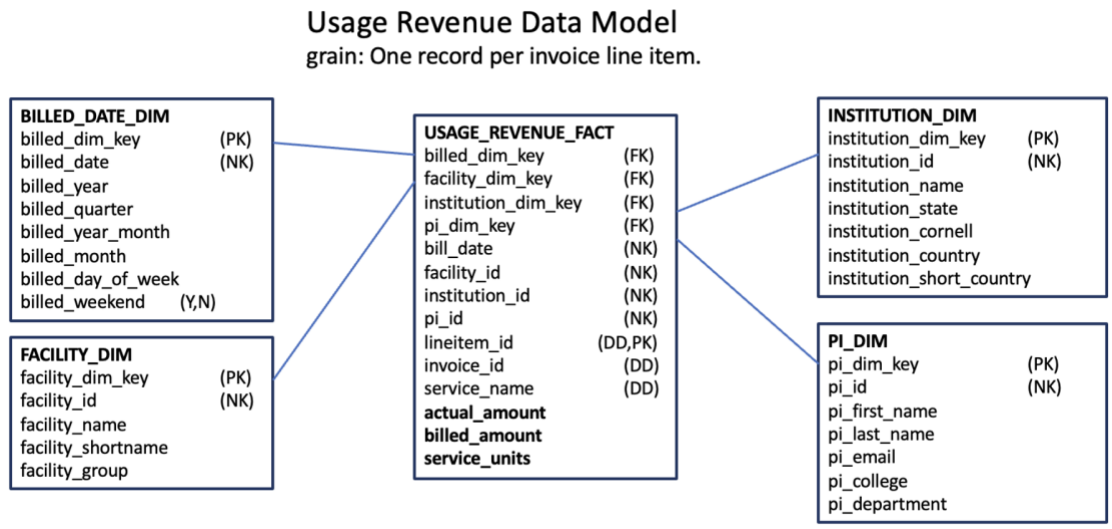
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Assignment 6: Tableau Dashboard

Background

This Tableau assignment is based on a prior year's class project and a dashboard developed by one of the project teams. The source system is a lab management system for the Cornell Biotechnology center. The requirements were to build a data model and dashboard to allow analysis of usage and revenue by various lab facilities, institutions, Pis, over time.

Data Model



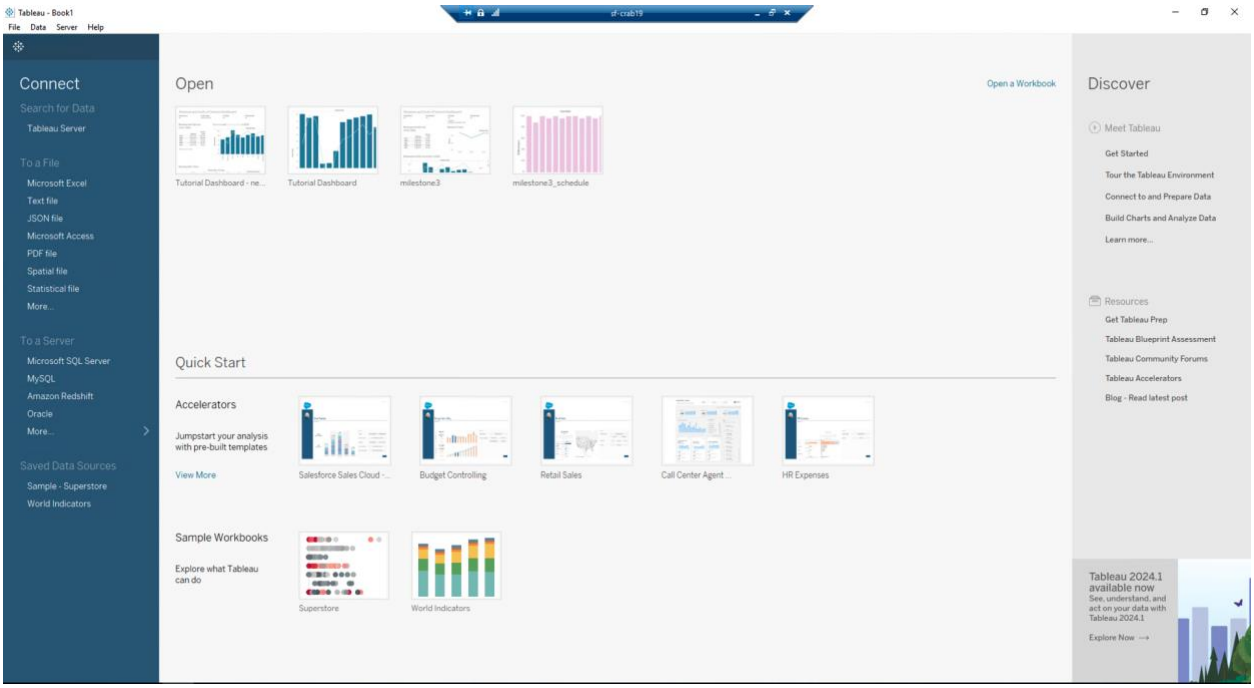
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Assignment 6: Tableau Dashboard

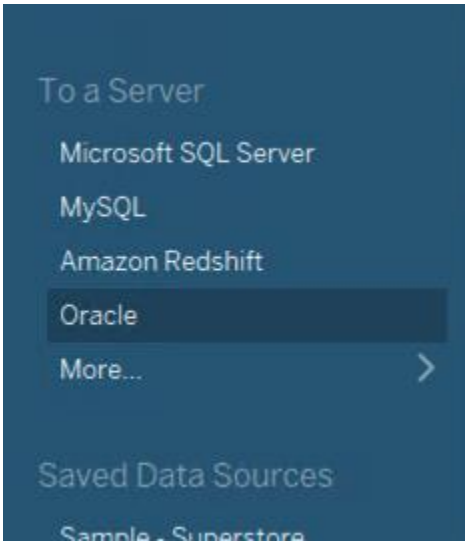
Database Connection Setup

You will need to set up the server.

1. Open Tableau desktop from Apps on Demand



You will see the screen above. Click on Oracle.

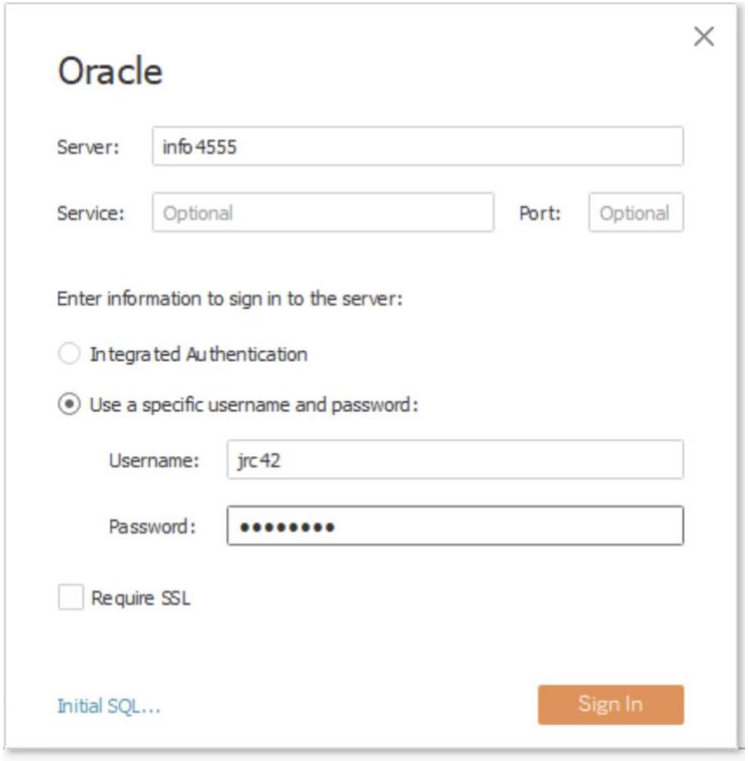


Input to “Server” info4555

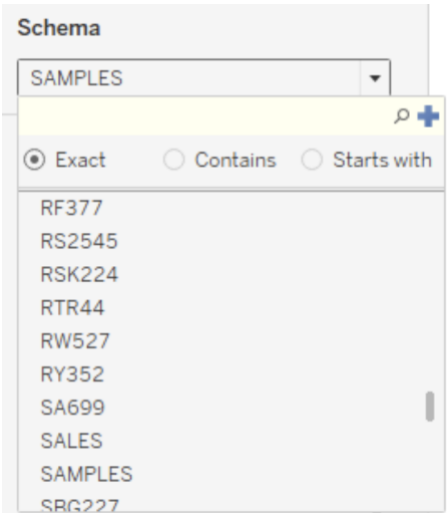
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Assignment 6: Tableau Dashboard

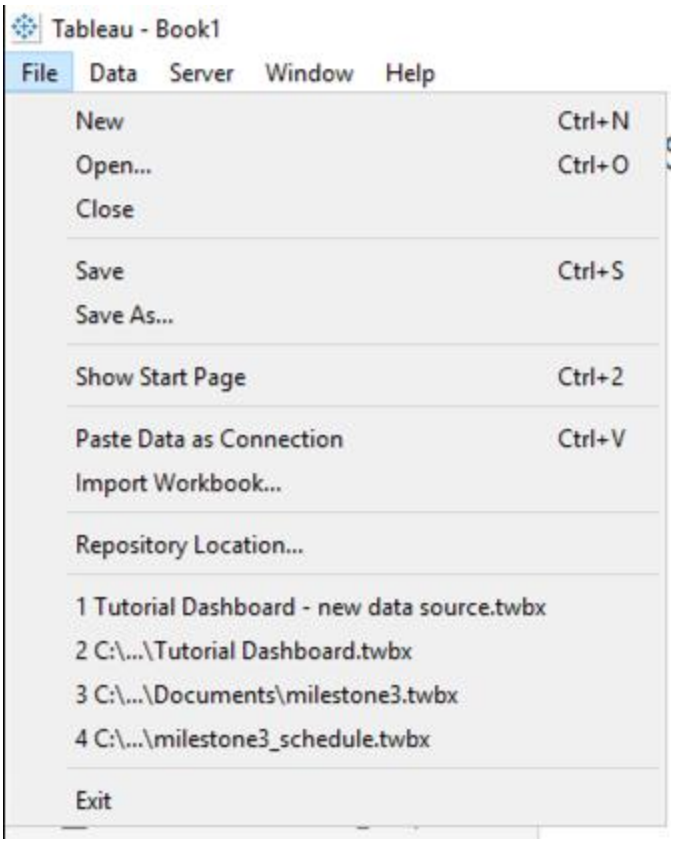
Input your netid database username and password and click “Sign In”



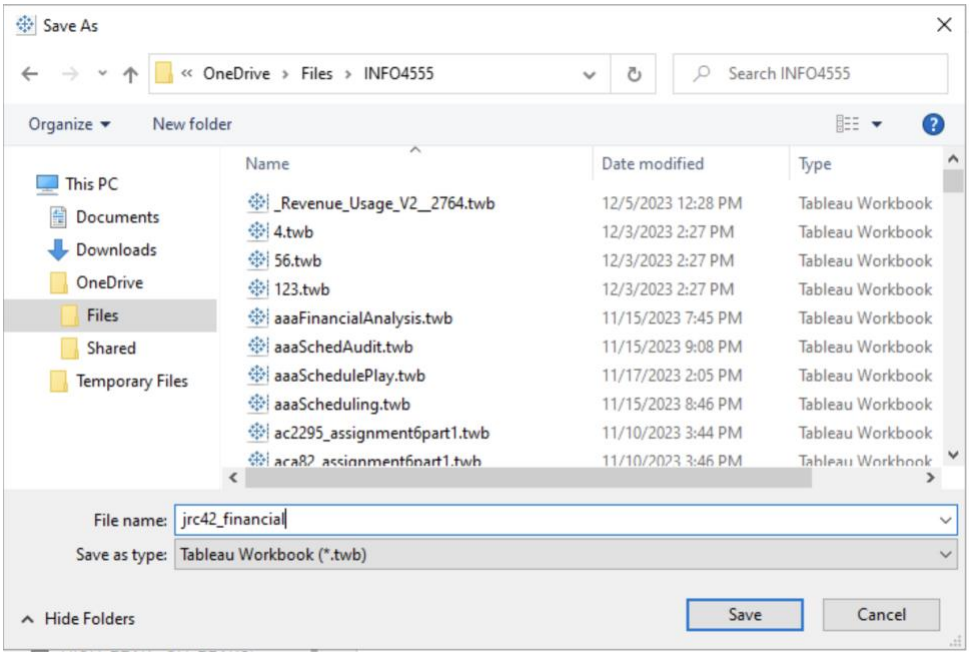
Select from the “Schema” dropdown the SAMPLES schema.



Now save your Tableau workbook. Click “File”, then “Save As…”



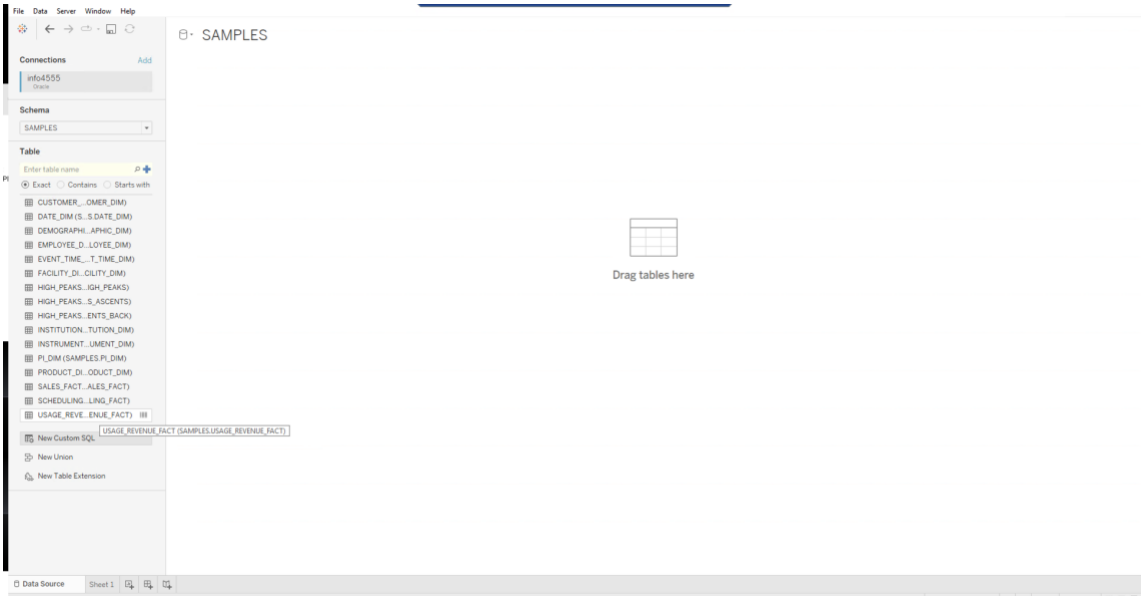
You will want to save it to your OneDrive. Save your file as a Tableau Workbook (.twb). Click save. Note that you should save your work often. Tableau will not auto save your work.



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Assignment 6: Tableau Dashboard

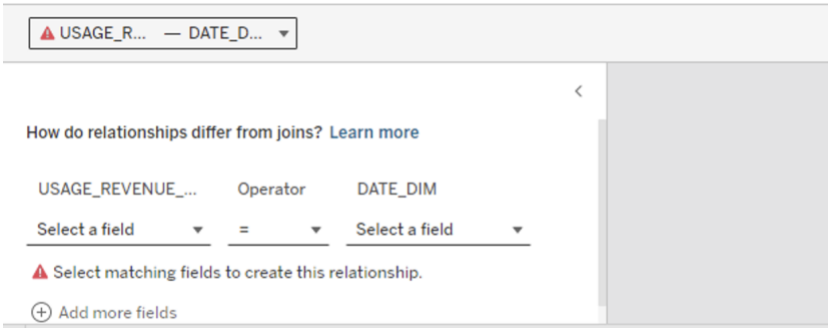
First, you will need to define the dataset that Tableau will be working with. Click on USAGE_REVENUE_FACT and drag it into the center pane.



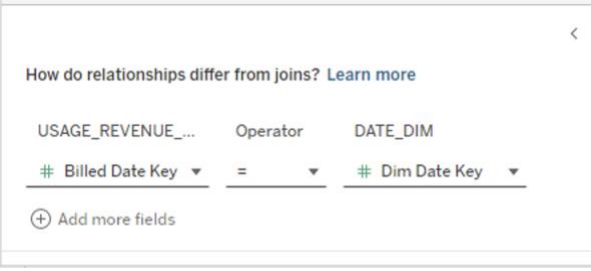
Repeat dragging and dropping the tables for **DATE_DIM**, **FACILITY_DIM**, **INSTITUTION_DIM**, and **PI_DIM**.

When dragging DATE_DIM, you will see warning signs because you need to indicate which field to join the tables on.





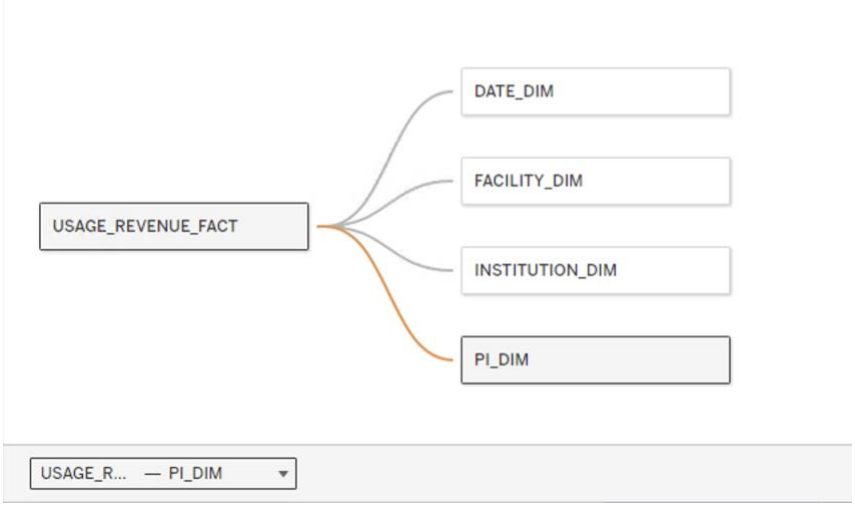
Select **Billed Date Key** in USAGE_REVENUE_FACT and **Dim Date Key** in DATE_DIM.



For FACILITY_DIM, select **Facility Dim Key = Facility Dim Key**
For INSTITUTION_DIM, select **Institution Dim Key = Institution Dim Key**
For PI_DIM, select **Pi Dim Key = PI Dim Key**

Be sure you have the correct keys for these joins, as shown above, or you will have problems with your results in later steps.

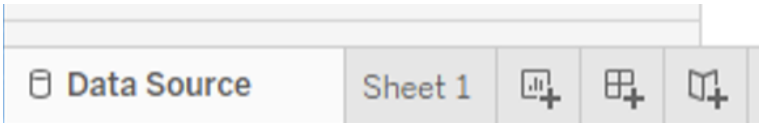
Once you've dragged all of the tables and entered the keys, you should see all of the tables connected. Select "Update Now" to load the data into Tableau.



Creating Worksheets

Sheet 1: Table

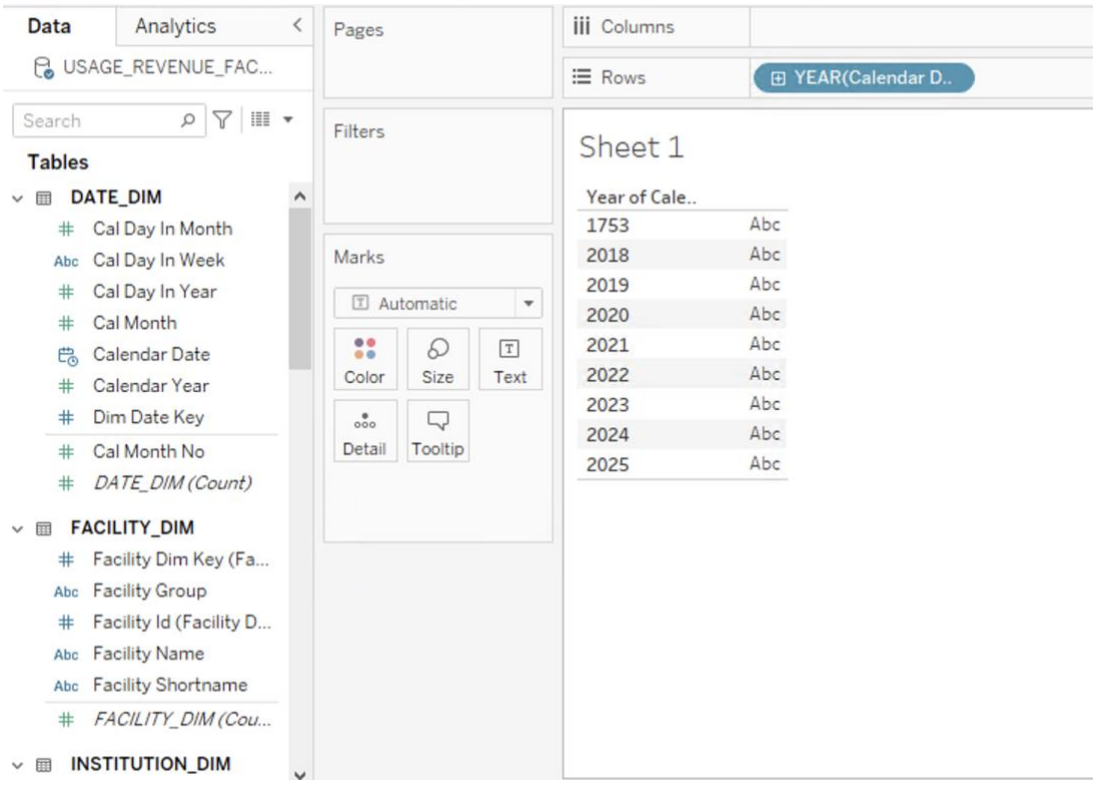
Now we will begin creating our Tableau worksheets. Each sheet will contain a visualization. Begin by clicking "Sheet 1".



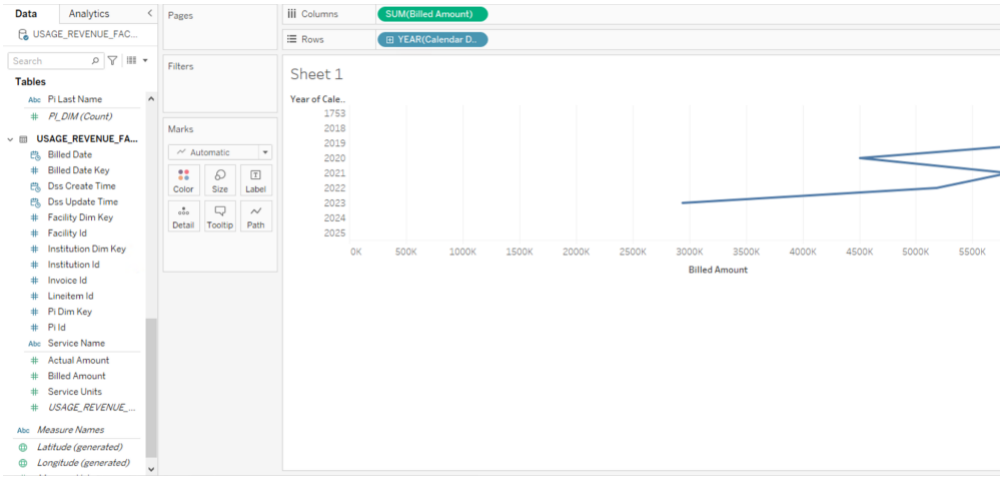
Our dashboard will focus heavily on examining the trends of revenue and service units used for biotechnology lab equipment at Cornell University. Drag the "**Calendar Date**" field under **DATE_DIM** to the Rows. Notice that Tableau automatically uses the YEAR() function on the field.

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Assignment 6: Tableau Dashboard

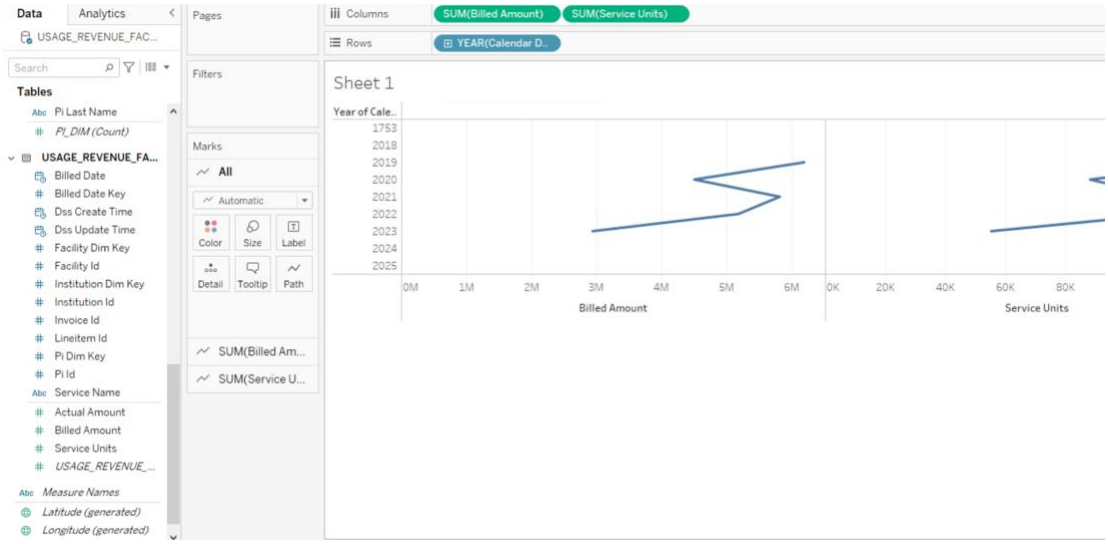


Next, we want to see the revenue displayed. Drag the "**Billed Amount**" field under USAGE_REVENUE_FACT to the columns. Notice how Tableau automatically applies the SUM() function. Repeat this for "**Service Units**" to see this field displayed.



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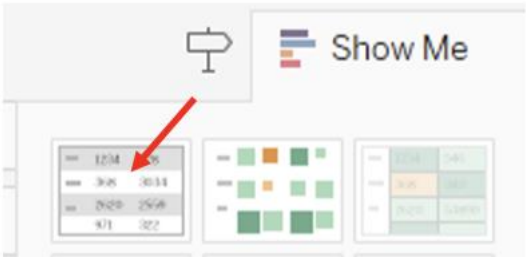
Assignment 6: Tableau Dashboard

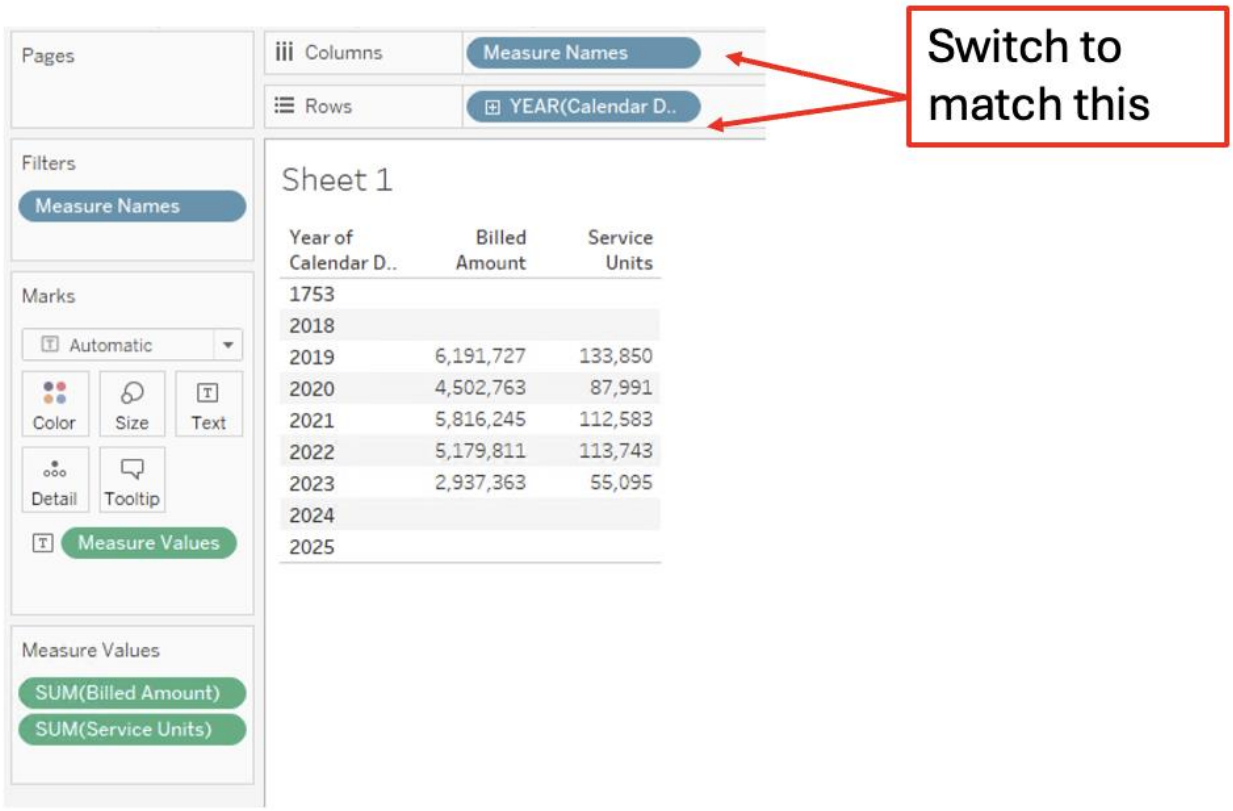


A table would be a better view to see this data. Click “Show Me” in the top right to see more visualization options.

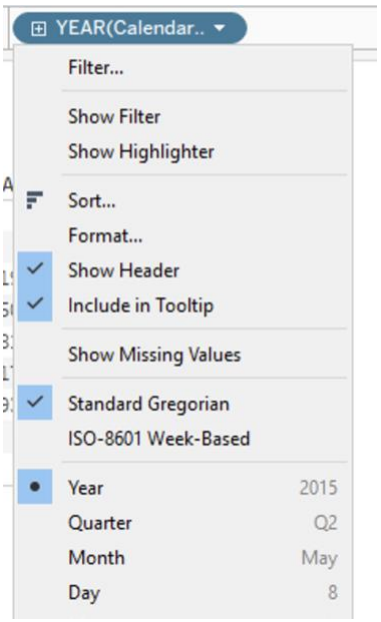


Select the first option in the menu, which is a “text table”. This will display the values in a concise form. If the rows and columns switched when you clicked on the menu option, drag the “Measure Names” to the columns and the “**YEAR(Calendar Date)**” to the rows.

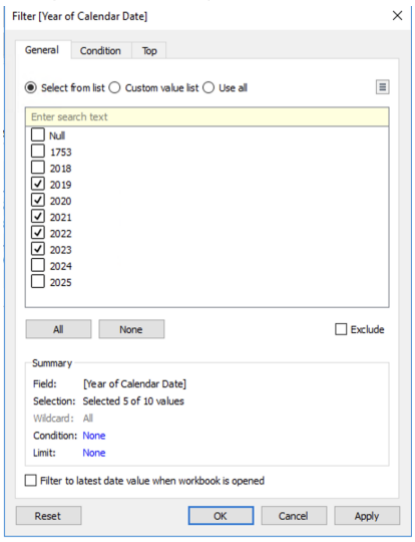




Let’s filter out certain years, since only 2019-2023 has data. Hover over the year field. Select the down facing arrow to view more options. Select “Filter”.

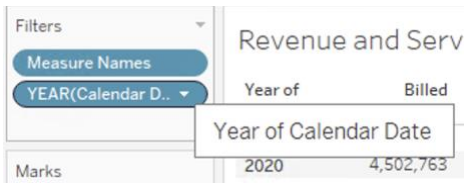


Only select the years 2019, 2020, 2021, 2022, and 2023.

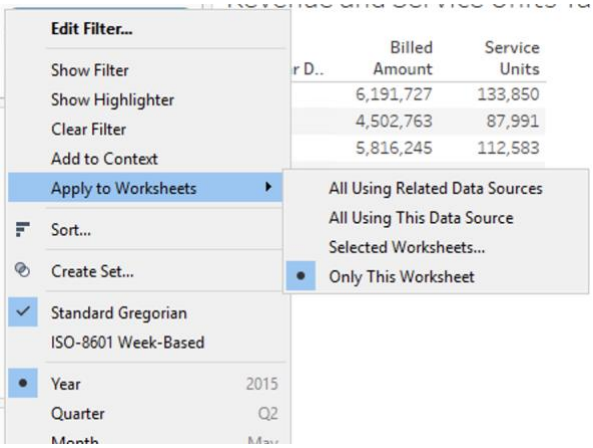


Click OK.

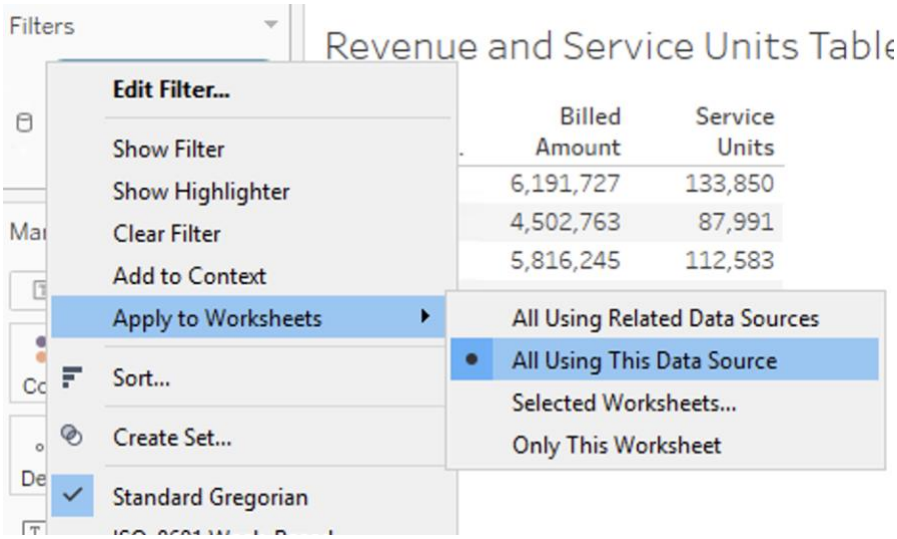
This applied the filter to this worksheet. We want to apply these years to every sheet. Under filters, select "**YEAR(Calendar Date)**".



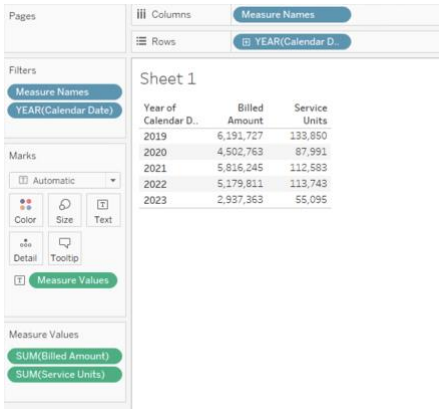
Select “Apply to Worksheets”.



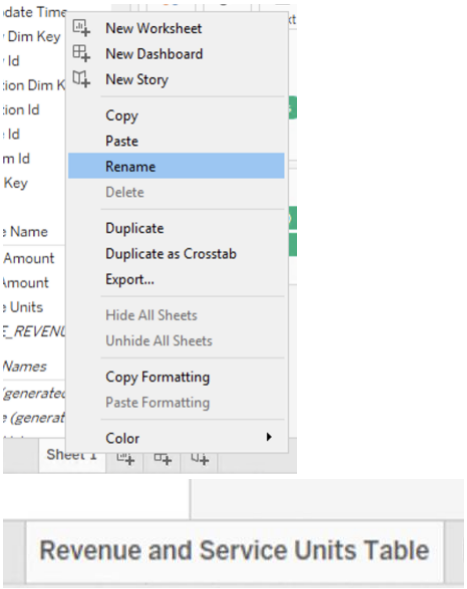
Then select “All Using This Data Source”. This will apply the filter to all sheets using the data source we setup earlier.



Now we have our Revenue and Service Units Table! Notice if you hover over values, a tooltip will appear with the year and value.



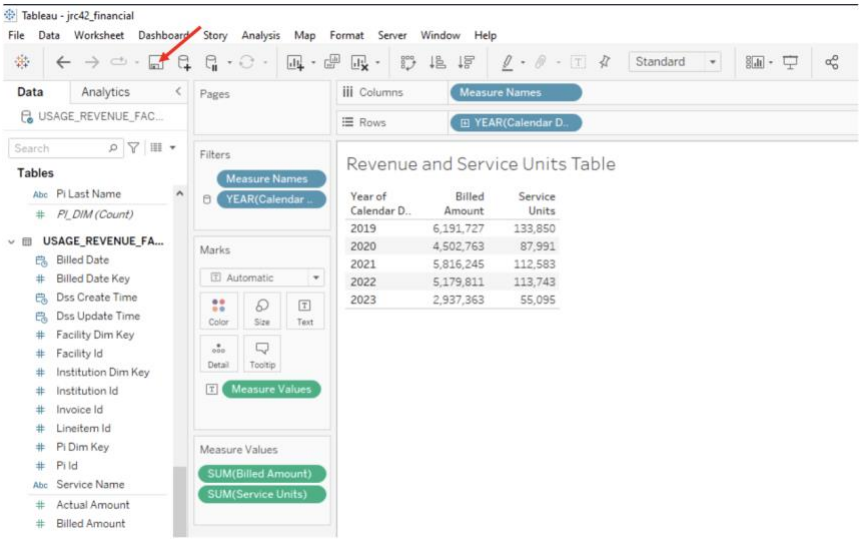
Now we will rename the sheet. Right click on the "Sheet 1" tab on the bottom and select "Rename". Type in "Revenue and Service Units Table" in the highlighted area and select "Enter". You should see the sheet name change.



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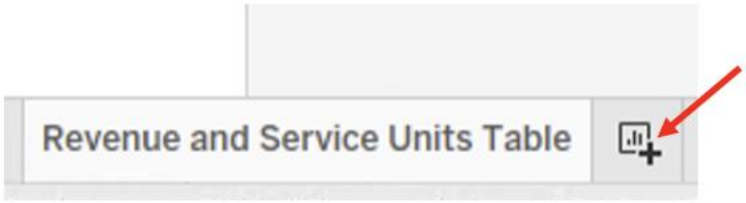
Assignment 6: Tableau Dashboard

Remember to save your work frequently by clicking on this icon.



Sheet 2: Bar and Line chart

Let's create another worksheet. Select the icon with the bar chart and plus sign. This is the button to create a new worksheet.

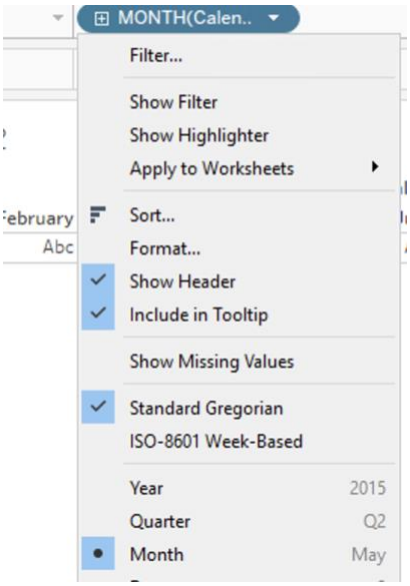


This time, we want to view the trends over a specific month. Similarly, drag **Calendar Date** from the **DATE_DIM** field to columns.

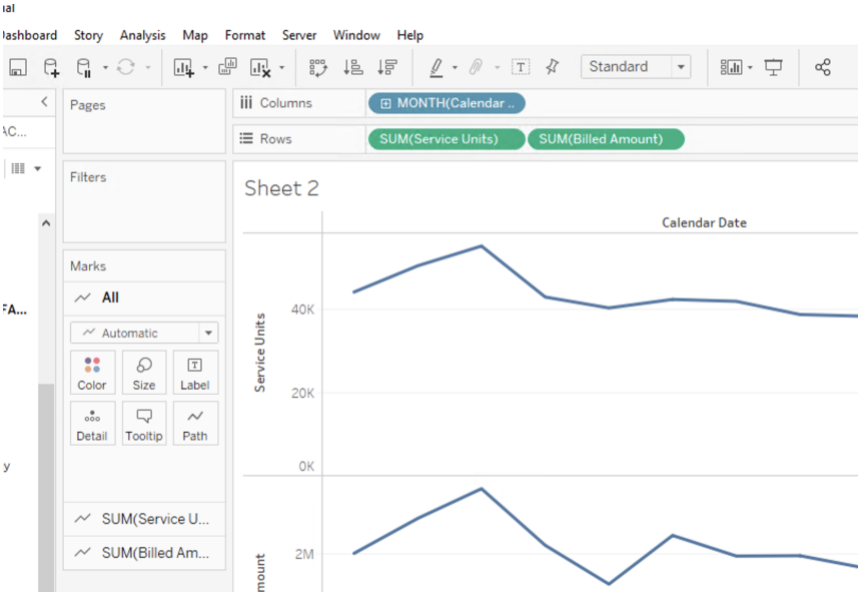
Click on the field for the more options dropdown. Select **Month**.

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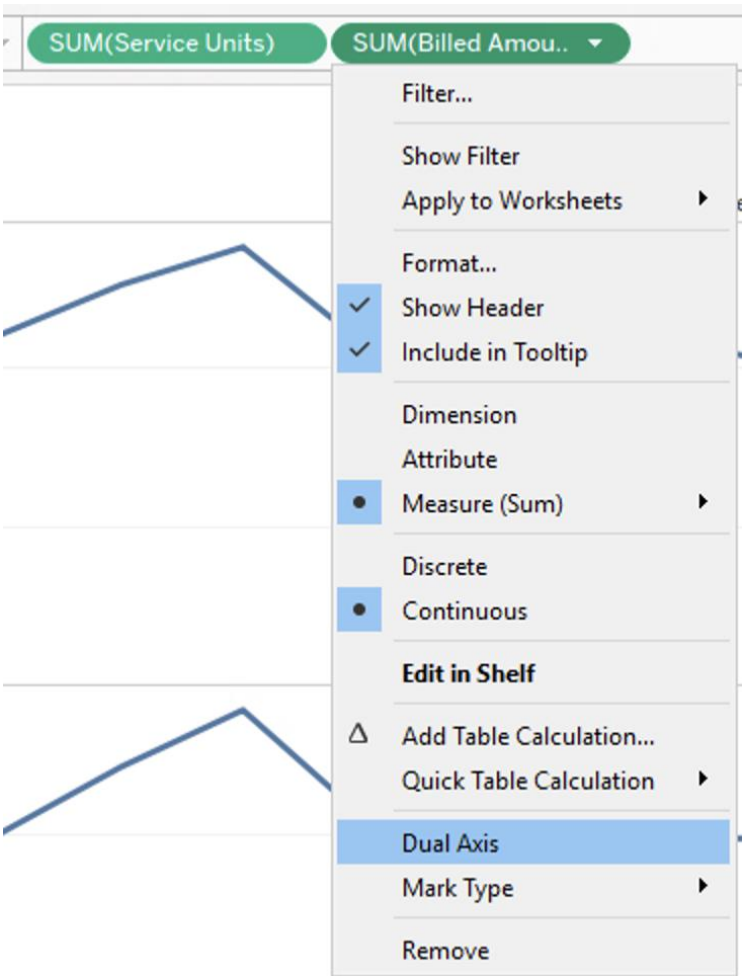
Assignment 6: Tableau Dashboard



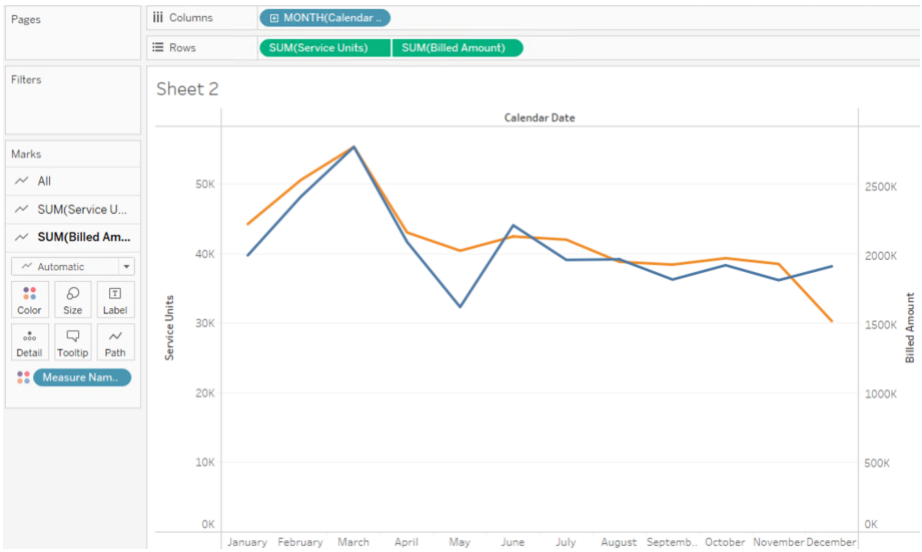
Drag “Service Units” and “Billed Amount”, from the usage_revenue_fact, to the rows.



We want to see the trends on one chart. We will create a **dual axis**. A chart with a dual axis has one axis for a field and an axis with a different scale for another field. Select the “Billed Amount” dropdown menu.



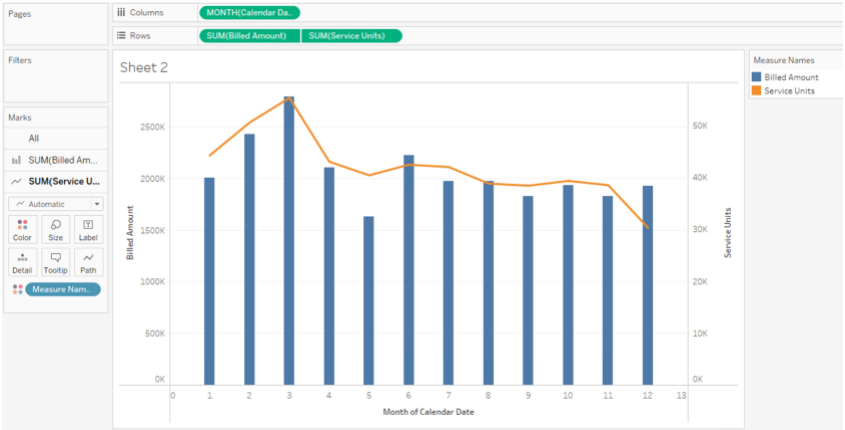
Click “Dual Axis”.



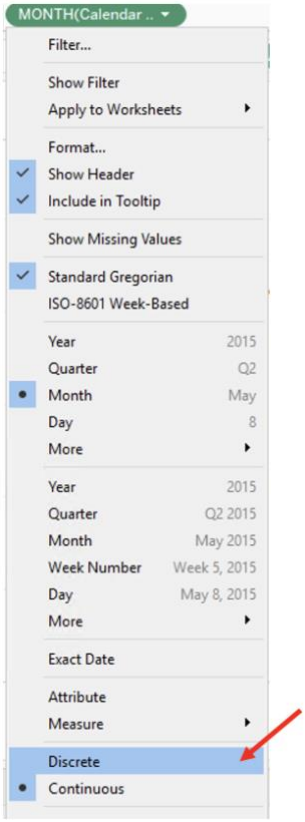
Now both trends are on the same chart. Let's change one of the measures to a bar chart. Click "Show Me".



Select the "dual combination" option. This will graph a bar chart and a line chart on the same axes.



Let's add the month names to the x-axis. Currently, our month field is continuous, so it is continuing past 12. We want to only show the months January-December. Click the "MONTH(Calendar Date)" field to view the dropdown.



Select "Discrete".



Great! Now we have a bar and line chart.

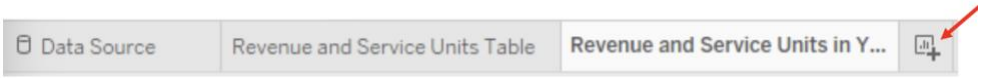
Rename the sheet by right clicking on “Sheet 2”.

Right click on sheet 2 tab and type in “Revenue and Service Units in Year”. We will change this name later.

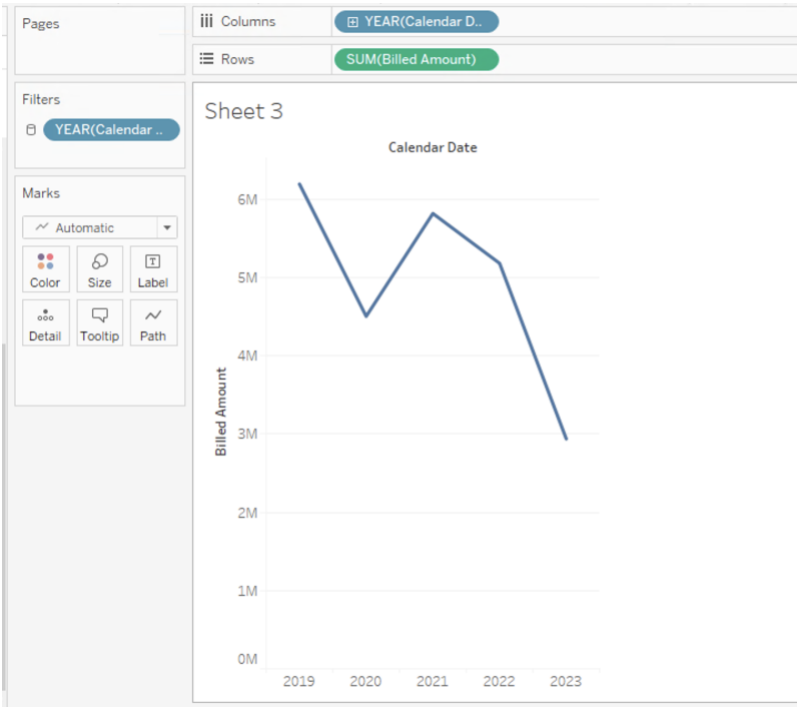
Sheet 3: Multiple Bar Charts

We’ve created some charts with a general overview of trends. Let’s dive deeper into specific colleges and departments within Cornell.

Add a new sheet by clicking the worksheet icon.



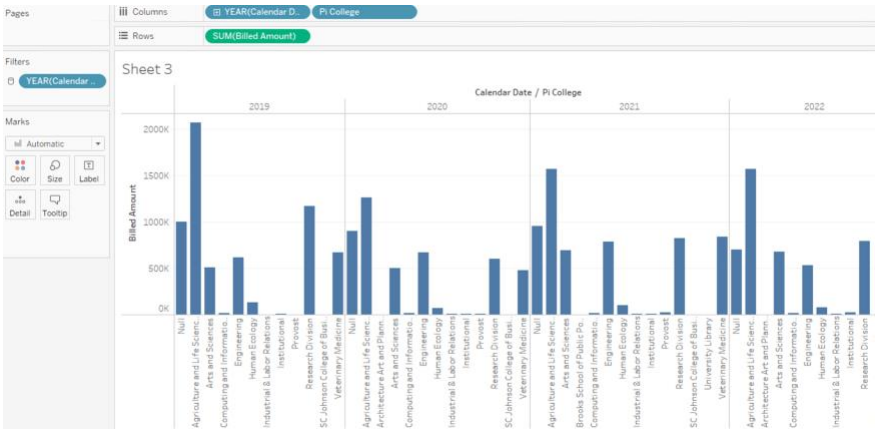
Once again, drag the “Calendar Date” field from the DATE_DIM table to the Columns. Drag “Billed Amount” from USAGE_REVENUE_FACT to the Rows.



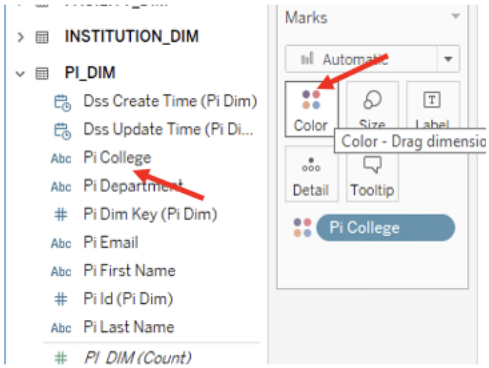
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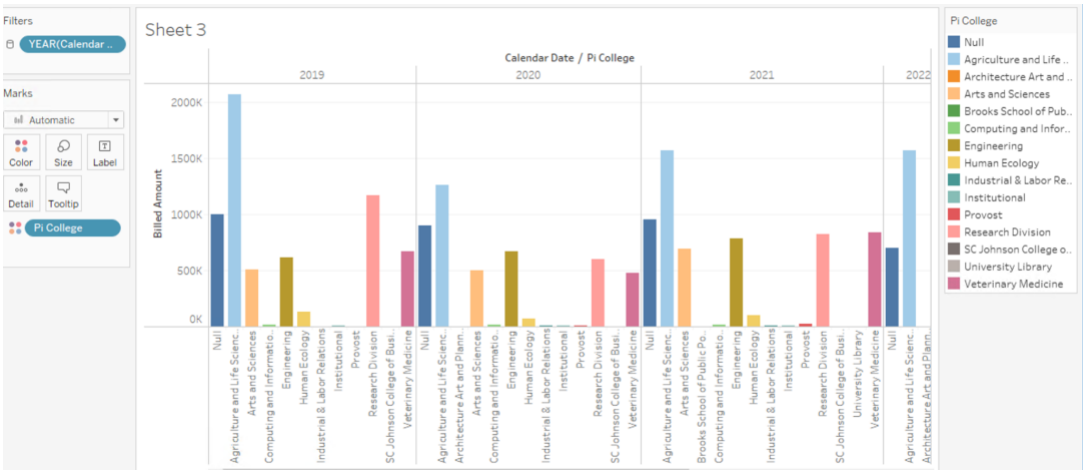
Assignment 6: Tableau Dashboard

We want to see the revenue trends for specific colleges. Now drag “Pi College” from PI_DIM to Columns.



Let’s differentiate each PI college with different colors. From PI_DIM, drag Pi College to the “Color” icon.

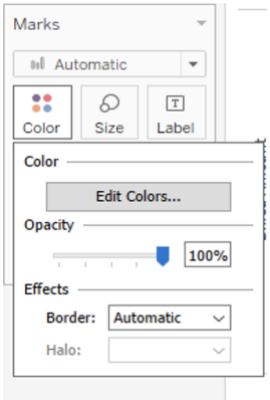




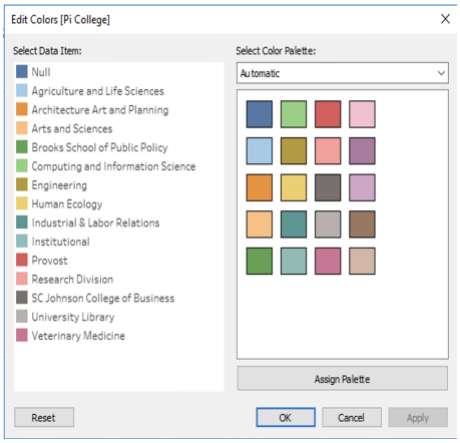
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Assignment 6: Tableau Dashboard

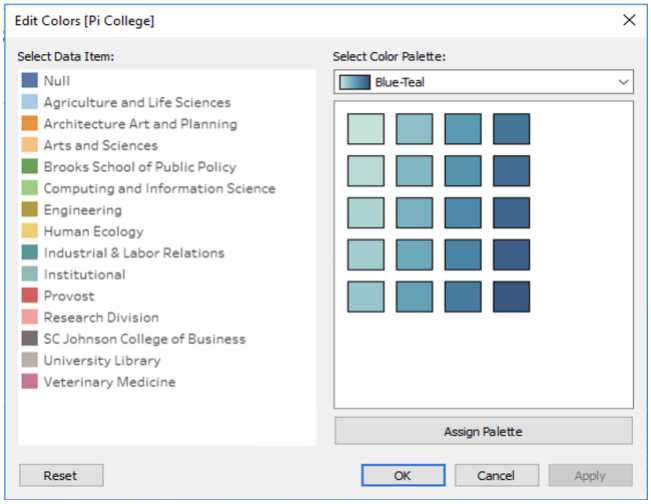
Tableau has built in color schemes. To change the color scheme, select the “Color” icon.



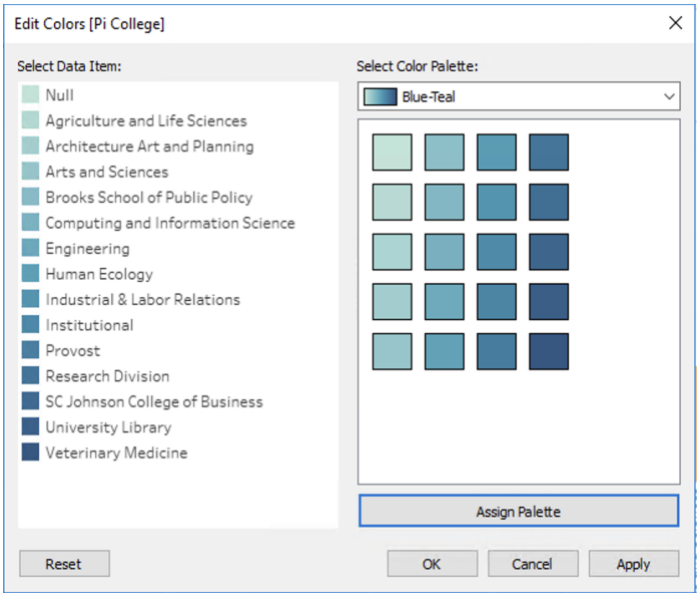
Select “Edit Colors…”.



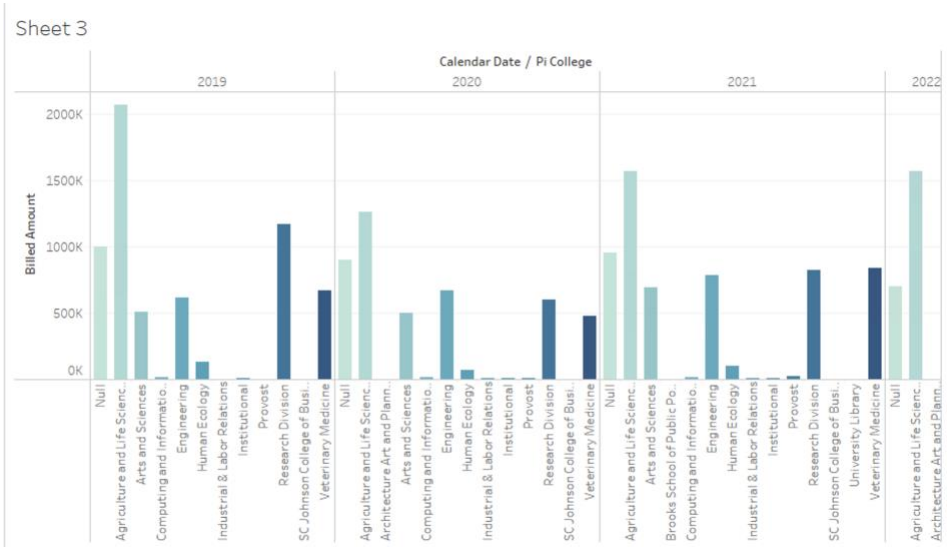
Click the dropdown menu under “Select Color Palette”. Let’s select “Blue-Teal.”



Click “Assign Palette” to apply the palette to the data.



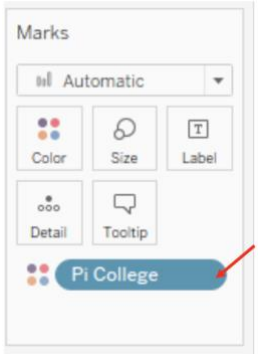
Click “Apply”, then OK.

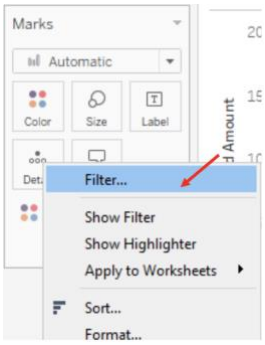


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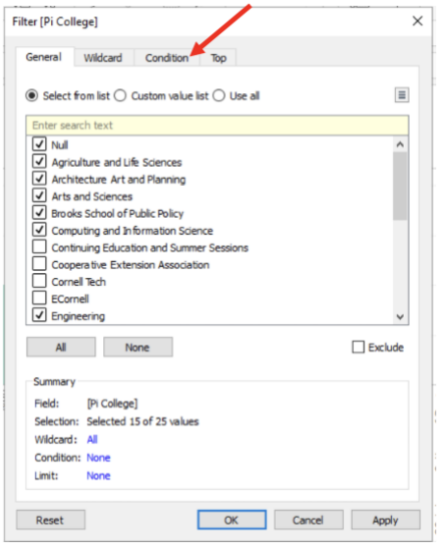
Assignment 6: Tableau Dashboard

We want to filter for the more active colleges. Select Pi College under Marks and then “Filter…”.





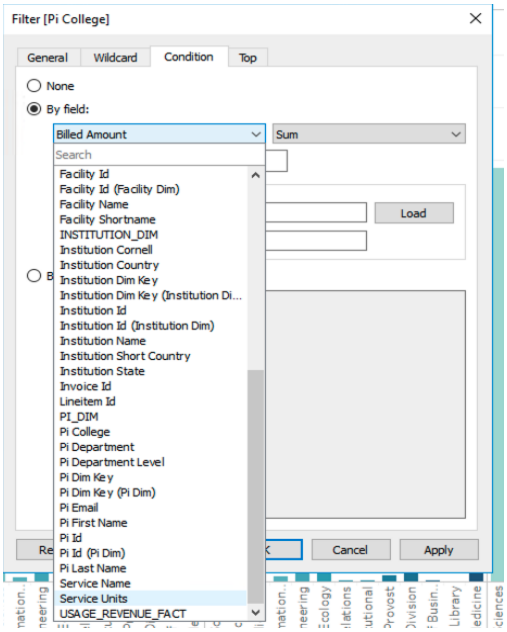
You should see the popup below. Click the “Condition” tab. This will allow you to filter based on a certain criteria.



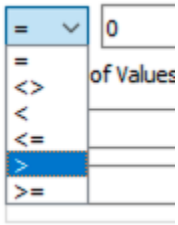
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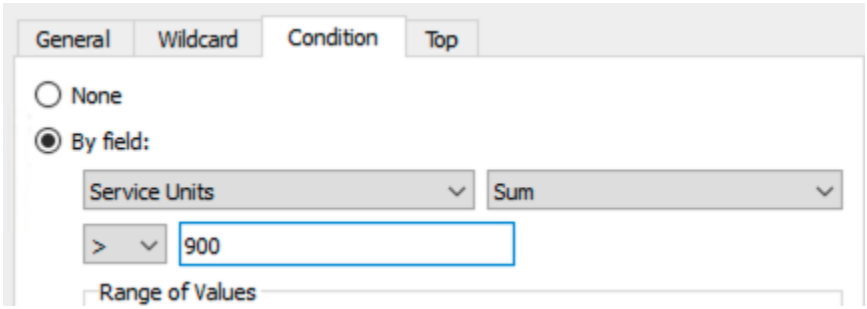
Assignment 6: Tableau Dashboard

For our purposes, we will filter for colleges that have over 900 service units. Select the “By field” button. In the dropdown, select “Service Units”.



Change the operator to greater than and enter 900.

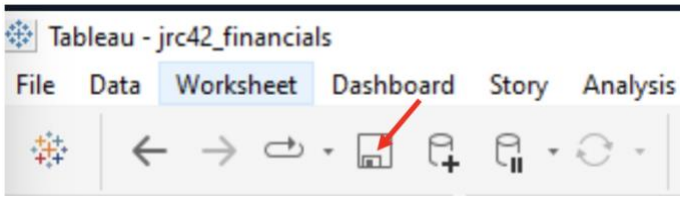




Select “Apply” then “Ok”.

Rename the sheet to “Pi College Bar Chart”

This is a good time to save your Tableau worksheet again.

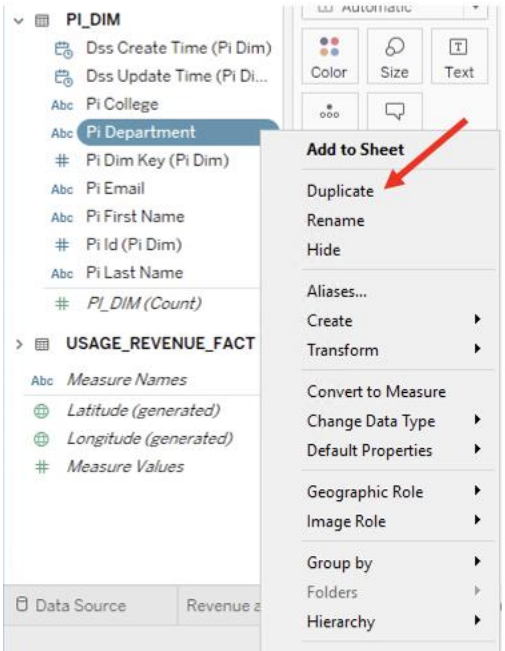


Sheet 4: Hierarchy Table

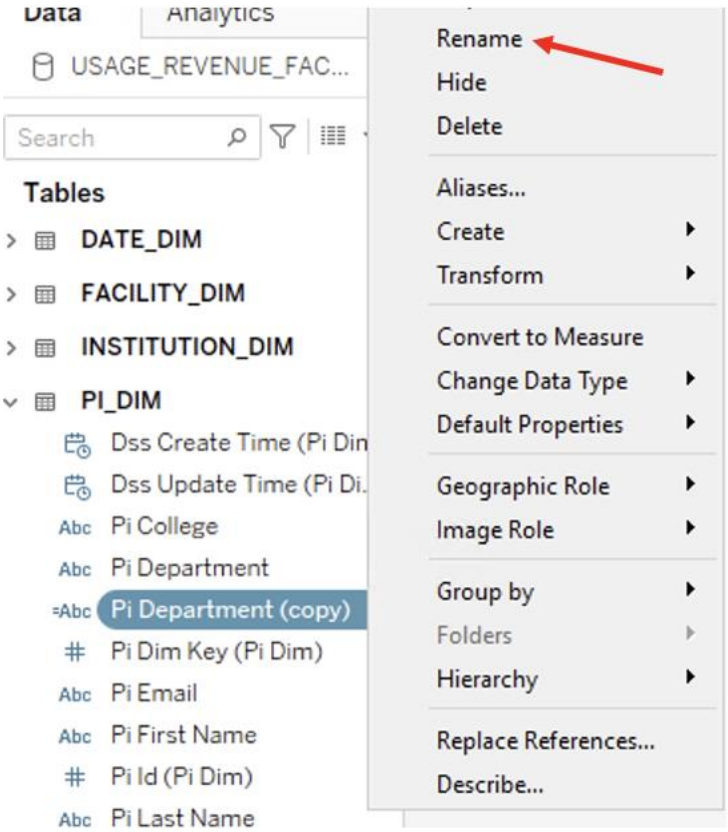
A hierarchy is a system that has levels from highest to lowest. This enables an entity to be present at different levels in a visualization. We are going to use this to look further into Pi departments and facility groups.

Create a new worksheet.

Make a duplicate copy of the fields **Pi Department** and **Facility Group**. To make a duplicate, click on the field and click Duplicate in the dropdown.

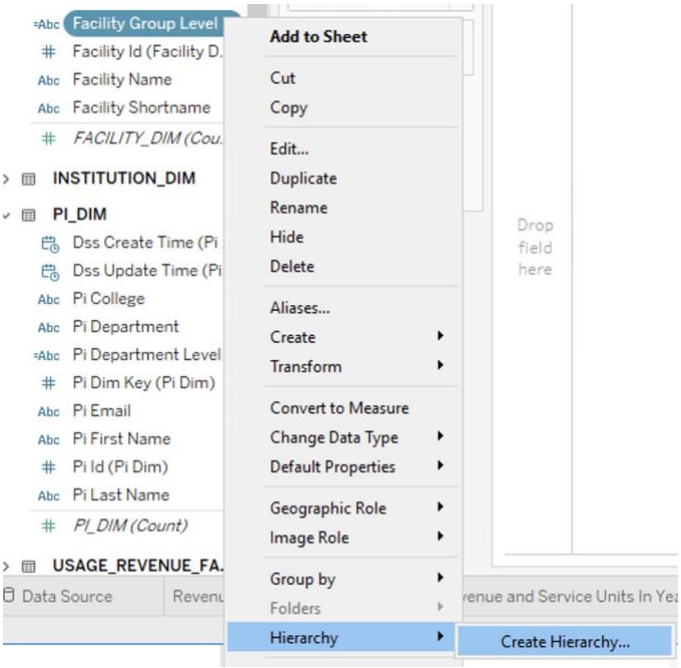


Rename the new field to Pi Department Level.

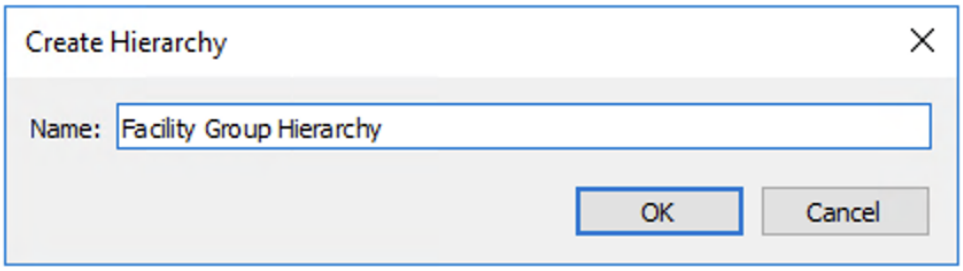


Do the same for the “Facility Group” and rename the duplicate field to “Facility Group Level”.

To create the hierarchy, click on the new field “Facility Group Level” and select “Hierarchy”, “Create Hierarchy” in the dropdown.



A popup will appear. Name the hierarchy “Facility Group Hierarchy” and click “OK”.



Now you’ve created a hierarchy with Facility Group Level Inside. Drag the new field “Pi Department Level” into the Hierarchy. Make sure the field is above Facility Group Level.

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Assignment 6: Tableau Dashboard



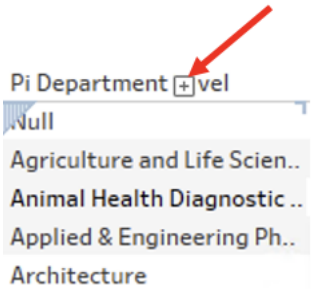
Create the hierarchy table by dragging the "Facility Group Hierarchy" field into the rows and drag " Billed Amount" and "Service Units" into the columns. Your table should look like this. If not appearing as a table, select "text tables" from the "show me" options.

Note that if Tableau tries to add ALL of the measures to your table, so many extra columns. It is probably easiest to delete your sheet 4, add a new then, then drag Pi Department Level Hierarchy and the two measures over again. Tableau did some funny stuff the first time I attempted sheet 4.

Sheet 4

Pi Department Level	Billed Amo..	Service Uni..
Null	3,883,631	55,127
Agriculture and Life Scien..	664	10
Animal Health Diagnostic ..	1,454	342
Applied & Engineering Ph..	26,659	2,282
Architecture	1,430	31
Atkinson Center for a Sus..	155,442	281
Baker Institute	291,516	8,400
Biomedical Engineering	2,289,828	34,873
Biomedical Sciences	923,861	22,222
CALS Animal Science	130,044	6,284
CALS Biological & Environ..	114,861	3,316
CALS Biology Stats & Com..	2,825	2
CALS Computational Biolo..	97,629	3,296
CALS Earth and Atmosphe..	5,228	20
CALS Ecology & Evolution..	382,535	2,675
CALS Entomology	341,646	12,334
CALS Food Science	460,001	23,597
CALS Lab of O - Administr..	396,408	4,307
CALS Microbiology	237,885	18,328
CALS Molecular Biology & ..	1,024,250	39,425
CALS Natural Resources	136,175	2,341
CALS Neurobiology & Beh..	93,560	698
CALS Nutritional Sciences	208,847	4,559
CALS Office of the Assoc D..	12,111	103

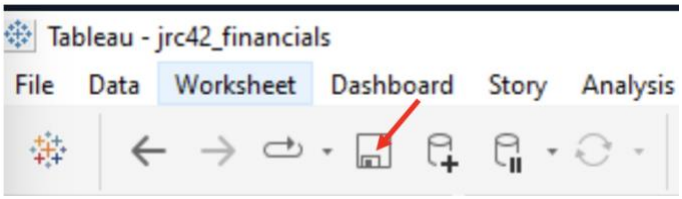
Notice if you hover over the column, “Pi Department Level”, a plus sign appears. Click on the plus sign. The plus sign is the symbol for expanding to the next level in the hierarchy. Notice how it adds a Facility Group column.



Pi Department Level	Facility Group Level	Billed Amo..	Service Uni..
Null	Bioinformatics	149,846	503
	BIP	3,322	16
	Epigenomics	17,982	125
	Flow	4,379	60
	Genomics Facilities	1,314,212	21,962
	Imaging	142,574	1,035
	Parking	63	21
	Preoteomics-Metabolomi..	713,413	12,258
	TREx	20,197	127
	Unknown	1,517,644	19,021
Agriculture and Life Scien..	Genomics Facilities	664	10
Animal Health Diagnostic ..	Genomics Facilities	1,454	342
Applied & Engineering Physics	Flow	1,796	30
	Genomics Facilities	13,565	1,905
	Imaging	7,306	309
	Unknown	3,992	38
Architecture	Imaging	638	24
	Unknown	793	7
Atkinson Center for a Sustainable Future	Bioinformatics	138,438	149
	Genomics Facilities	10,590	95
	Unknown	6,414	37
Baker Institute	Bioinformatics	73,778	842
	Epigenomics	1,495	9
	Flow	8,814	165

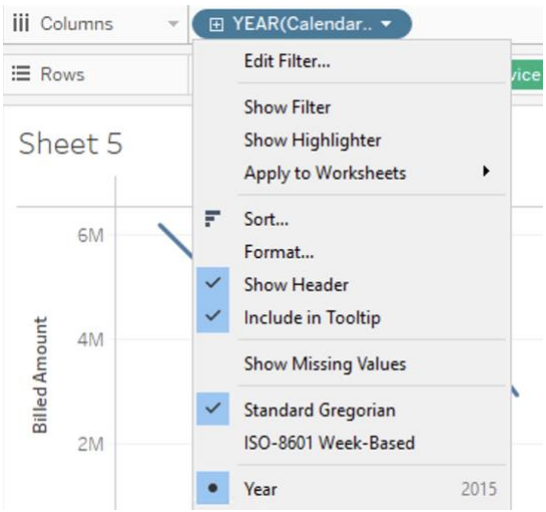
Rename the sheet “Facility Group Hierarchy Table”.

This is a good time to save your Tableau worksheet again.



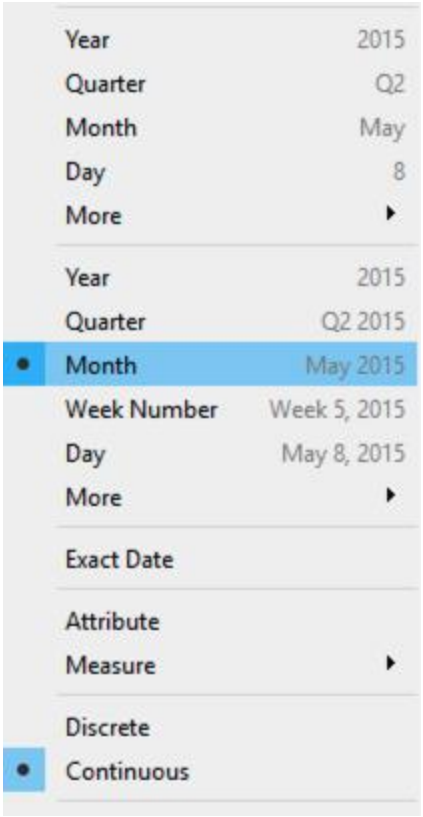
Sheet 5: Multiple Line Graph

Add another sheet.
Drag the fields “Billed Amount” and “Service Units” to the rows. Drag “Calendar Date” to the columns. It will automatically apply the YEAR() function to the field. Click on the field and change it to Month. Note there are two month values. Choose the second one. Additionally, select “Continuous” instead of “Discrete”.



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Assignment 6: Tableau Dashboard



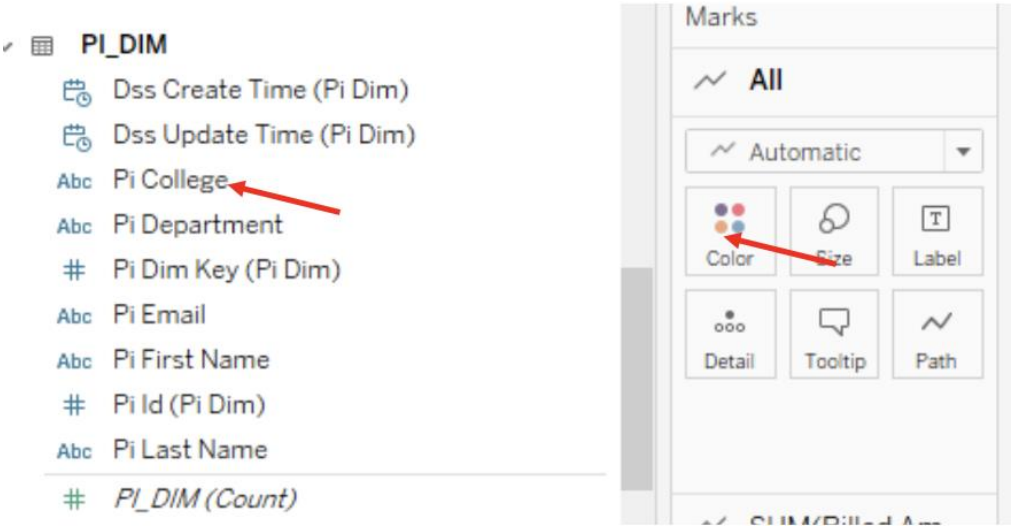
You should have what looks like two line graphs.

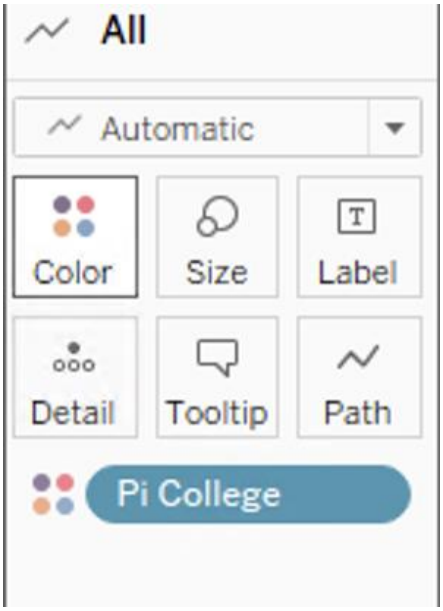


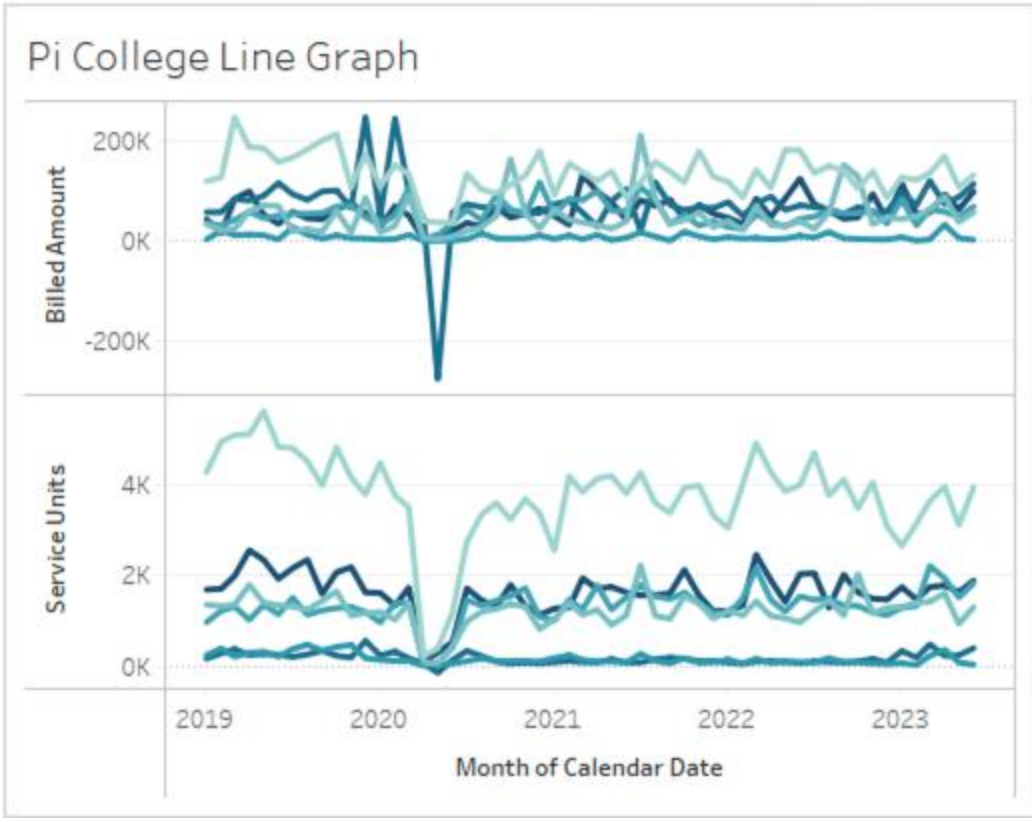
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Assignment 6: Tableau Dashboard

We want to see the trends for different Pi colleges, with a line representing each college. To do this, drag the field “Pi College” to the colors icon. If color palate does not default to blue-teal, set it to blue_teal.







Now we can view the trends for each college for both the revenue and service units. Rename the sheet “Pi College Line Graph”.

Now we’ve finished all 5 sheets. We can begin building the dashboard.

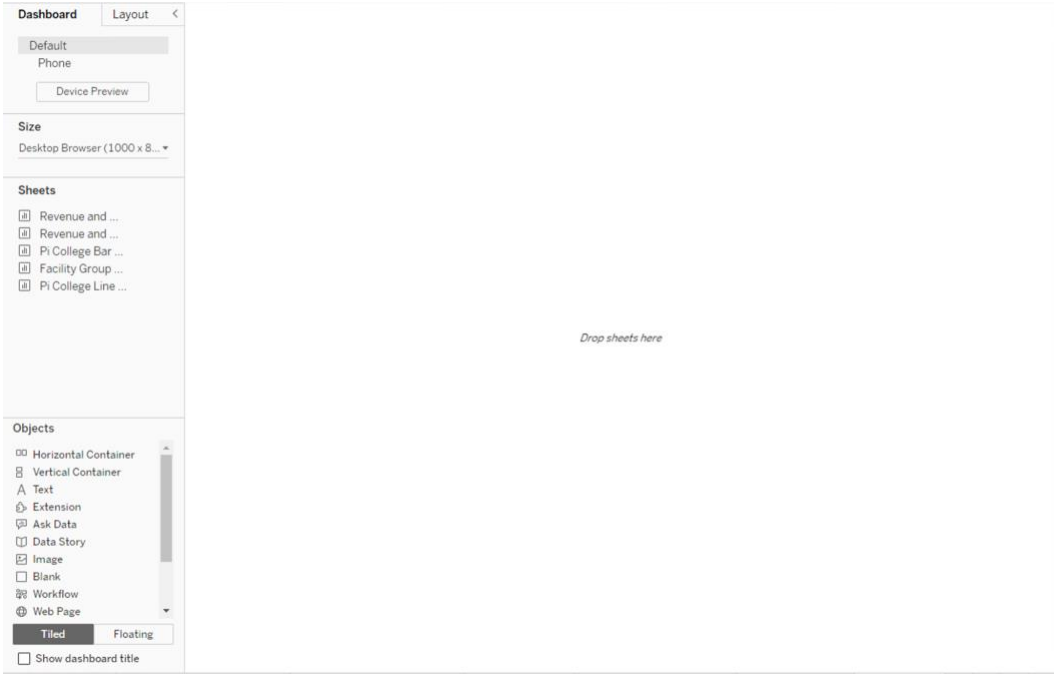
Building the Dashboard

A Tableau dashboard is a collection of multiple sheets that you can view at the same time.

To create a dashboard, select the window icon with a plus sign.

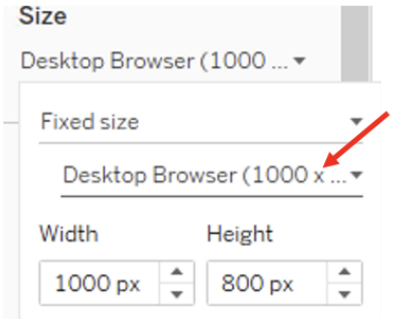


You should have a screen that appears like this with the 5 sheets we created on the left side panel.



Click on the “Dashboard 1” tab and rename the dashboard to “Revenue and Units of Service Dashboard”.

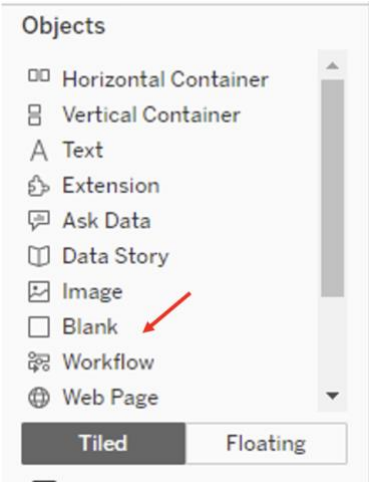
To change the size of the dashboard, click on the Desktop Browser dropdown under Size.



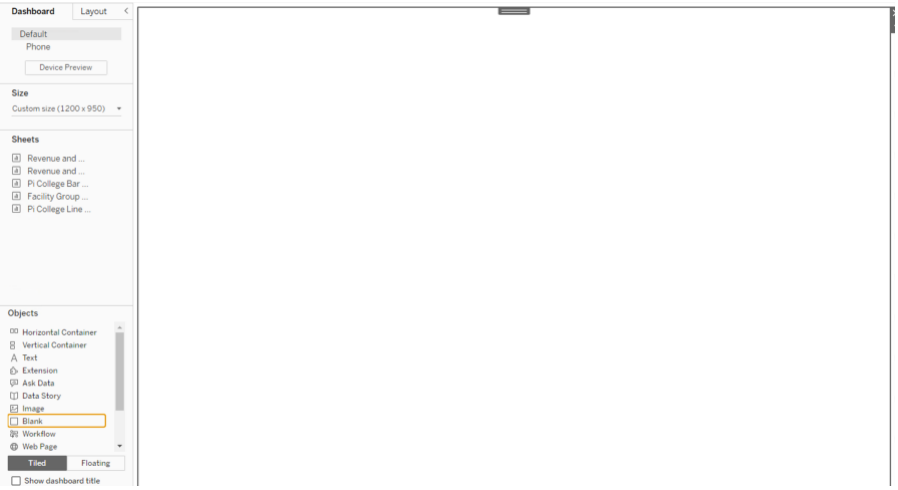
We can customize the size by changing the width and height. Change width to 1200 and height to 950.



At the bottom of the panel, you will notice objects. We will add these to the dashboard for alignment.

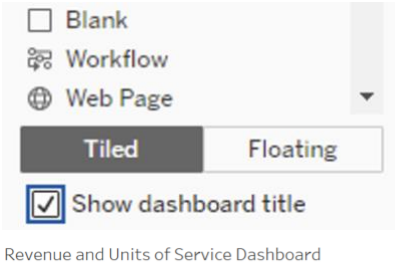


Drag a “Blank” object onto the dashboard. This will be the box that contains the objects of the dashboard.



Notice how a gray rectangle appears.

Select the “show dashboard title” checkbox. This is found in left pane, scroll to bottom. This will make the name of the dashboard appear at the top of the dashboard.

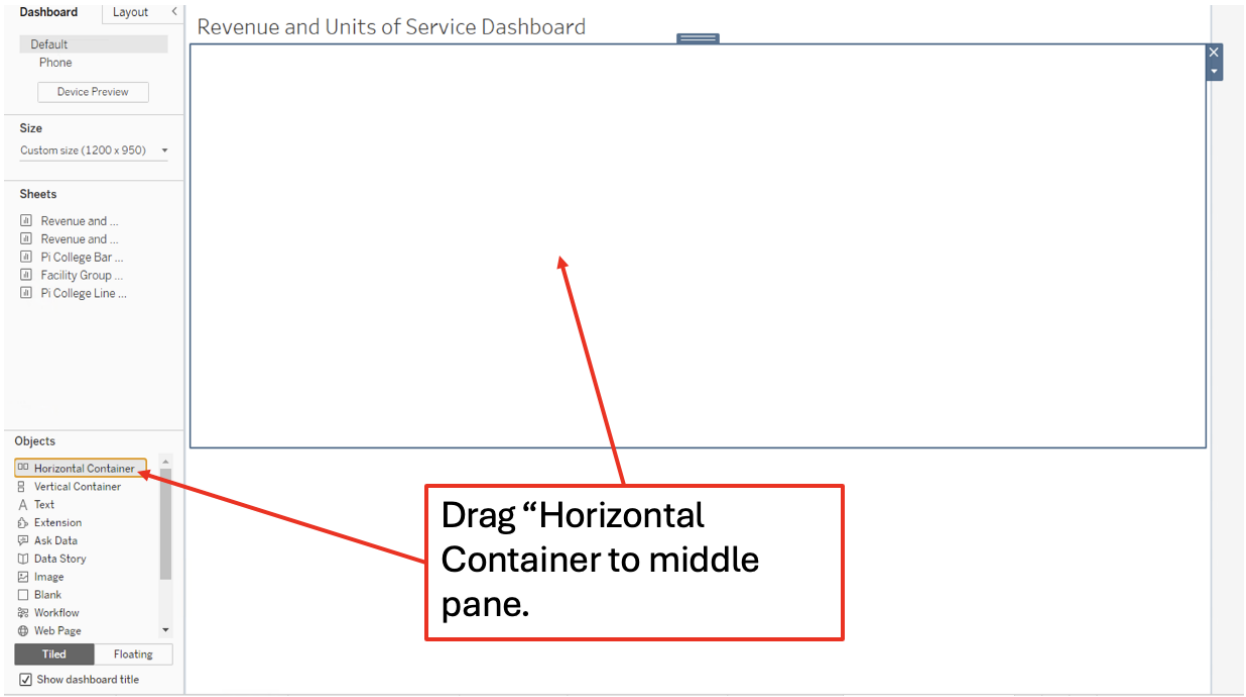


Adding Containers

Containers help to align items vertically and horizontally. A horizontal container can have items next to each other, while a vertical container allows for items to easily be aligned above and below each other. When you drag a container, you will see part of the dashboard get shaded in gray, which indicates which part of the dashboard you want the container.

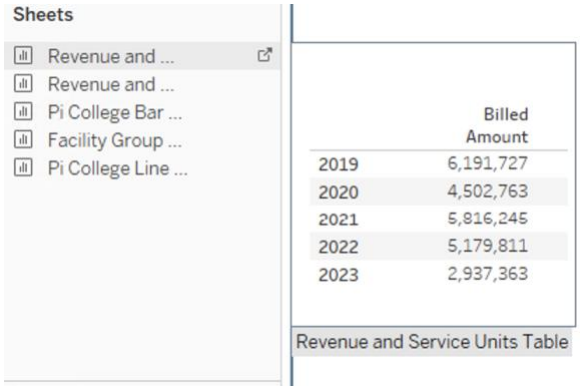
Watch this tutorial to learn more about layout containers before proceeding:
https://www.youtube.com/watch?v=96371LvULXM

First drag a horizontal container to the dashboard. Release the container when the shaded horizontal rectangle appears.

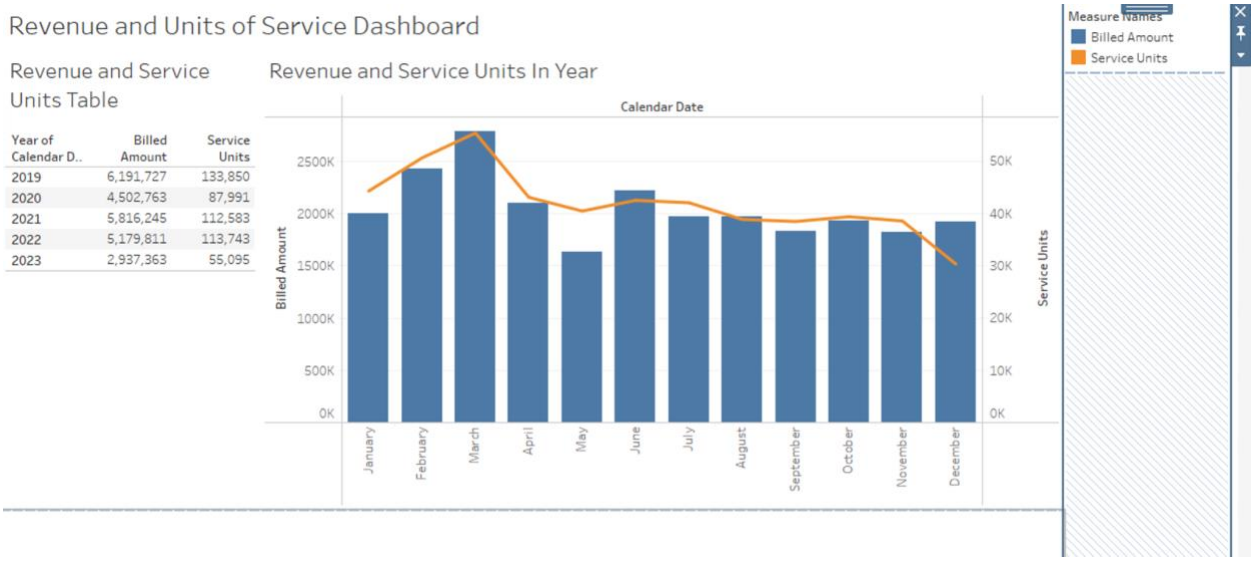


Notice how containers have a blue border.

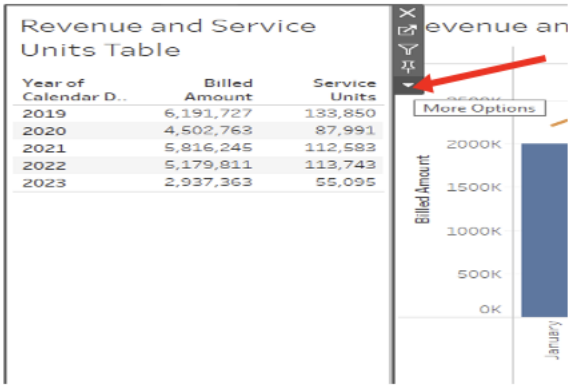
Next we are going to add sheets into the container. First drag the “Revenue and Service Units Table" to the left side of the container. Since there is nothing in the container, you can release it when the entire container is shaded gray. You can hover over the sheet names to see the full name and chart.



Next, drag the “Revenue and Service Units in Year” to the right side of the container. You want to release the chart when the entire border of the container is highlighted and there is a small gray rectangle shaded on the right side.



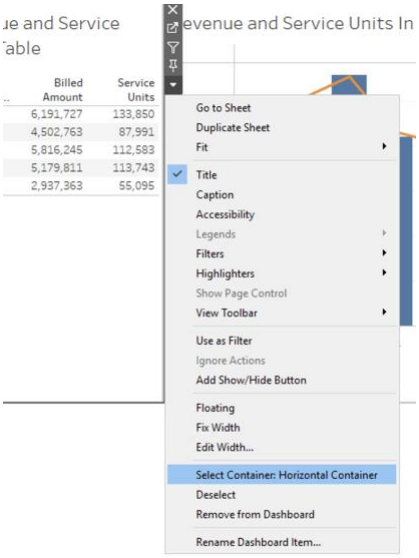
To check if the chart is actually inside a container, you can select a chart and click on the downward facing arrow.



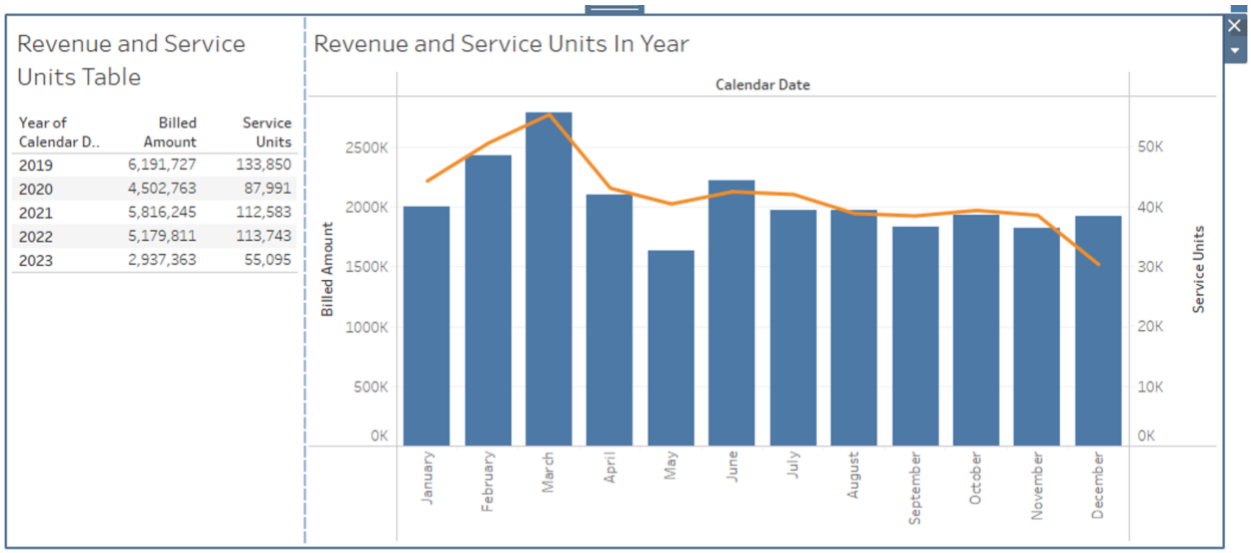
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Assignment 6: Tableau Dashboard

Select “Select Container: Horizontal Container”.

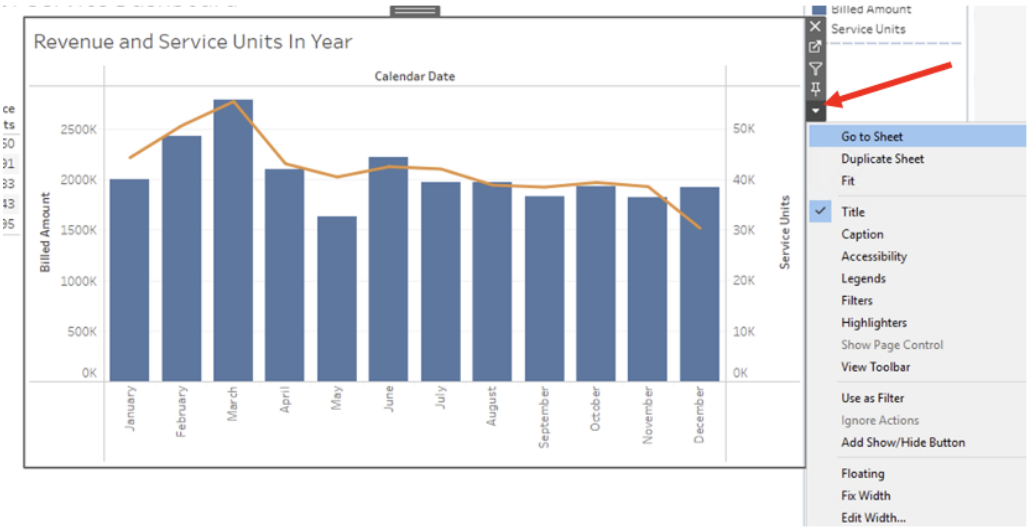


If done correctly, you should see the entire container border highlighted with a dashed line between the sheets.

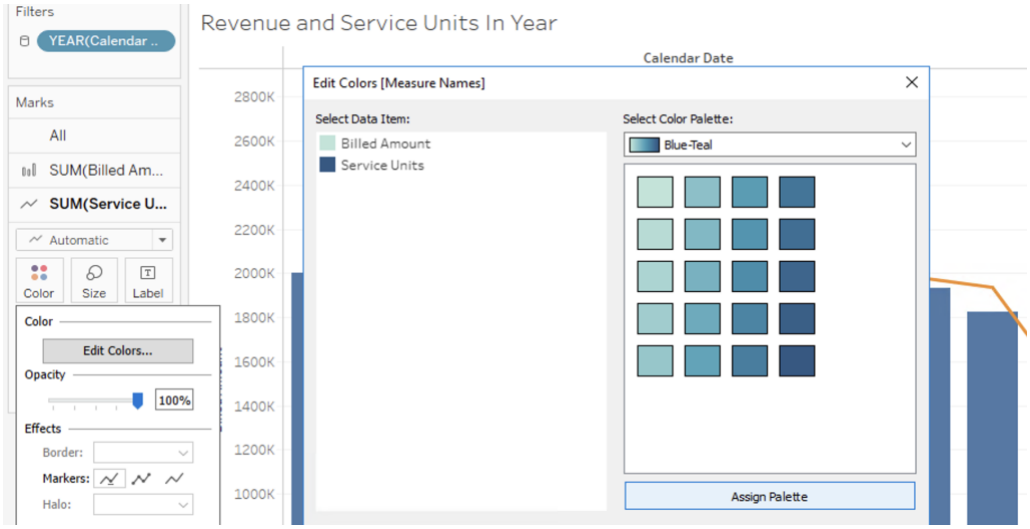


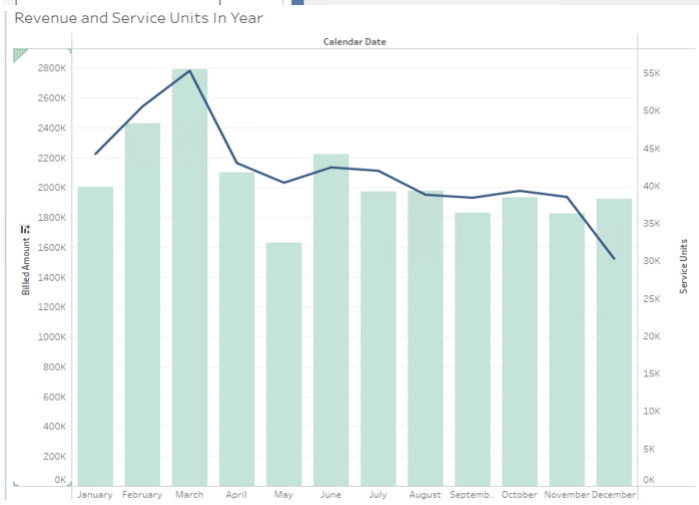
Notice how a legend appeared when adding the bar and line chart. First, let’s change the color of the sheet to match the theme. Select the sheet and the downward facing arrow, and click “Go to

Sheet". This allows us to go to edit the sheet.



Apply the Blue-Teal color scheme to the chart.

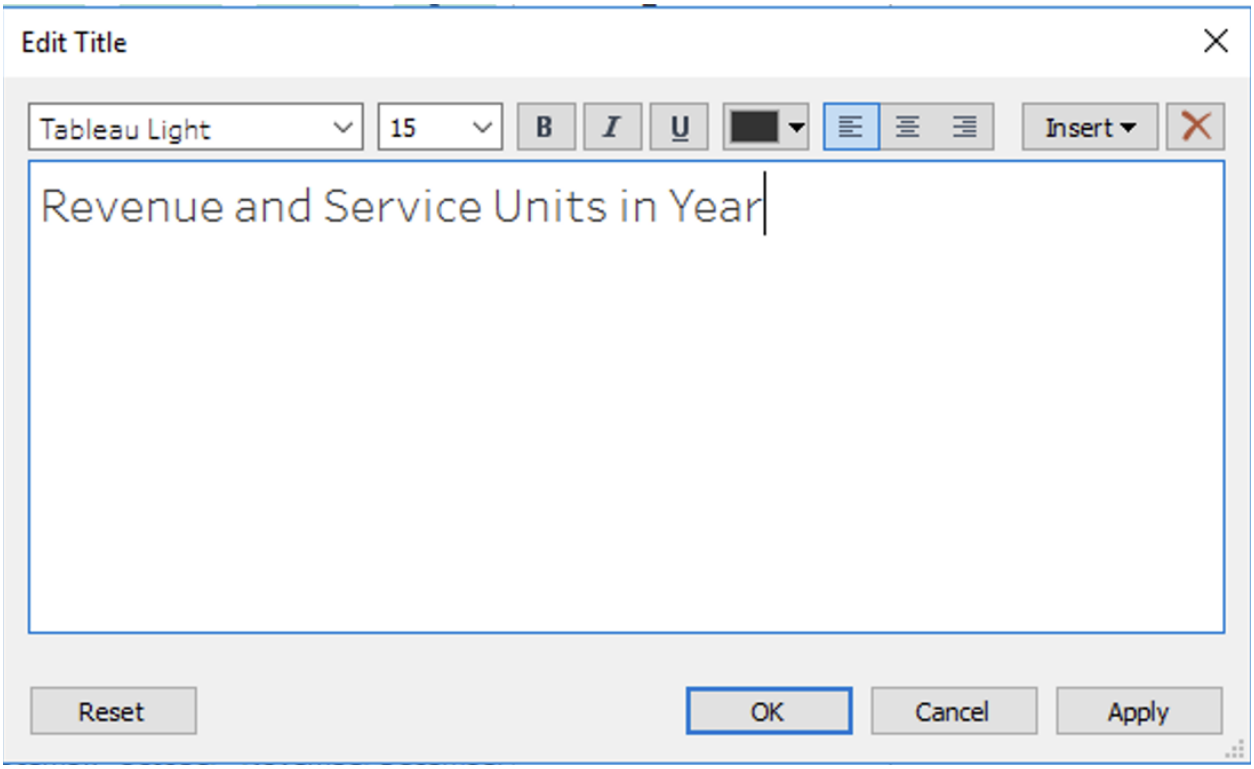




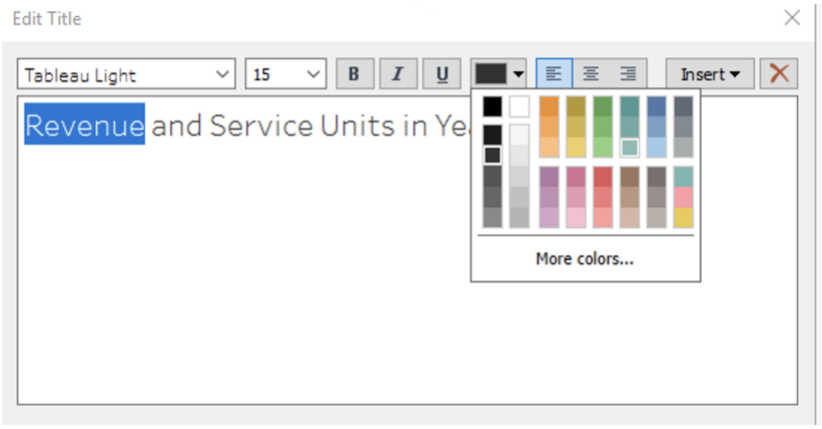
Let's make the title represent the legend. Double click the title of the sheet. *Note that the title is above the sheet, not the tab at the bottom.*

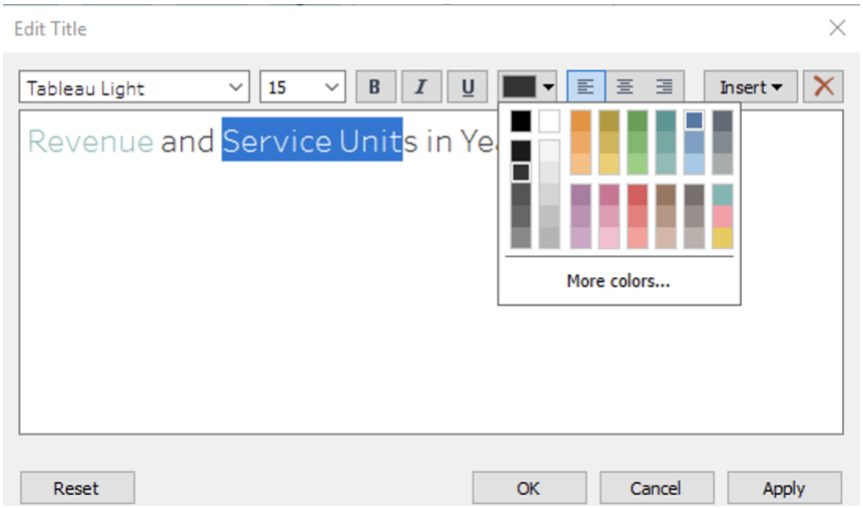


Currently the title is "<Sheet Name>", but let's rename it to the same name, "Revenue and Service Units in Year".

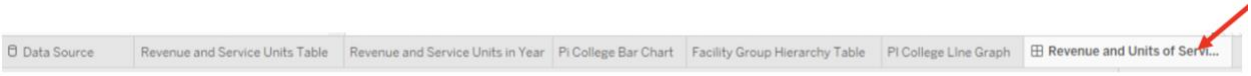


Highlight "Revenue" and change the color to the light teal. Repeat the same steps for "ServiceUnits" with the darkest blue. Hit "Ok".

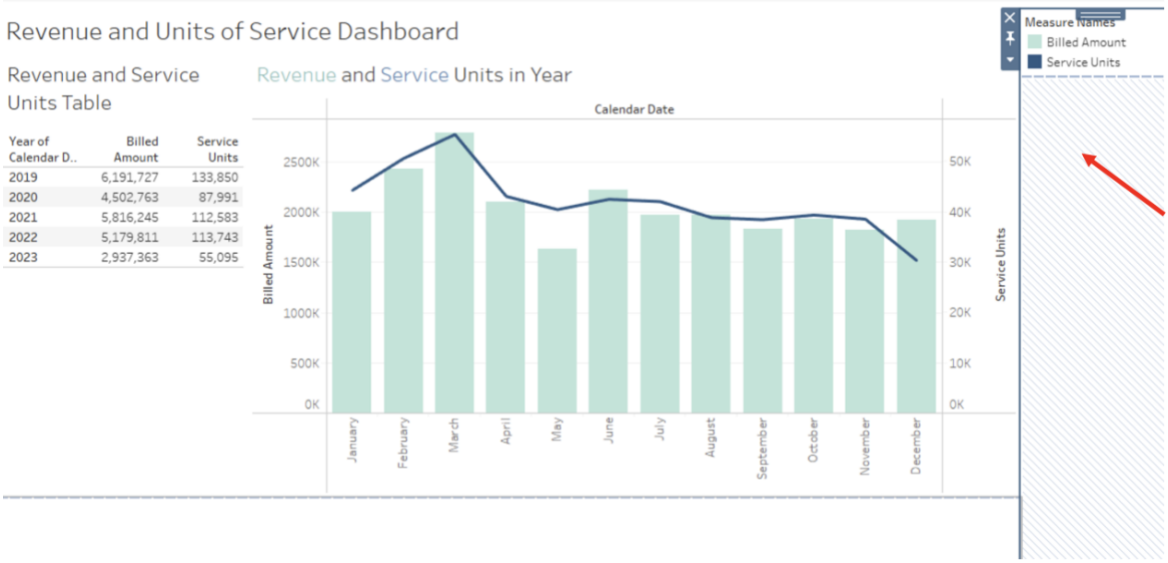




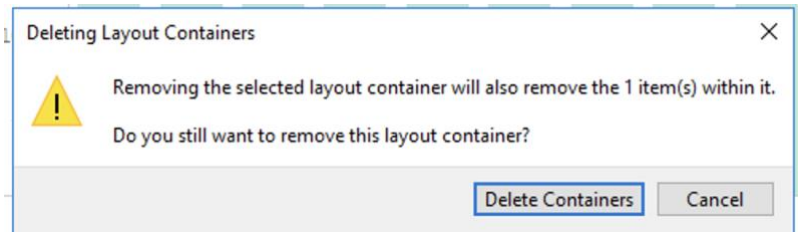
Click on the “Revenue and Units of Service” dashboard tab.



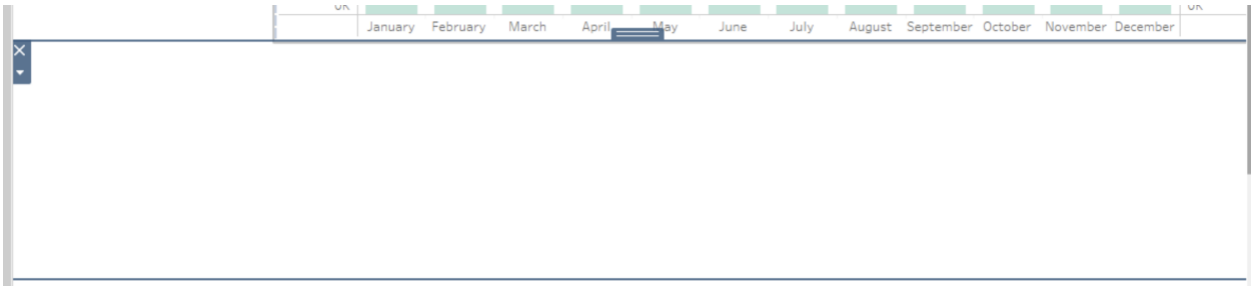
On the “Revenue and Units of Service” Dashboard, select the area below the legend. It should appear as a vertical blue rectangle.



Select the X icon. This will remove the legend from the dashboard. A popup will appear. Click “Delete Containers”.



Now, let's add another horizontal container below the first one. Drag the horizontal container below the charts.



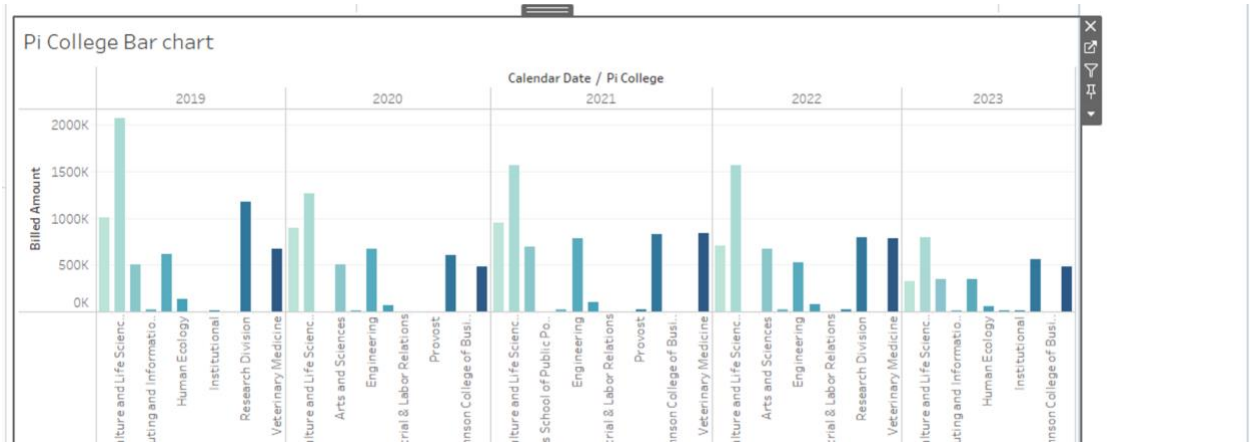
You can delete the blank container, found below the new horizontal container you just added. Simply click on the X.



Click on your new horizontal container, so it becomes highlighted.



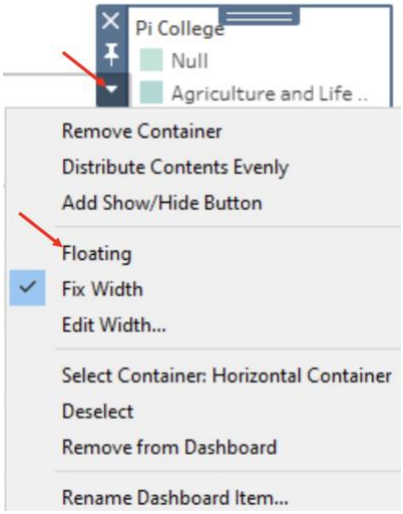
Drag the “Pi College Bar Chart” into the new container. Notice a legend appeared in a new container on the right.



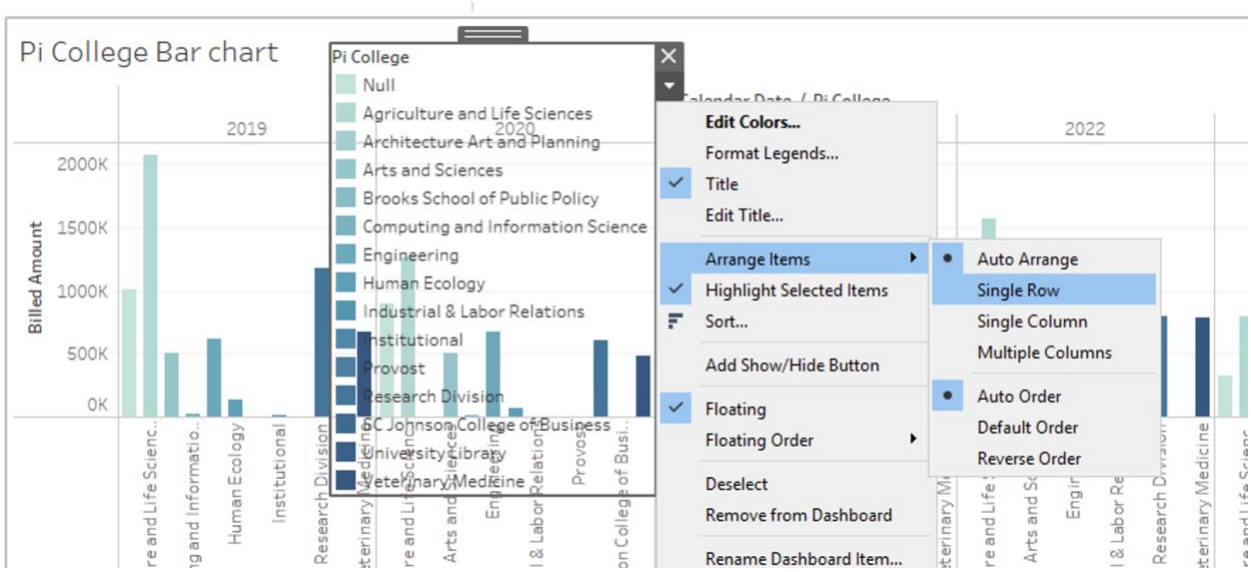
The legend looks like this.



Click the down arrow to the right of the legend, then select “Floating. This allows you to drag the item anywhere on the dashboard.



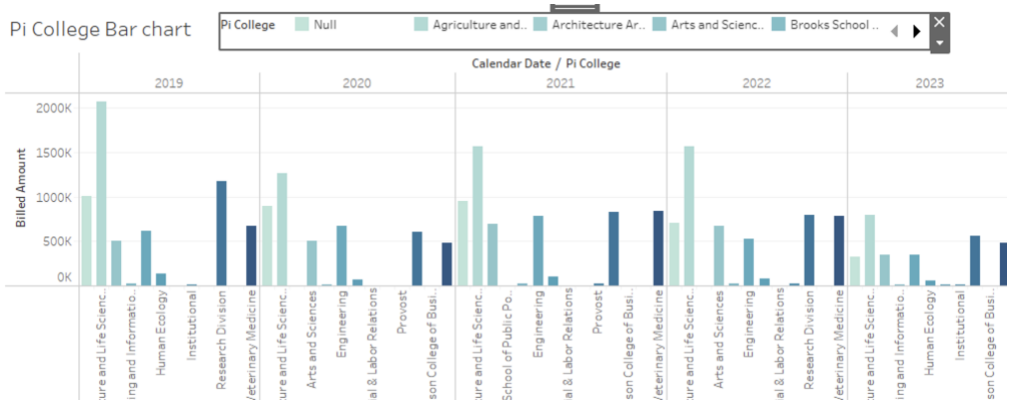
Drag the legend near the title of the chart. Select the dropdown and hover over “Arrange Items” and select “Single Row”.



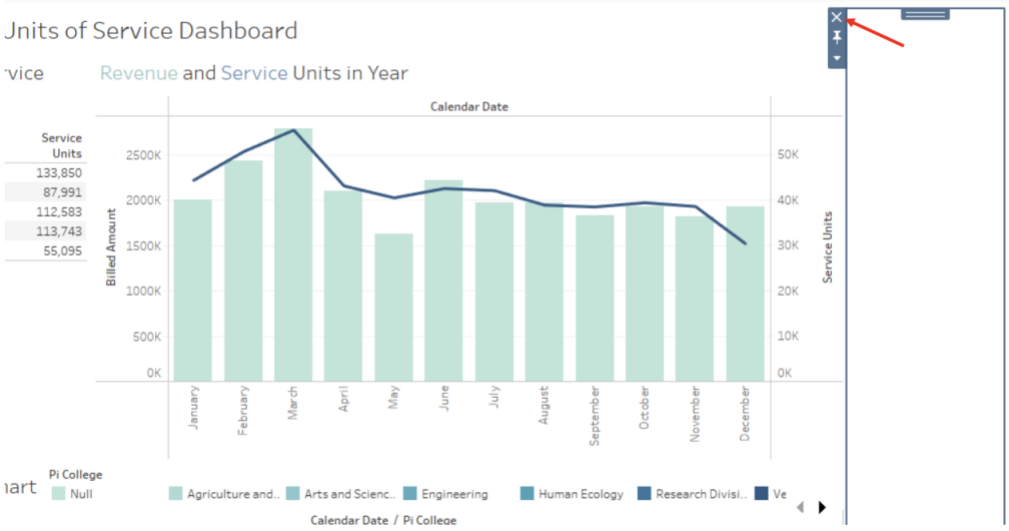
You can resize the legend to fit in the space above the chart by dragging the black border into the desired shape.

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Assignment 6: Tableau Dashboard



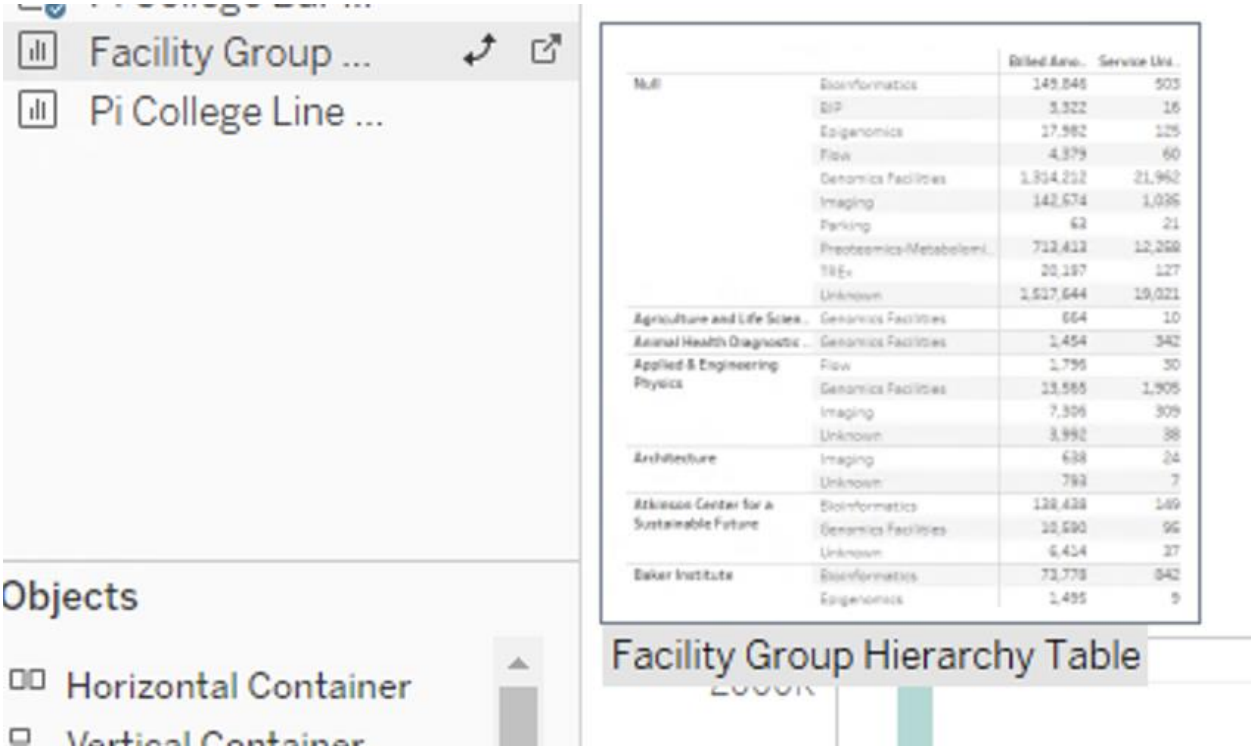
Click in the space to the right, where the legend was originally found. This will highlight the empty container left behind. Select the X to remove the empty container from the dashboard.



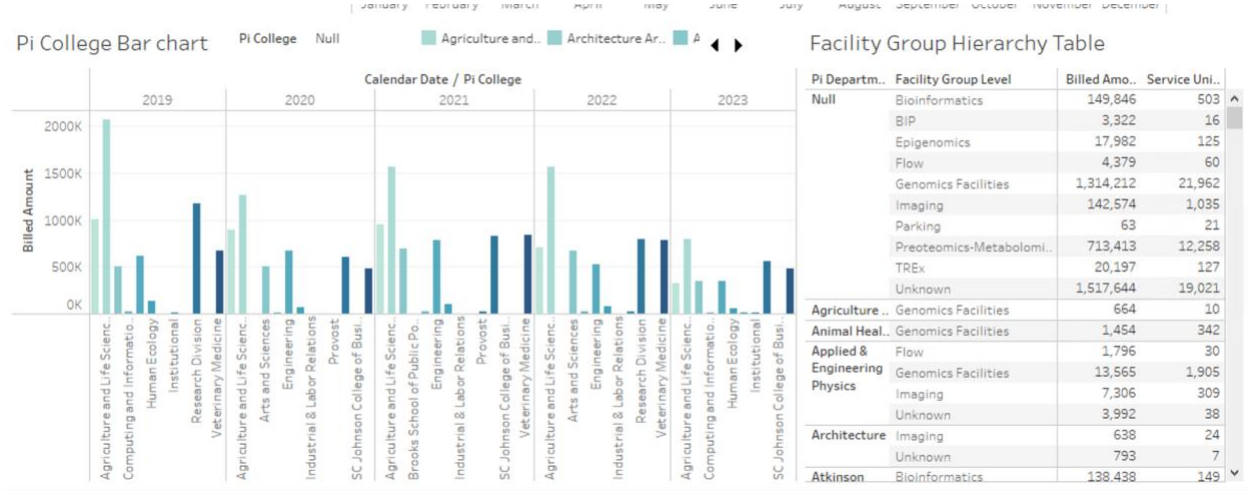
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Assignment 6: Tableau Dashboard

Drag the “Facility Group Hierarchy Table” to the right side of the bottom container.



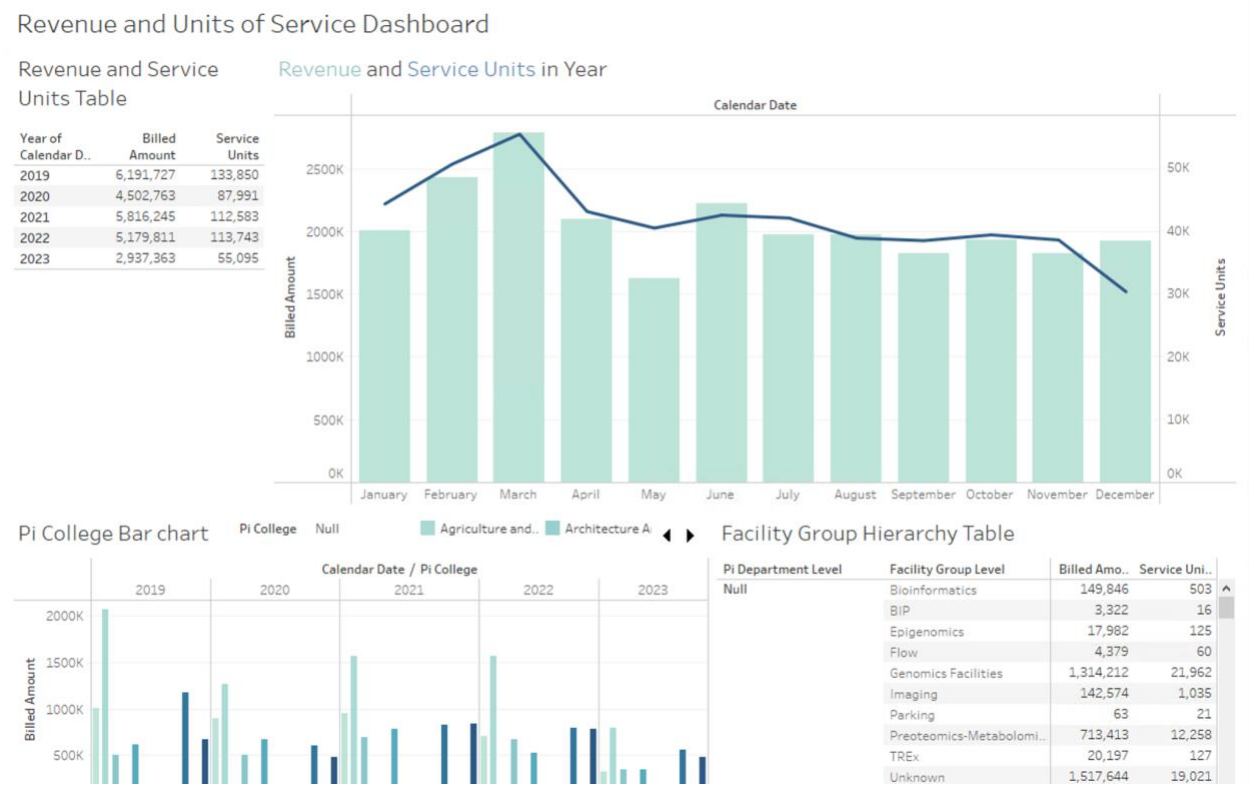
When you drag it to the container, release the chart when the container is highlighted in blue and there is a thin shaded vertical rectangle of the right border. You can re-size the legend and the charts by dragging their borders.



Your dashboard should look like this.

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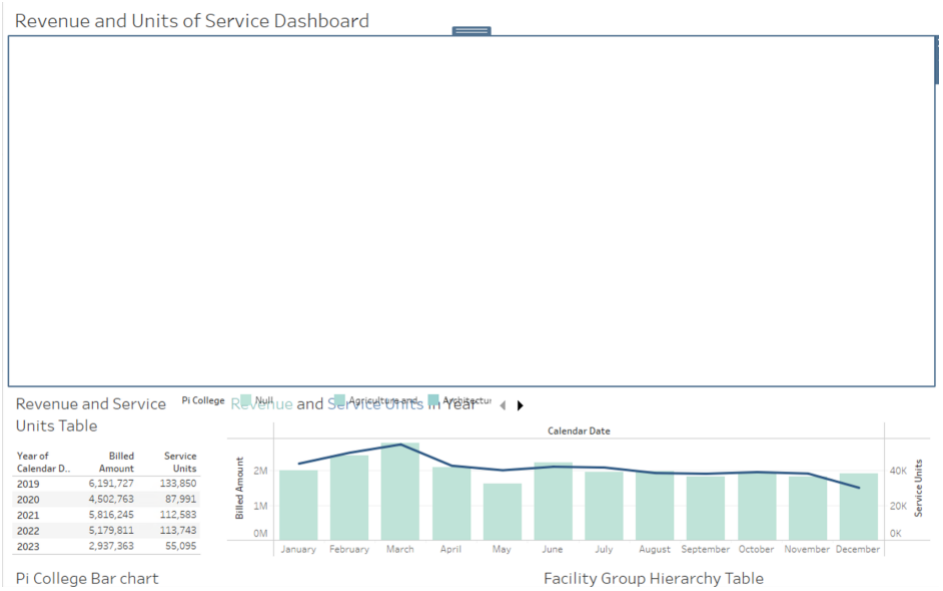
Assignment 6: Tableau Dashboard



Creating Filters

We will create a row of filters on our dashboard to control the behavior of our multiple visualizations.

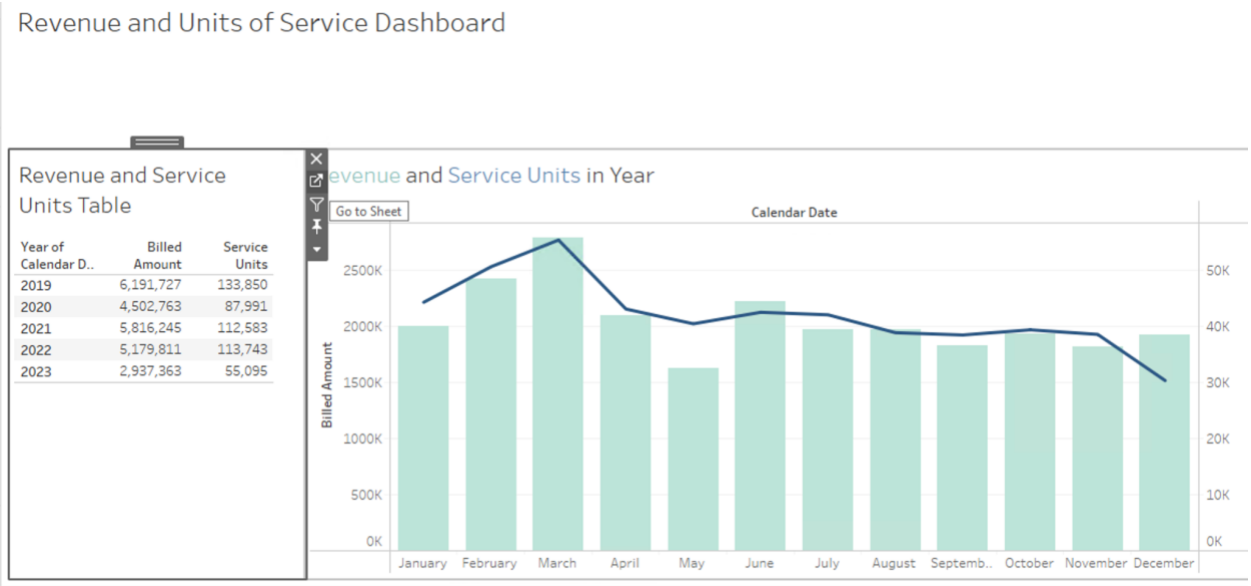
First drag a horizontal container below the title.



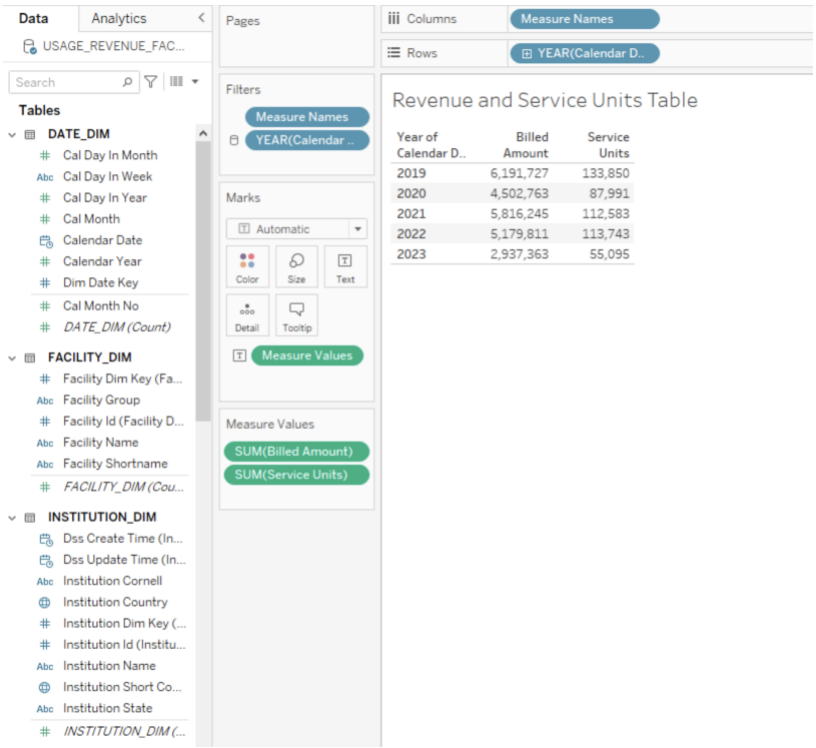
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Assignment 6: Tableau Dashboard

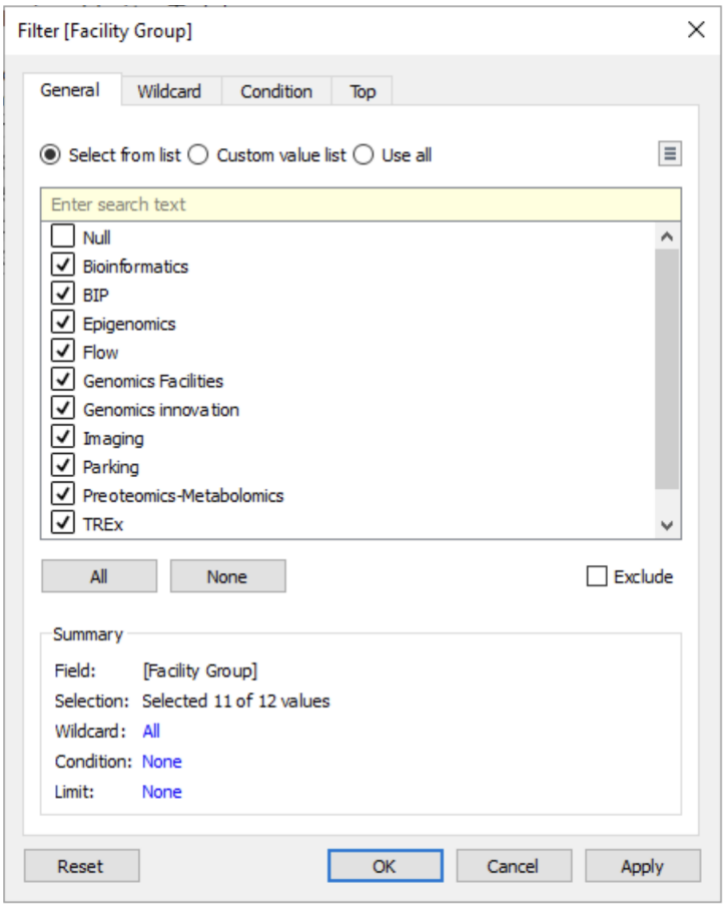
Resize the container so it's smaller. We need to add filters to every sheet so the filters will apply to all sheets. Select **any** visualization (ex. Revenue and Service Units Table) and hover over the arrow icon to go to the sheet, or navigate to a sheet using the bottom tab bar.



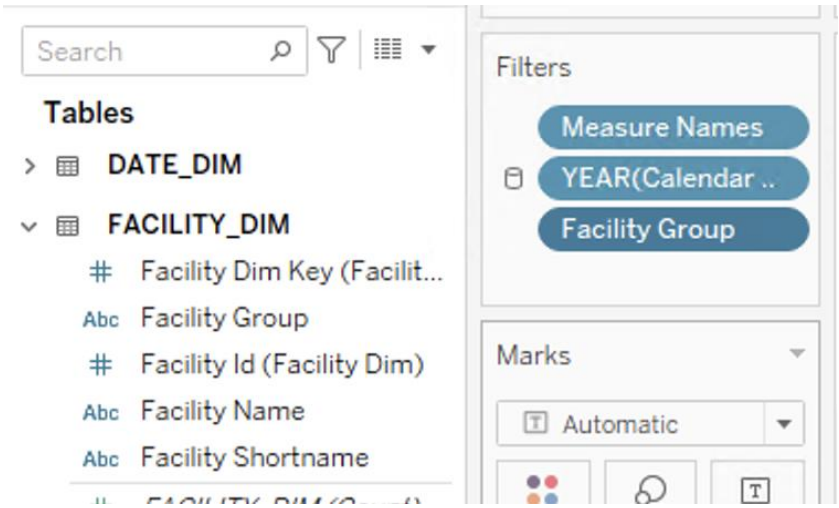
You should be on the sheet for the Revenue and Service Units Table (or your chosen sheet). First open up the left side panel to view all attributes in "**FACILITY_DIM**" and "**PI_DIM**".



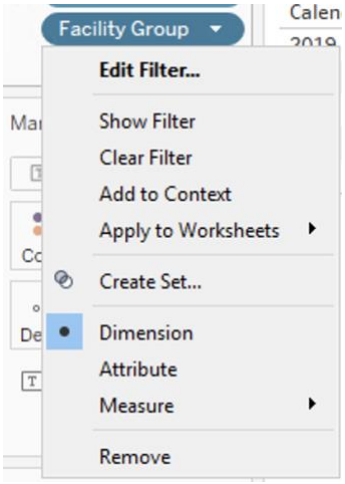
Drag "**Facility Group**" into the box that says filters. A popup will appear. Select "All", then deselect "null". Note this is a two step process. Click the "Use all", then select "select from list" and deselect the null checkbox".



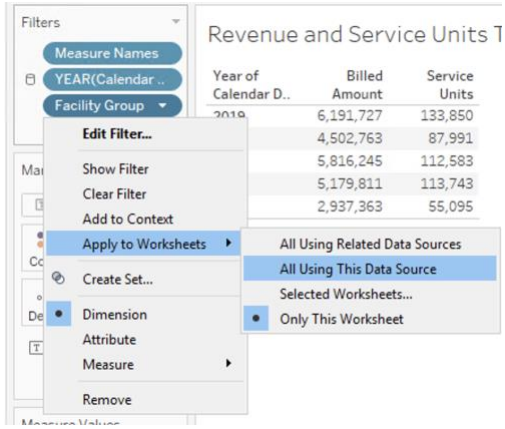
Click "OK" and you should see "Facility Group" in the filters box.



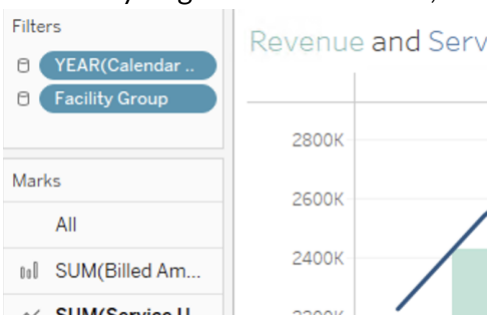
Hover over the filter and click to open the dropdown menu.



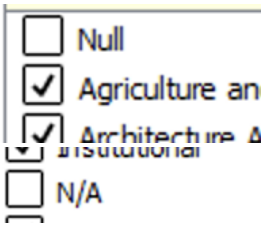
Hover over “Apply to worksheets” and select all using this data source.



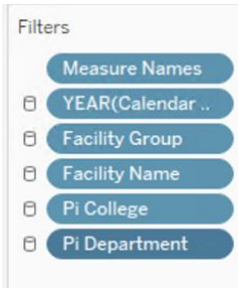
Notice if you go to another sheet, the facility group filter is automatically added.



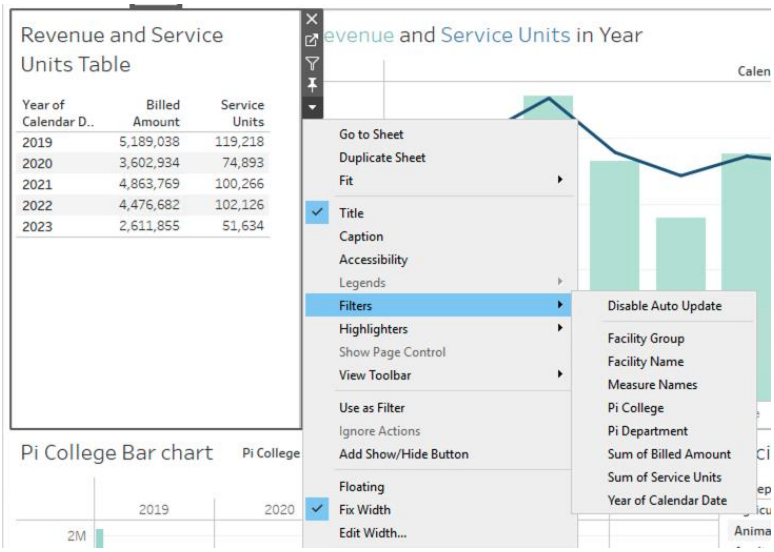
Repeat the same process for the attributes "**Facility name**", "**Pi_college**", and "**Pi_department**". Note, for "Pi_college" and "Pi_department", you should deselect Null and N/A.



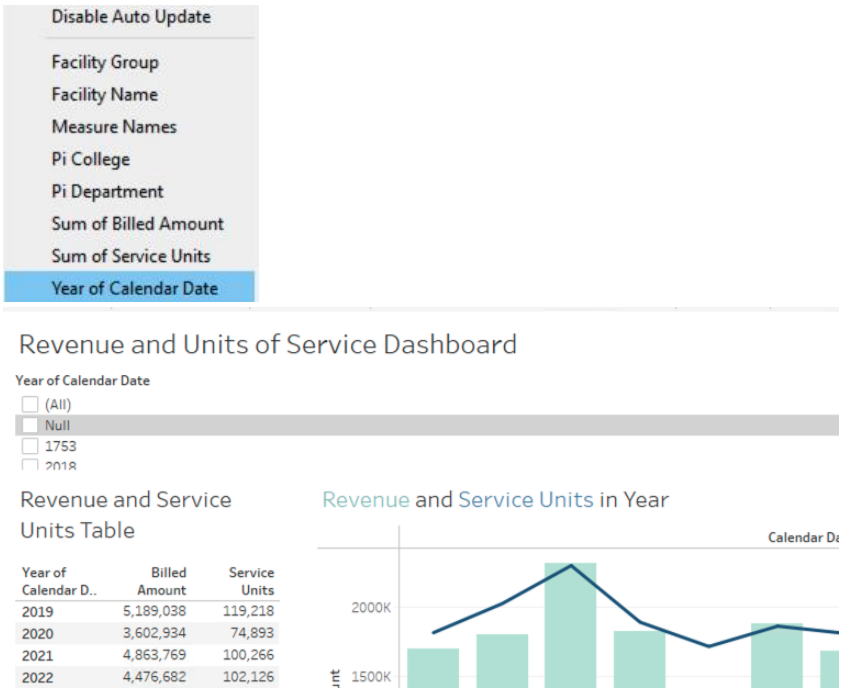
Once finished, your filters box should look like this.



Now, go back to the "Revenue and Units of Service" dashboard. Select any sheet and click on the dropdown menu. Hover over "Filters".

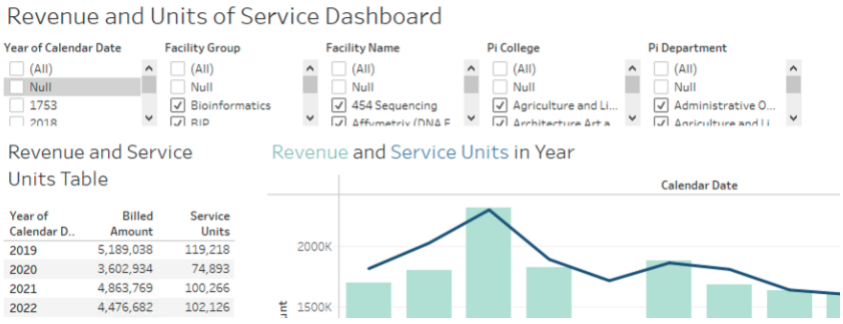


Select "**Year of Calendar Date**".

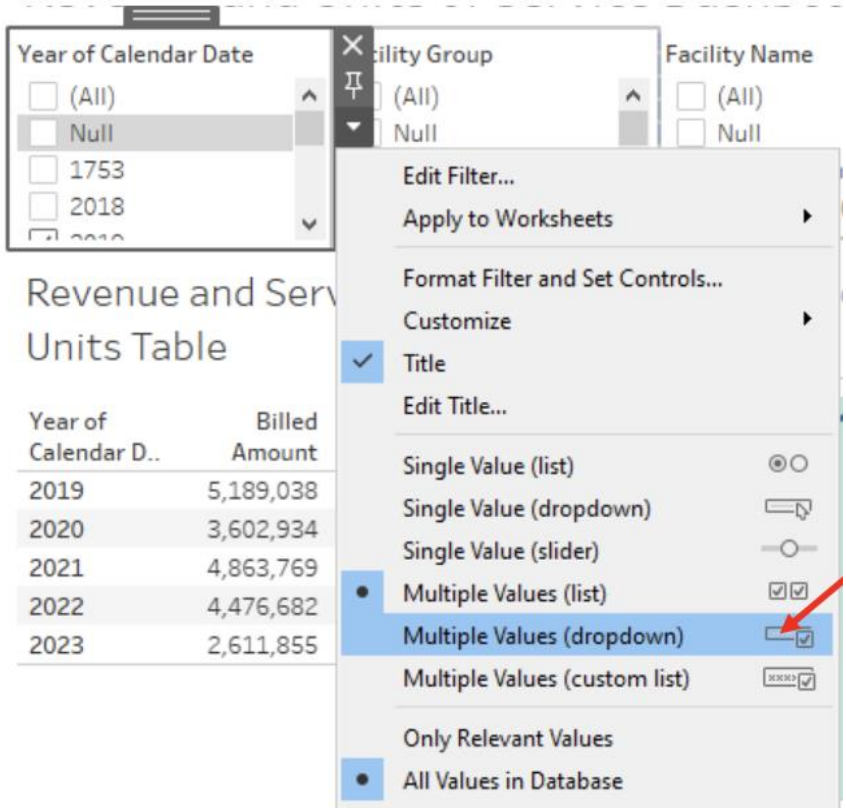


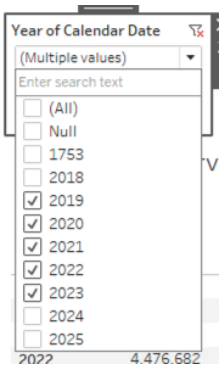
Notice it appears inside the horizontal container we added.

Repeat the same for "**Facility Group**", "**Facility Name**", "**Pi College**", and "**Pi Department**".



We can choose different formats for our filters. If you hover over the dropdown for a filter, you can make single or multiple value dropdowns, lists, or slides. Select “Multiple Values (dropdown).”



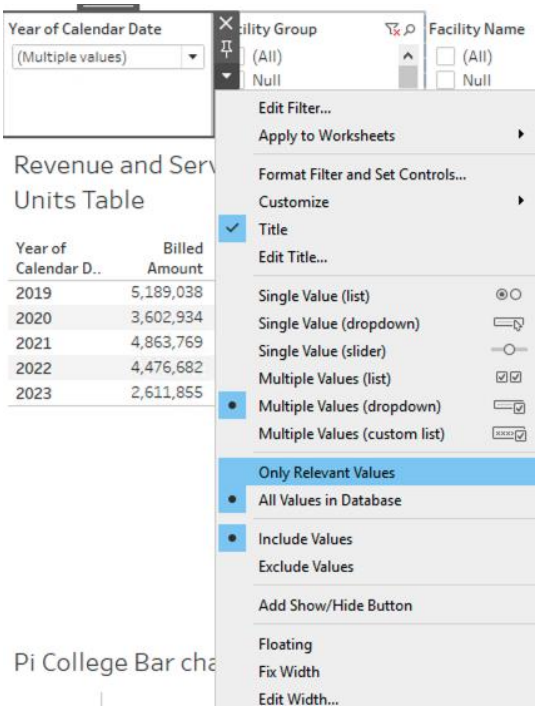


Notice how the filter changed to a dropdown menu. Additionally, the years are automatically selected based on what we selected previously. We don’t want the other years to be options since we are only interested in 2019-2023.

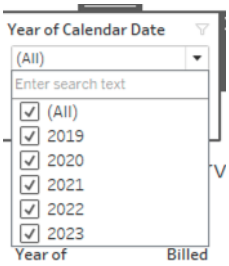
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Assignment 6: Tableau Dashboard

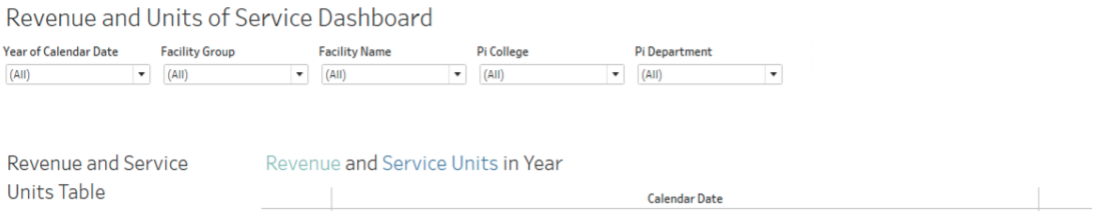
Select “Only Relevant Values” in the filter dropdown menu.



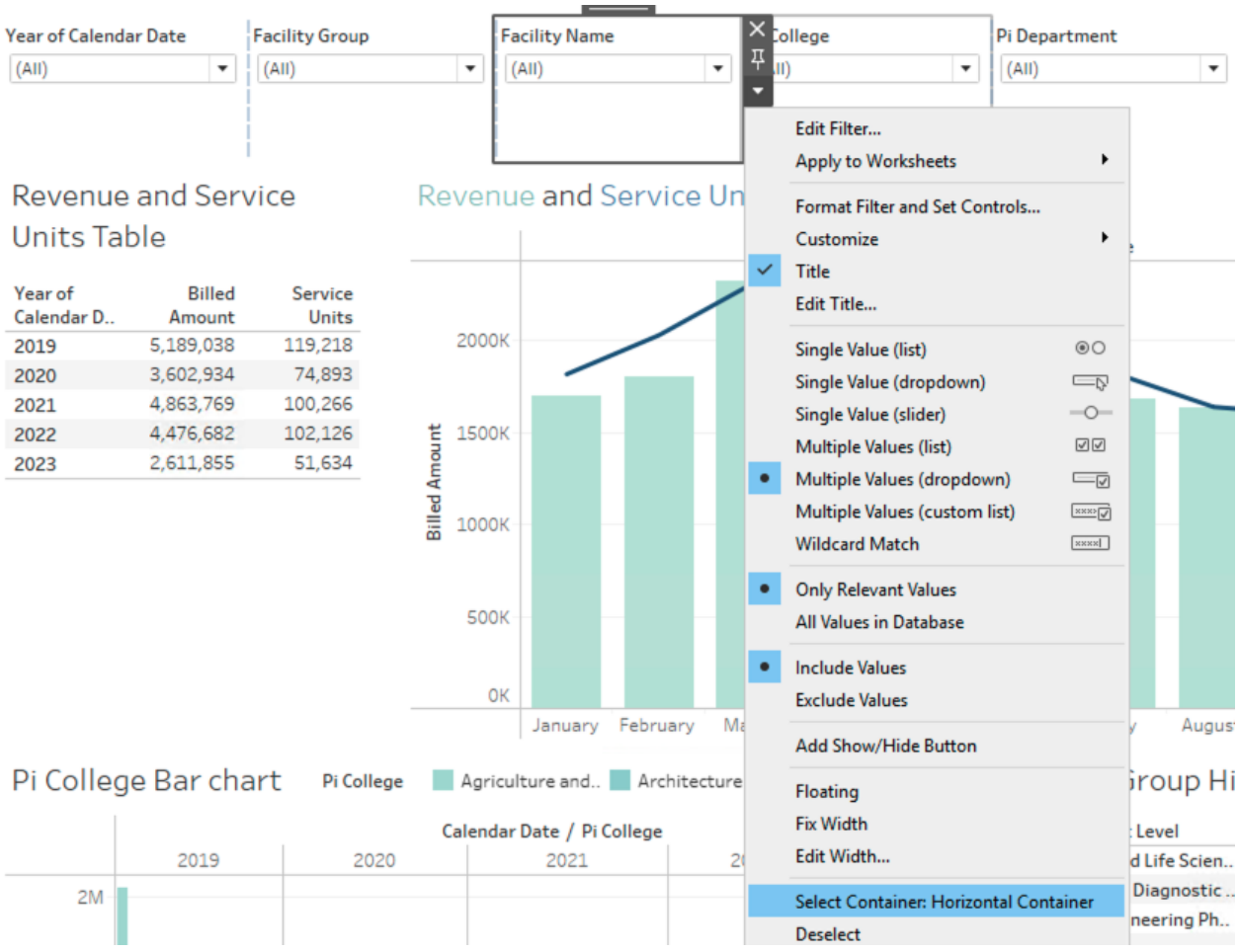
Now, all the unselected values are removed from the filter.

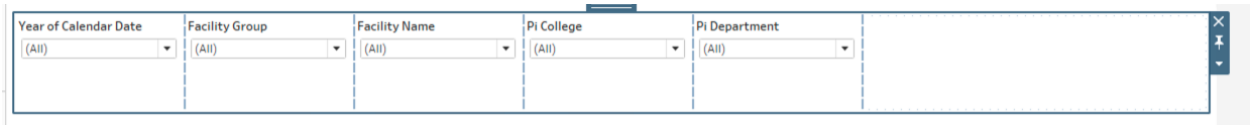


Repeat the same steps for the remaining filters. It should look like below.

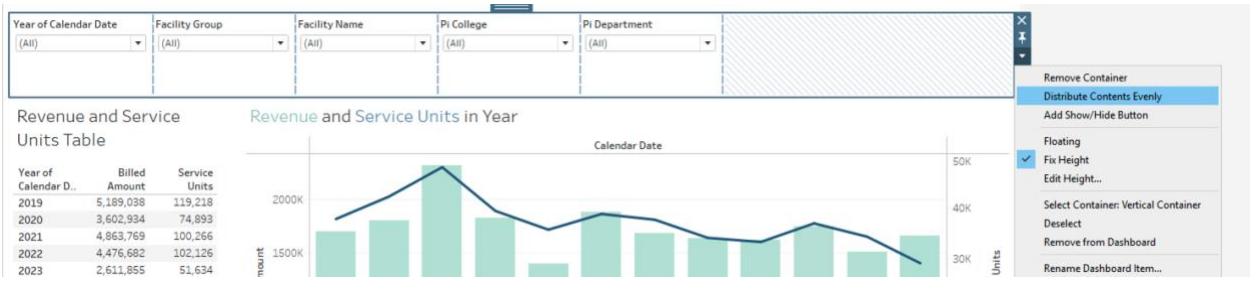


The filters are all clustered together to the left. To evenly space them across the page, select a filter and click “Select Container: Horizontal Container”. This will highlight the container in blue.





Then click on the dropdown for the container and select “Distribute Contents Evenly”.



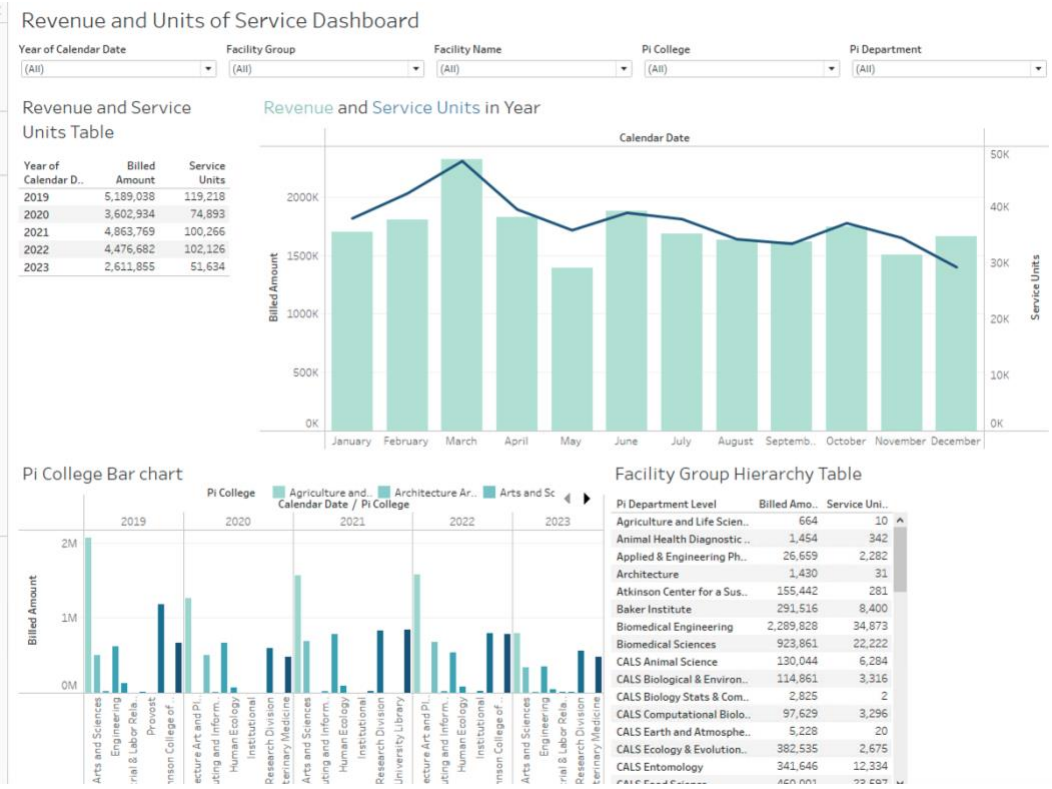
Now we’ve successfully added the filters! Feel free to play around with the filters to watch the graphs change.

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To resize the entire container, select any filter and click “Select Container: Horizontal Container” to when it's highlighted blue, and drag the border to the desired size.

This is what your dashboard should generally look like.



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Assignment 6: Tableau Dashboard

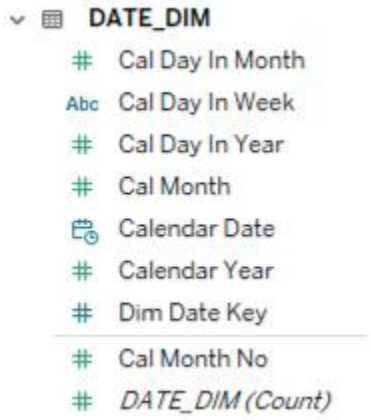
Adding Interactions

Parameters

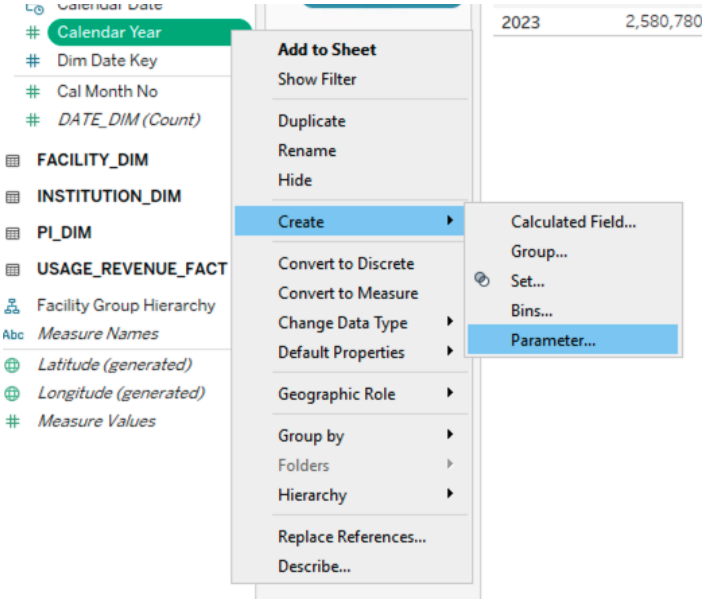
A parameter placeholder is a value that can replace a constant value. They are useful for increasing interactivity and flexibility. For example, if we were trying to indicate whether a field is above 90%, the user would be able to replace 90% with a different value.

A. Create the parameter

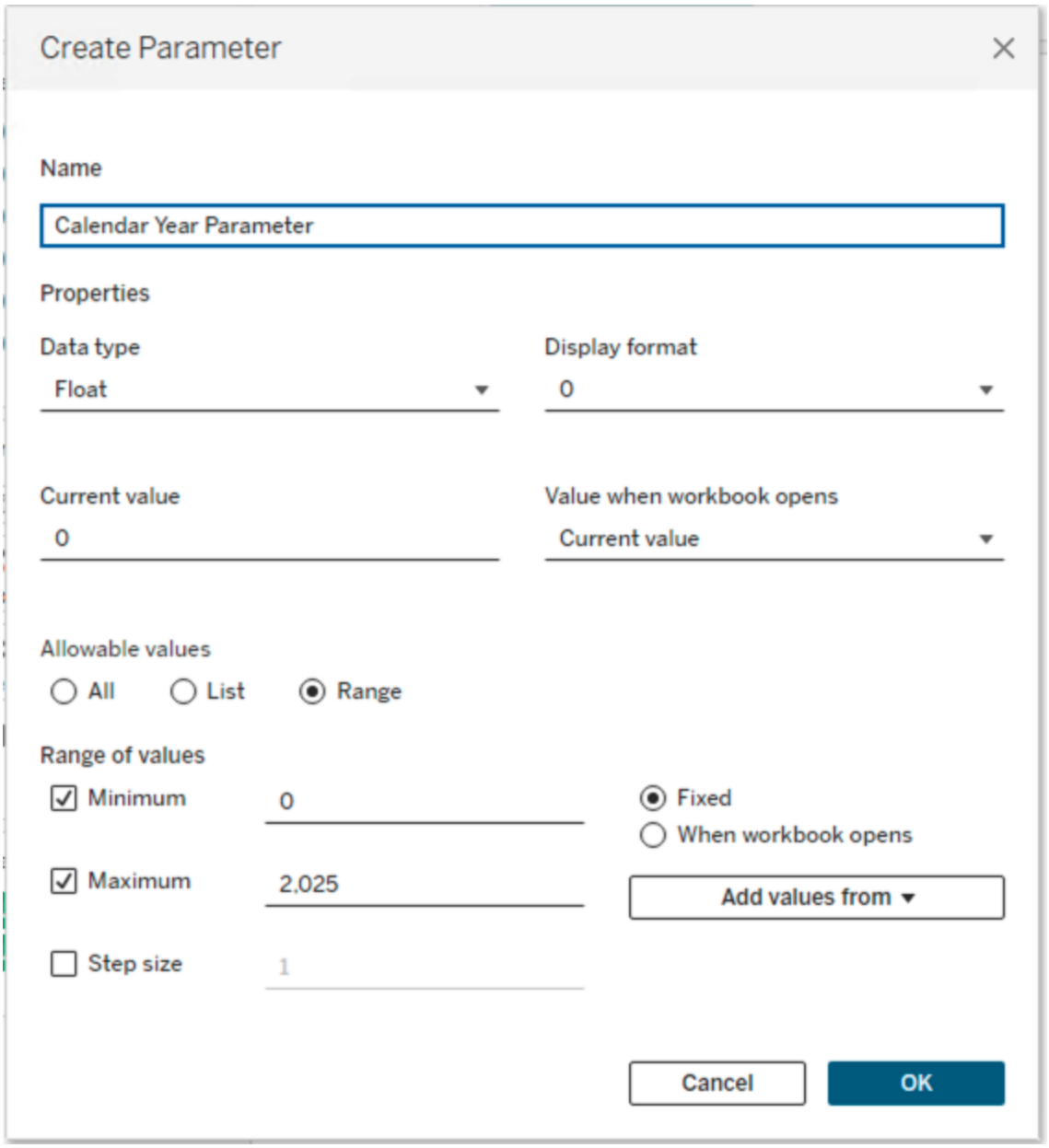
Go to the "**Revenue and Service Units Table**" sheet. Open **DATE_DIM** in the side panel.



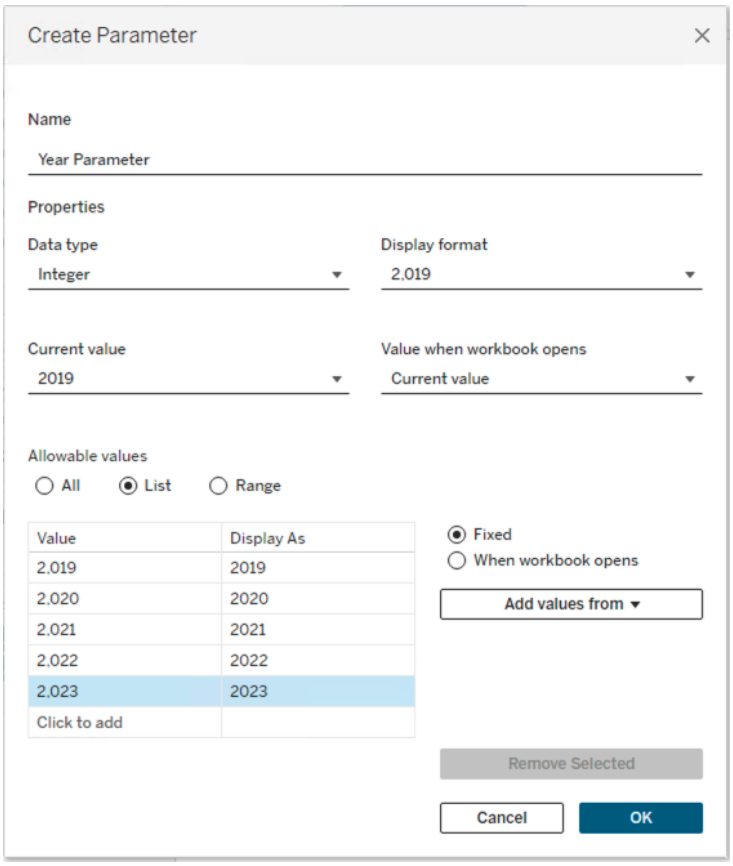
Select "**Calendar Year**" and then "Create", and then "Parameter …"



A popup should appear.



1. Enter “Year Parameter” for the Name.
2. Change the “Data type” to Integer.
3. Select “List” under “Allowable Values”.
4. Enter the years 2019 - 2023.
    a. For each year, under “Display As”, remove the comma.



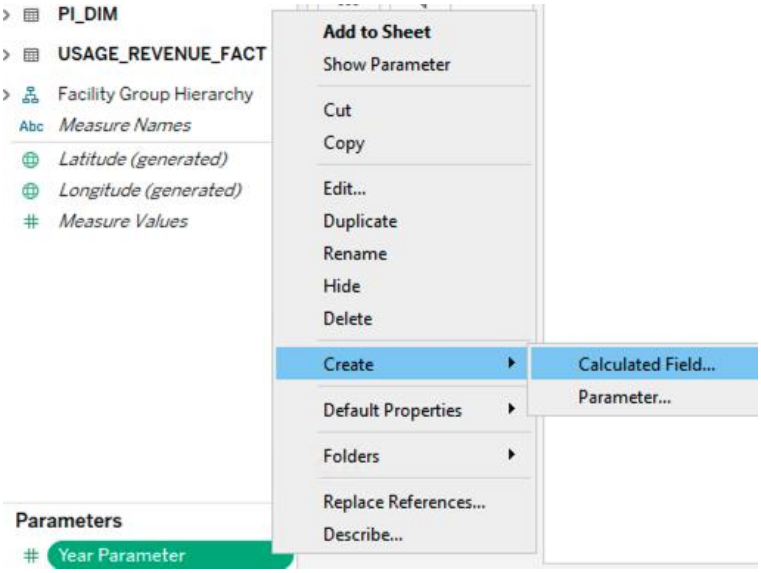
Click “OK”.

B. Create the Calculated Field using the Parameter

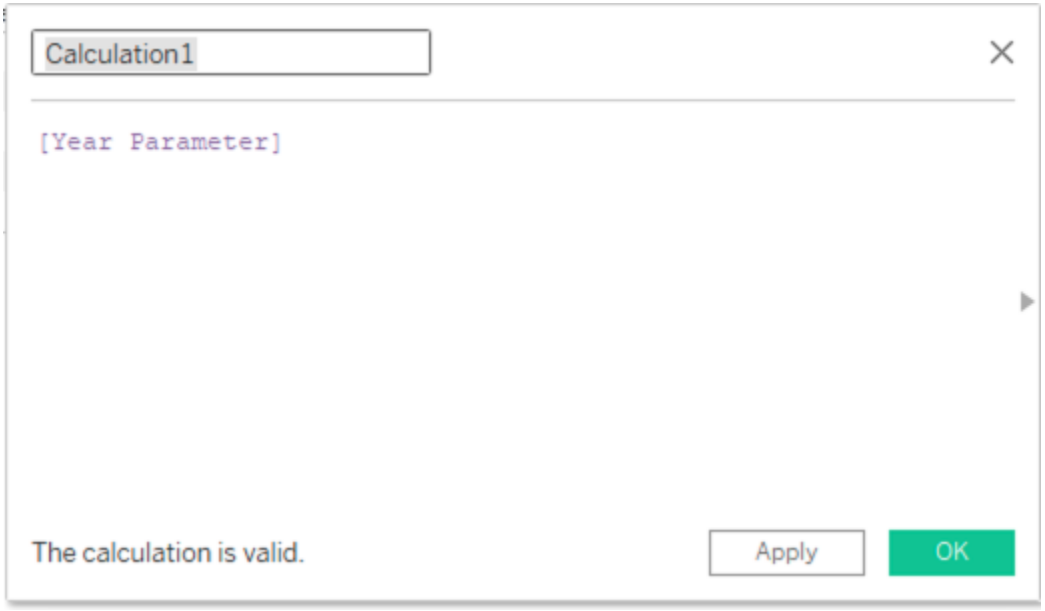
We are going to create a calculated field using the parameter we created. A calculated field can be created from an existing field to create new data from the existing data. An example would be calculating a ratio between two fields.

Select your “**Year Parameter**” (It should be located at the bottom of the left side panel) and select “Create” and then “Calculated Field”.

Assignment 6: Tableau Dashboard



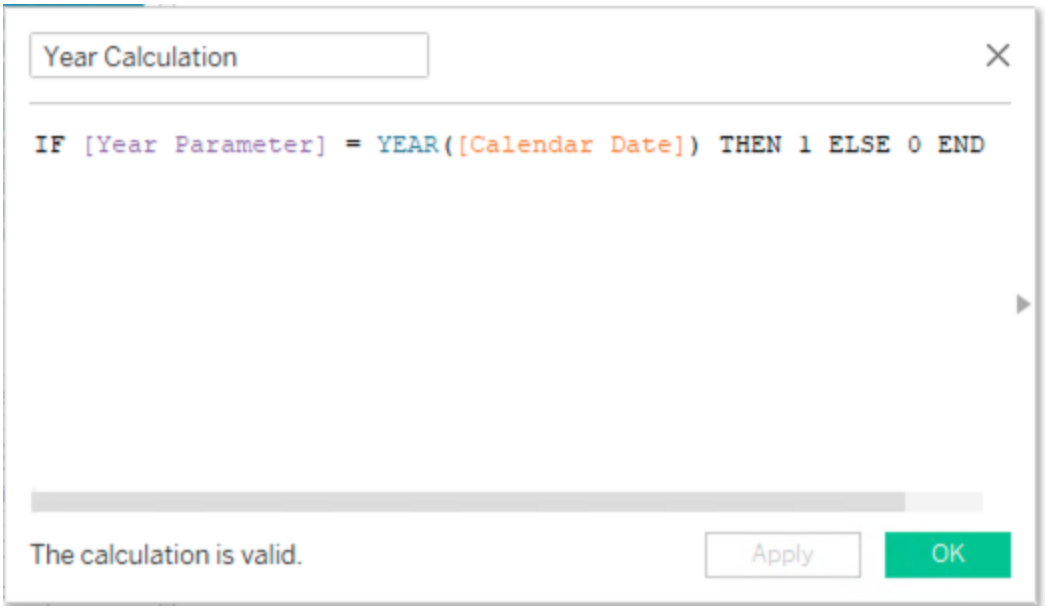
A popup will appear.



Rename the field to “Year Calculation”.

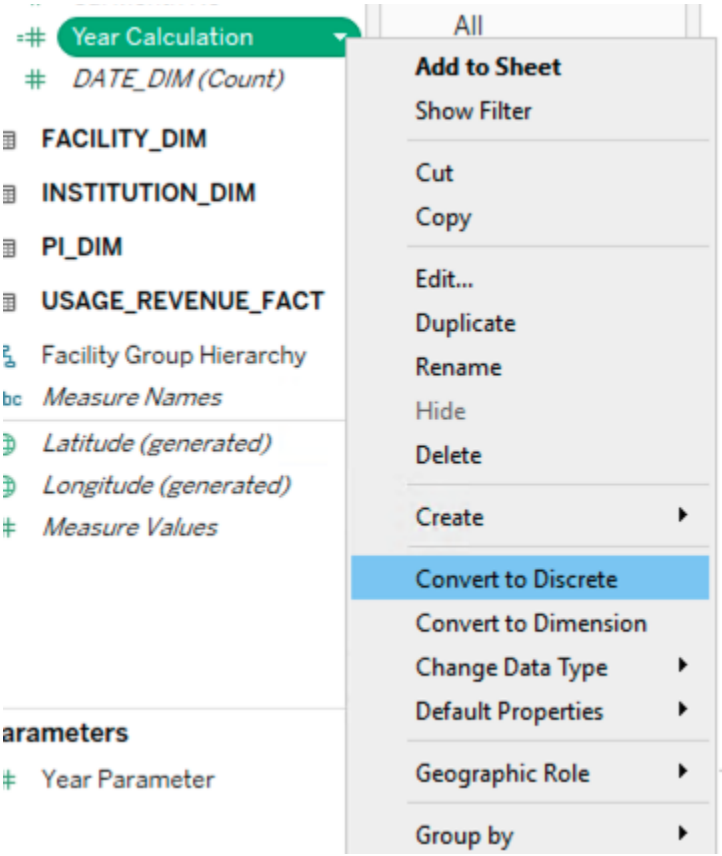
We want to create a field that will indicate whether the year parameter is equal to the year date field. Paste in the text below:

IF [Year Parameter] = YEAR([Calendar Date]) THEN 1 ELSE 0 END

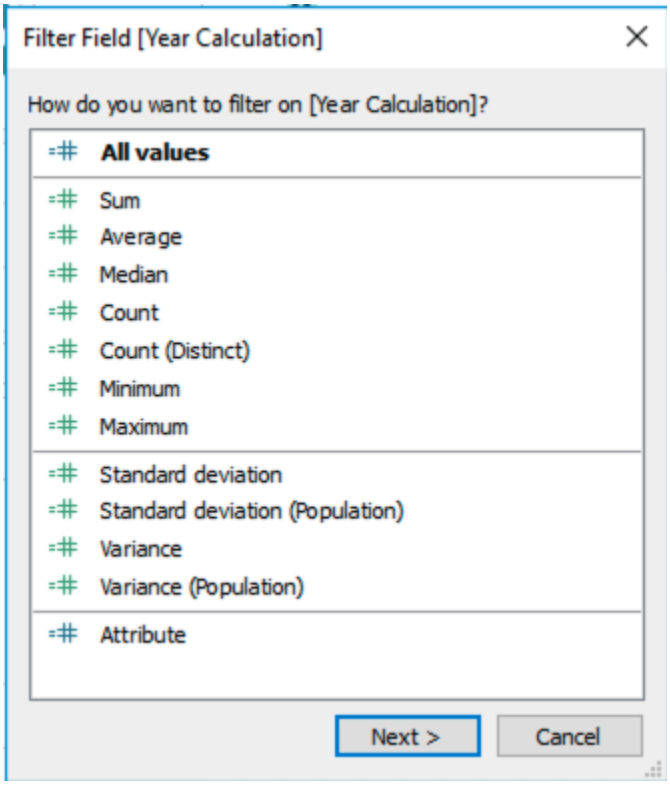


Click “OK”.

Go to the sheet named “**Revenue and Service Units in Year**”. Select “**Year Calculation**” and select “Convert to Discrete”.

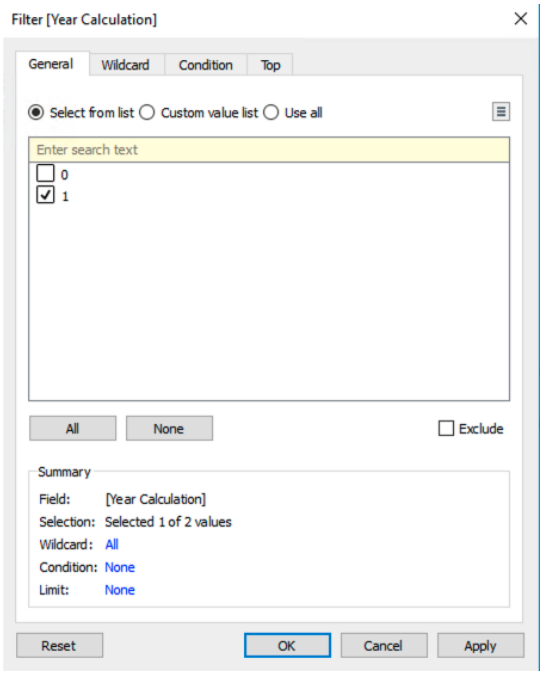


Drag the "**Year Calculation**" field to the filters panel. A pop will appear.



Select "All values". and click Next.

Select 1 from the dropdown. This will filter for only when the year parameter matches the year field.



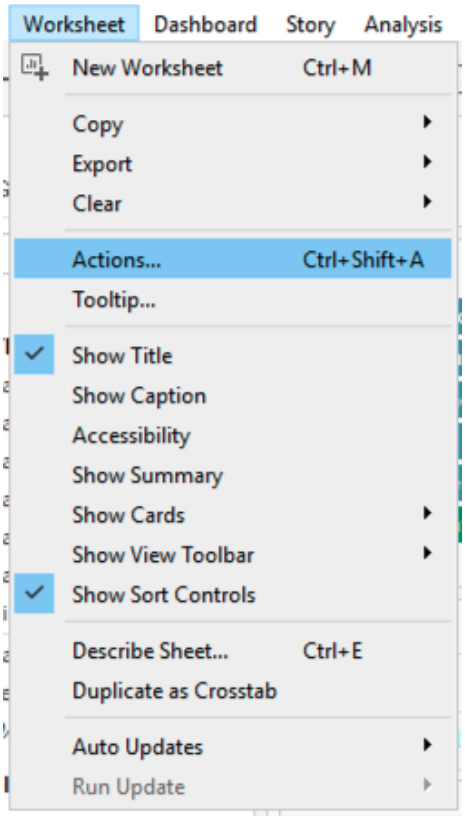
Click “OK”.

C. Add the Worksheet Action using the Parameter

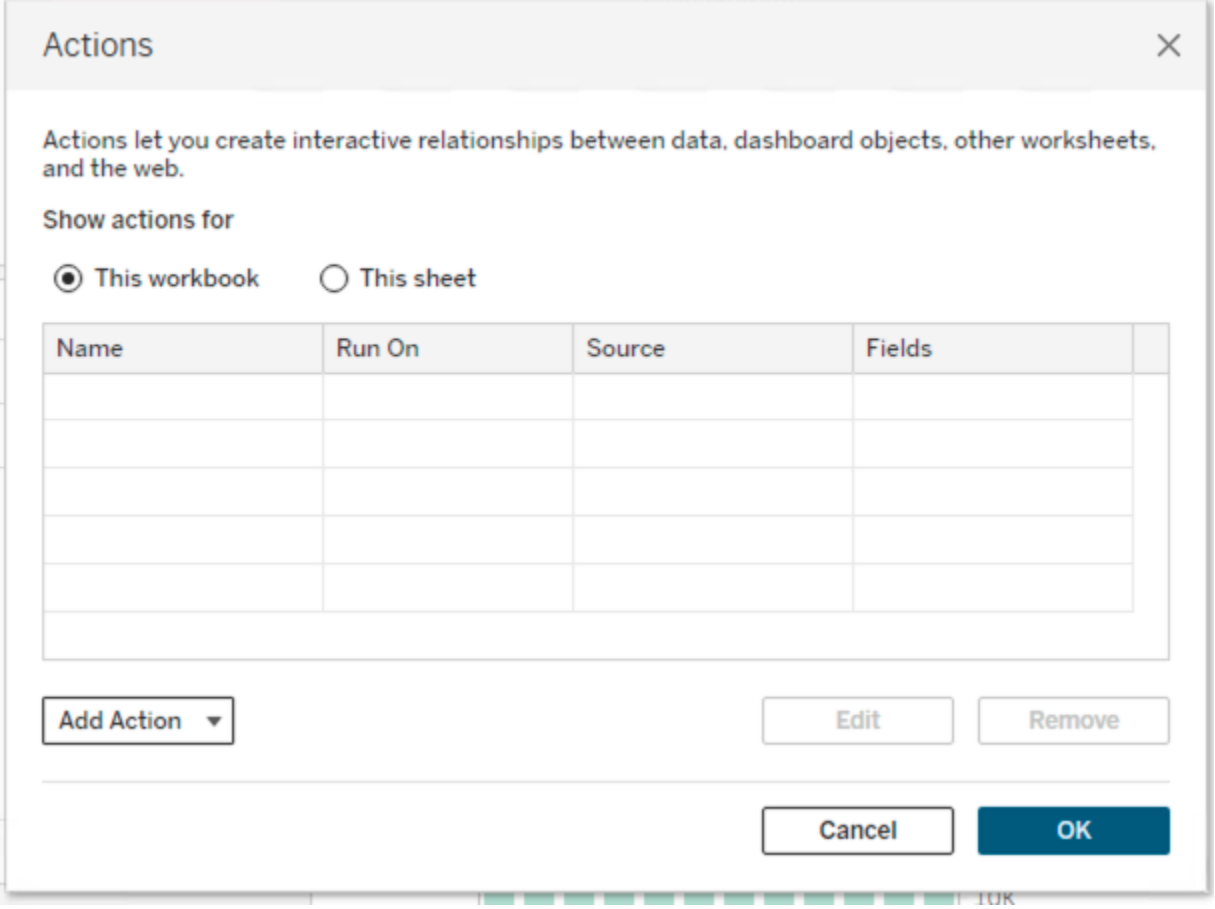
An action allows the user to interact with the dashboard and manipulate certain views. The possible actions are:

- Filter
- Highlight
- Go to URL
- Go to Sheet
- Change Parameter
- Change Set Values

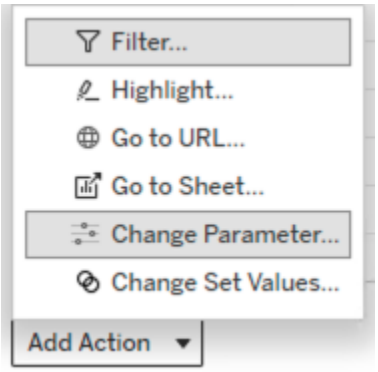
In the toolbar at the top, select “Worksheet”. Select “Actions…”

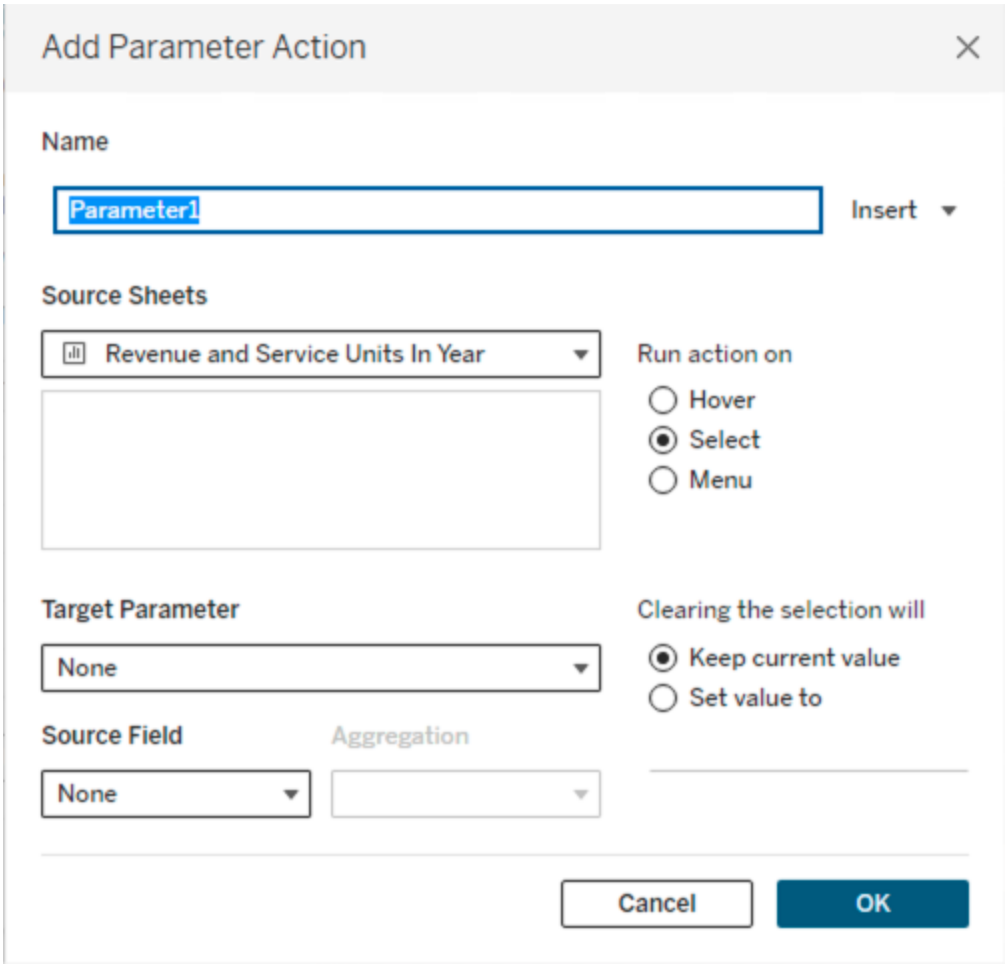


A popup will appear.



Select “Add Action” then “Change Parameter”. A popup will appear.



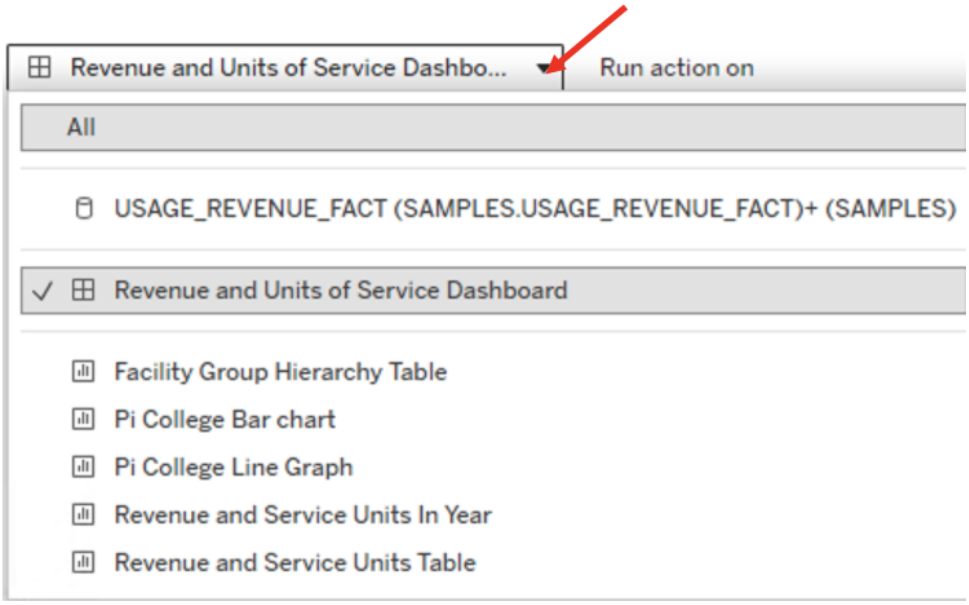


Name the Parameter to “Change Year Parameter”. It is important to rename the actions to a specific description to avoid confusion when there are many actions.

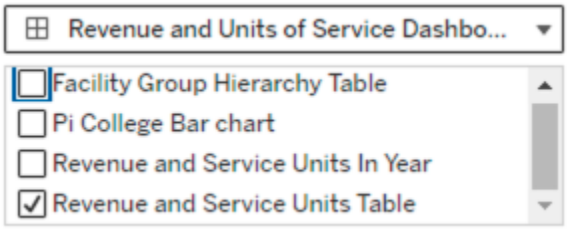
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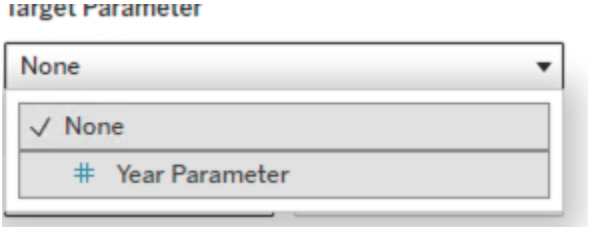
Switch to the "**Revenue and Units of Service Dashboard**" under "Source Sheets".



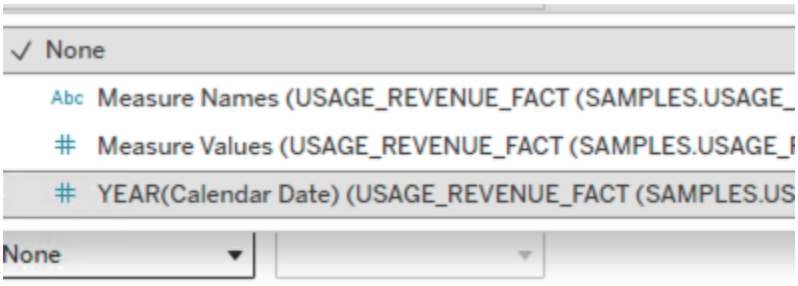
Only keep the "**Revenue and Service Units Table**" selected.



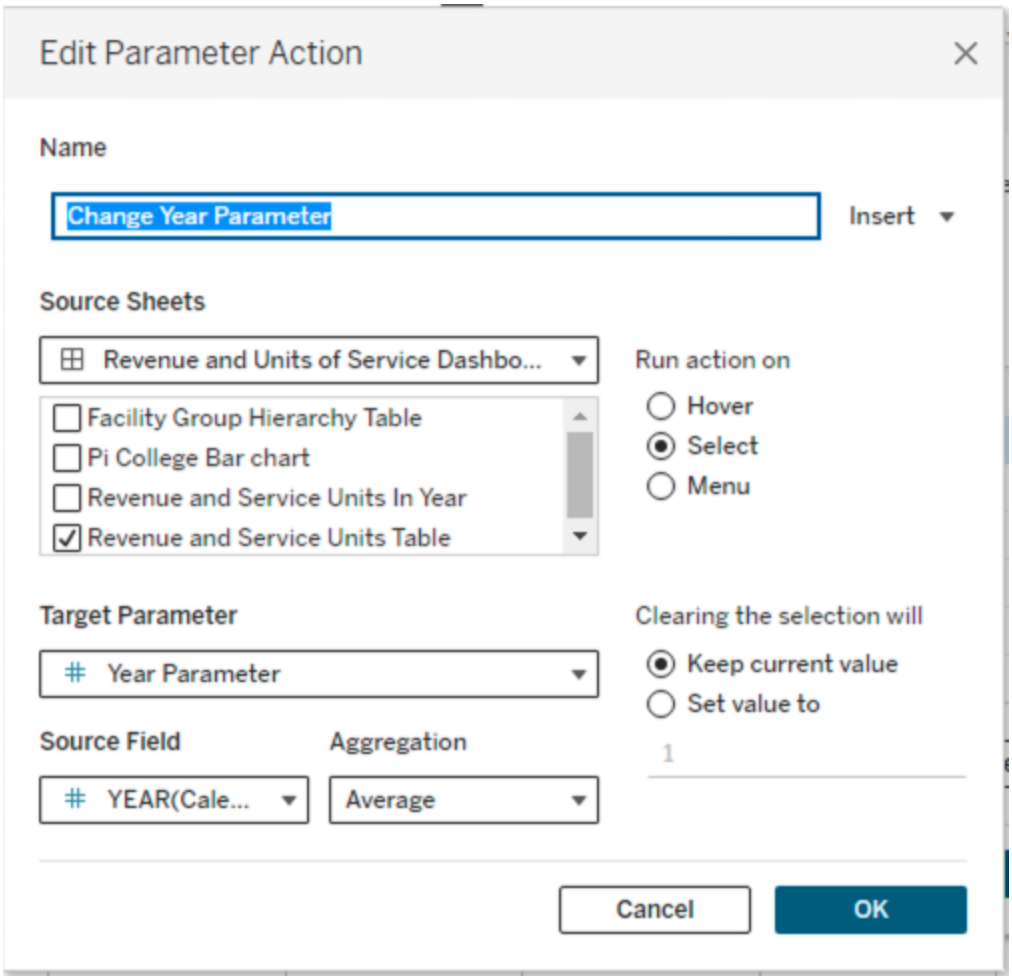
Change "Target parameter" to "**Year Parameter"**.



Under "Source Field", select **YEAR(Calendar Date)**.

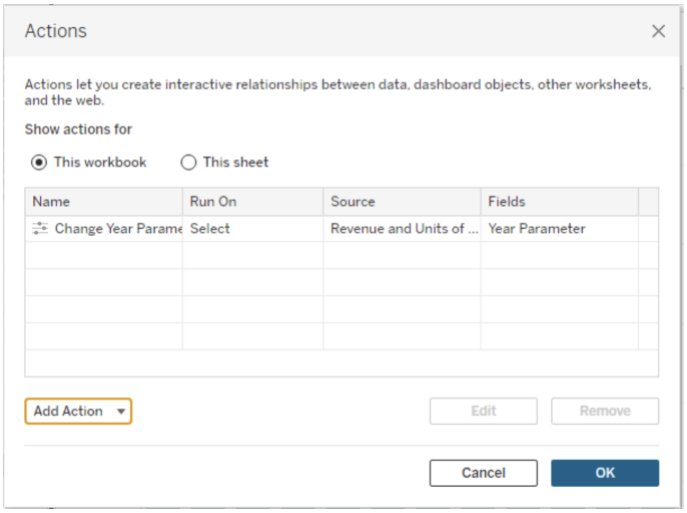


Select "**Average**" under "Aggregation".



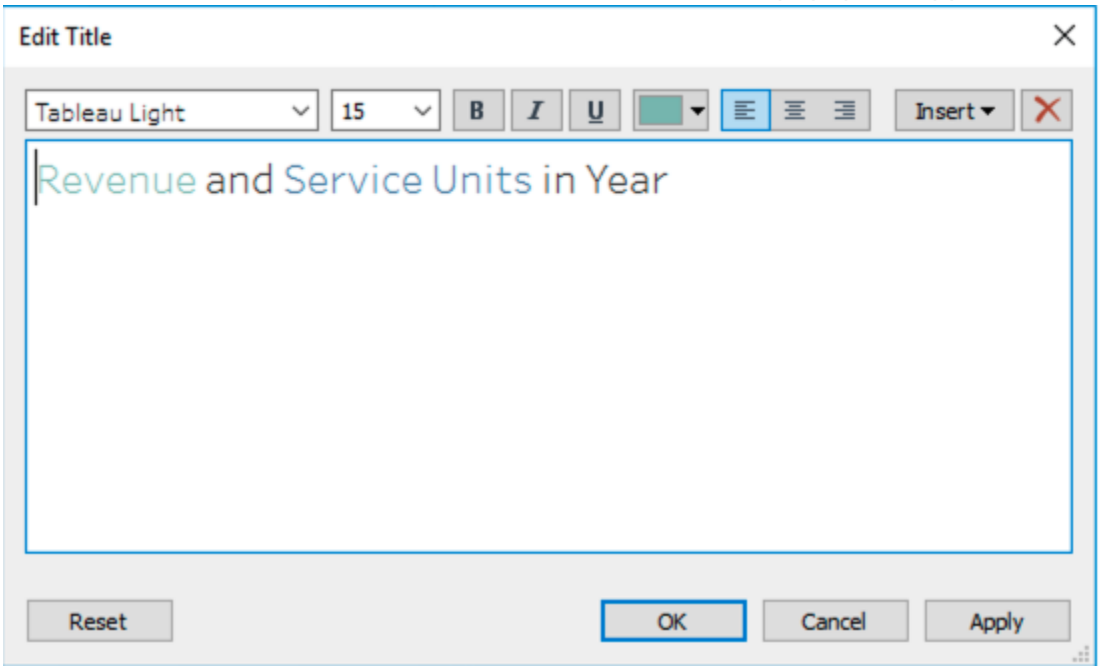
If the popup looks like the above, Click “OK”.

For Actions popup, also select OK.

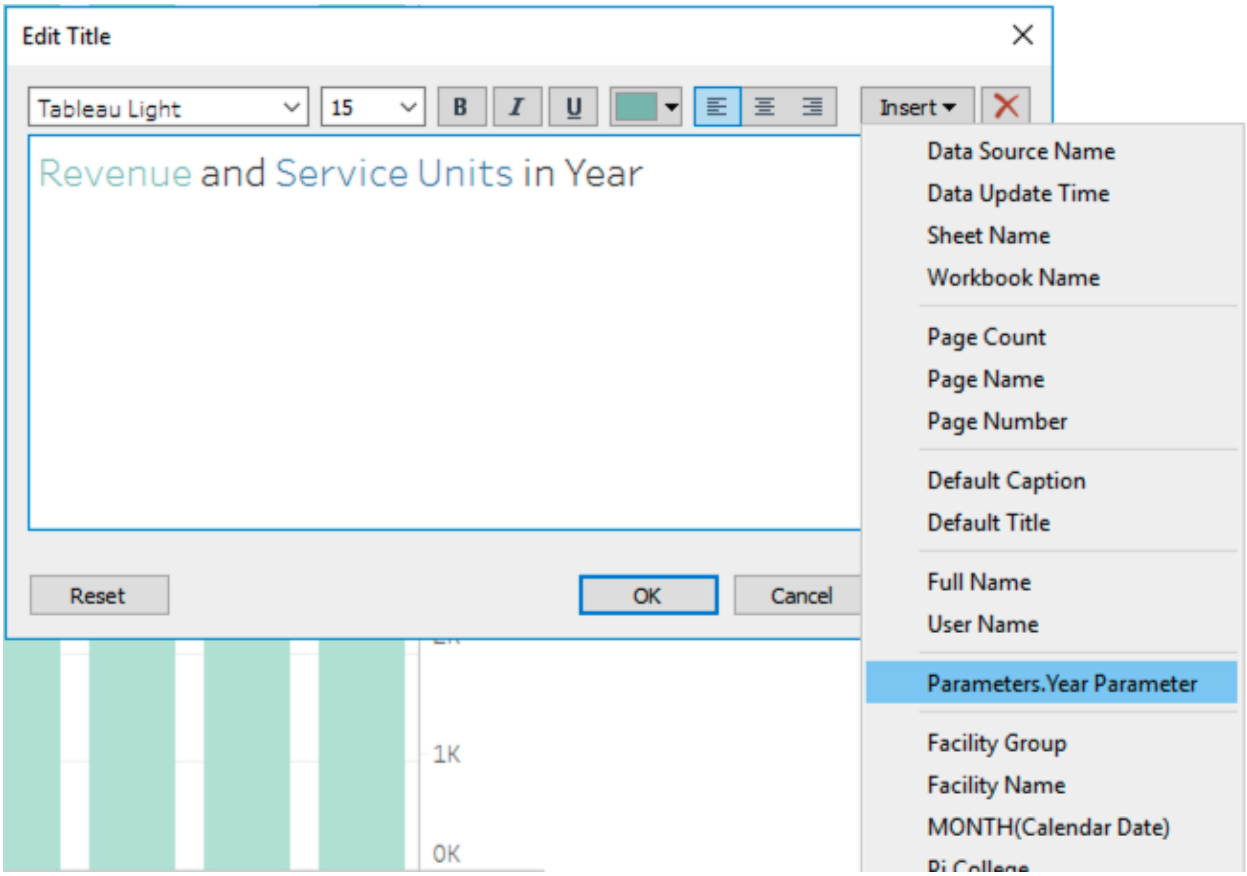


Great! You have created your first worksheet action. When you select a row on the "Revenue and Service Units Table", it will filter to the corresponding "Revenue and Service Units in Year" bar chart. Test it out to make sure the bar chart is changing when you click different years.

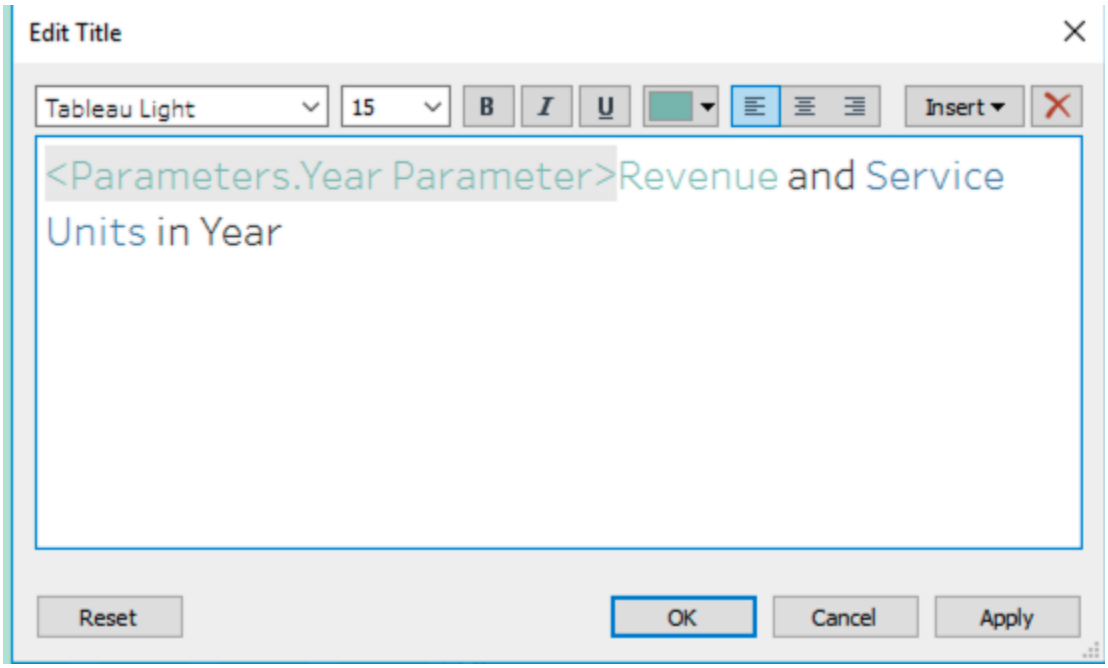
Let's update the title of the Revenue and Service by Year bar chart. Go to the "**Revenue and Service Units in Year**" sheet. Double click the title. This title popup will appear.



Select "Insert" in the top right corner. Select "**Parameters.Year Parameter**".



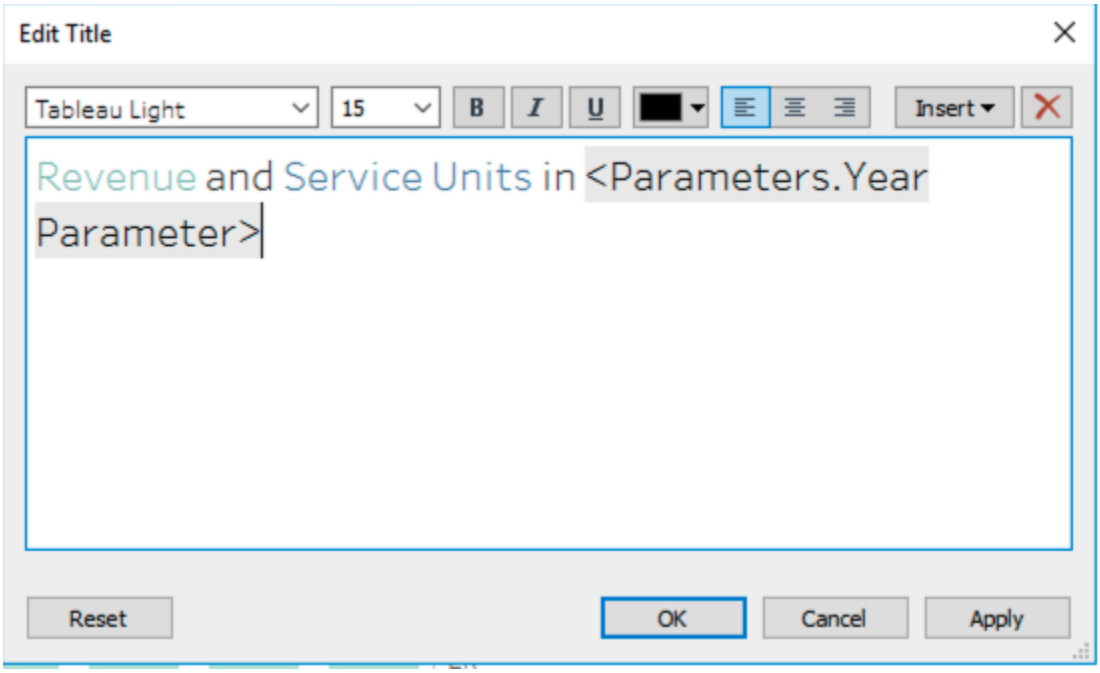
This selection should insert highlighted text into the box. Copy the new text and replace “Year” in the title. Additionally, change the color of the new text to black.



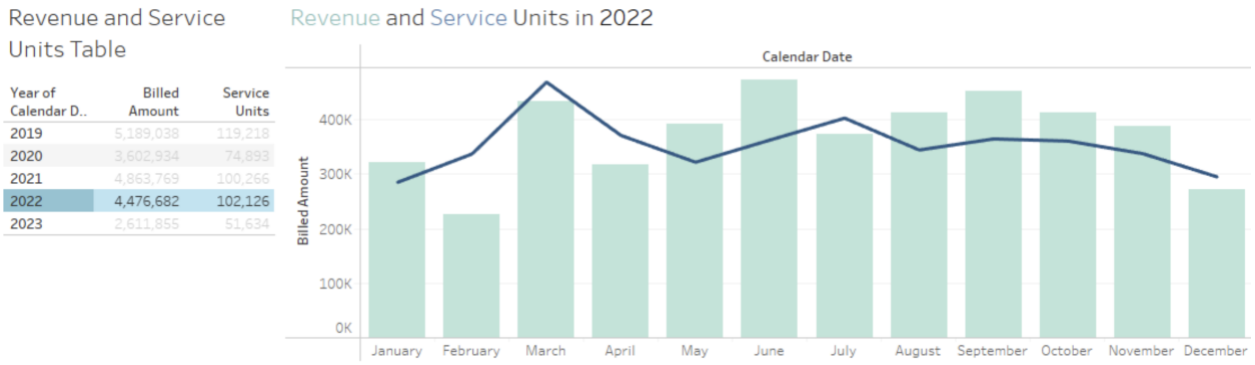
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Assignment 6: Tableau Dashboard

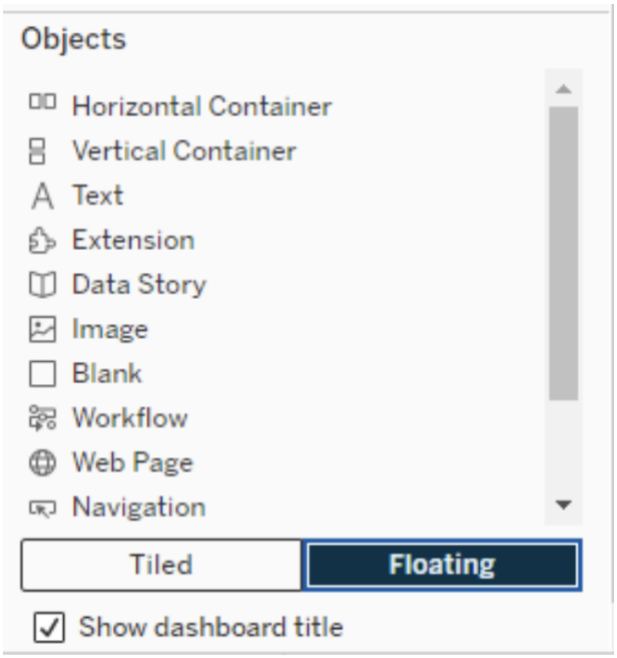
If the title looks like below, click “OK”.



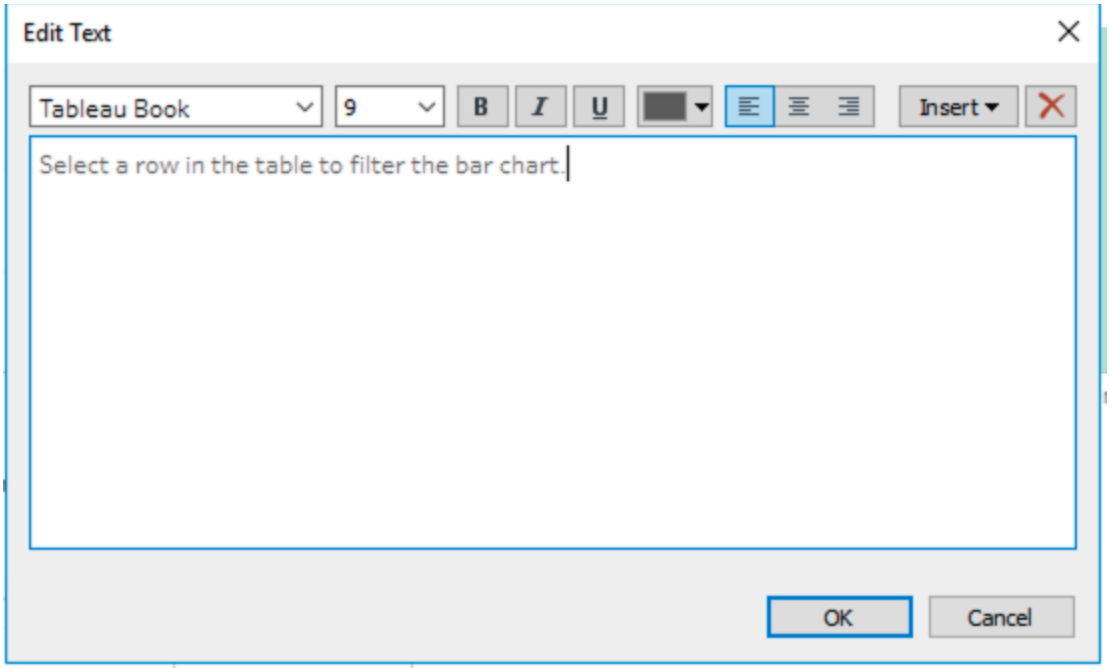
Go back to your “Revenue and Units of Service” Dashboard. The title will update dynamically with the selected year.



Let’s add a text box to help the user understand the interaction between the two charts. Select a “Text” object from the panel in the bottom left menu. Switch the option from “Tiled” to “**Floating”**.



Drag the text box below the table. A popup will appear.



Copy and paste: Select a row in the table to filter the bar chart.
Click “OK”.



Resize the text box to fit under the table.

Save your work, if you haven't recently…

Tooltip Chart

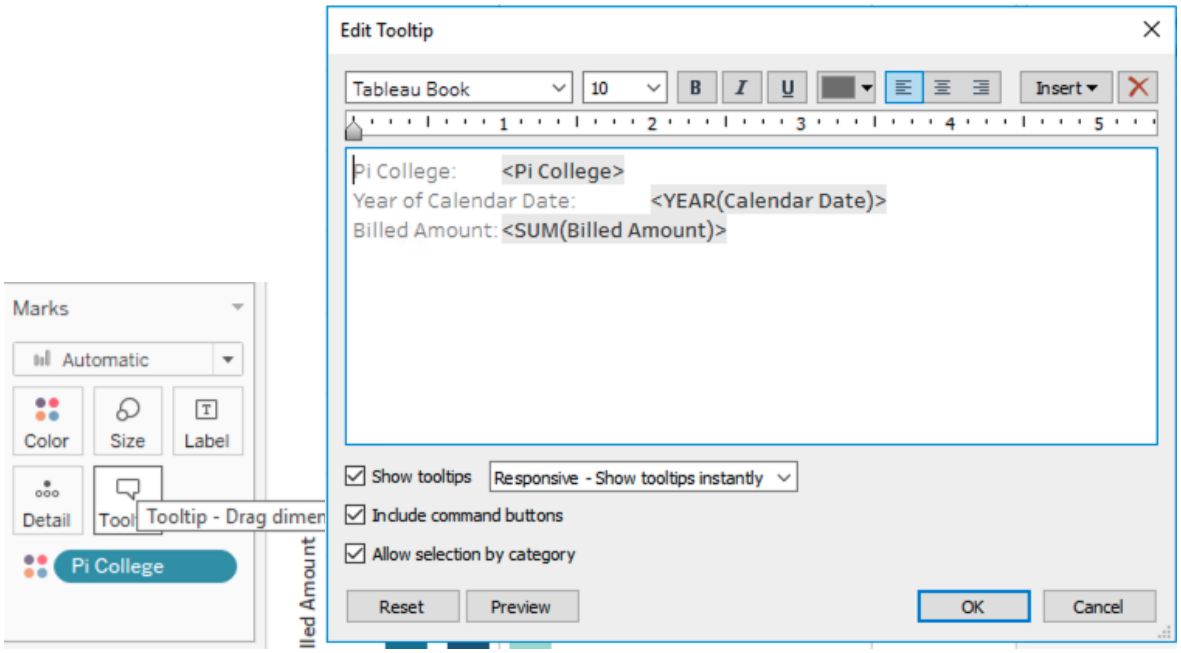
We can add charts into our tooltips.

Go to the sheet named "**Pi College Bar Chart**".

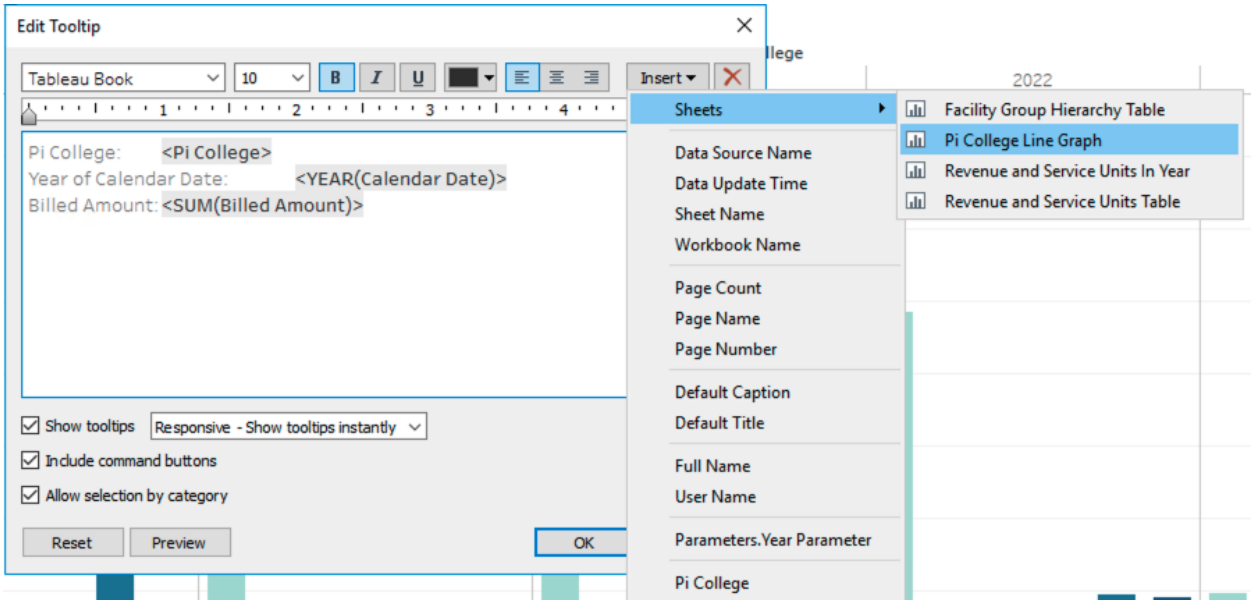
Select tooltip under "Marks".

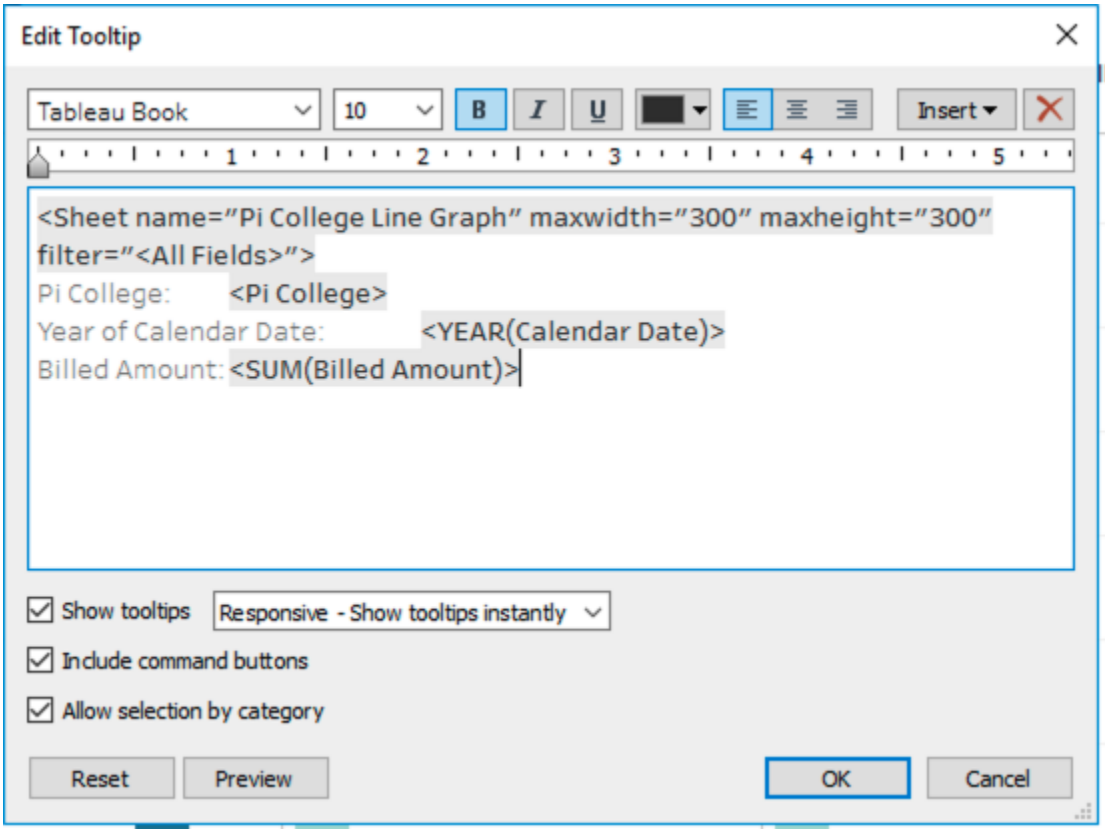
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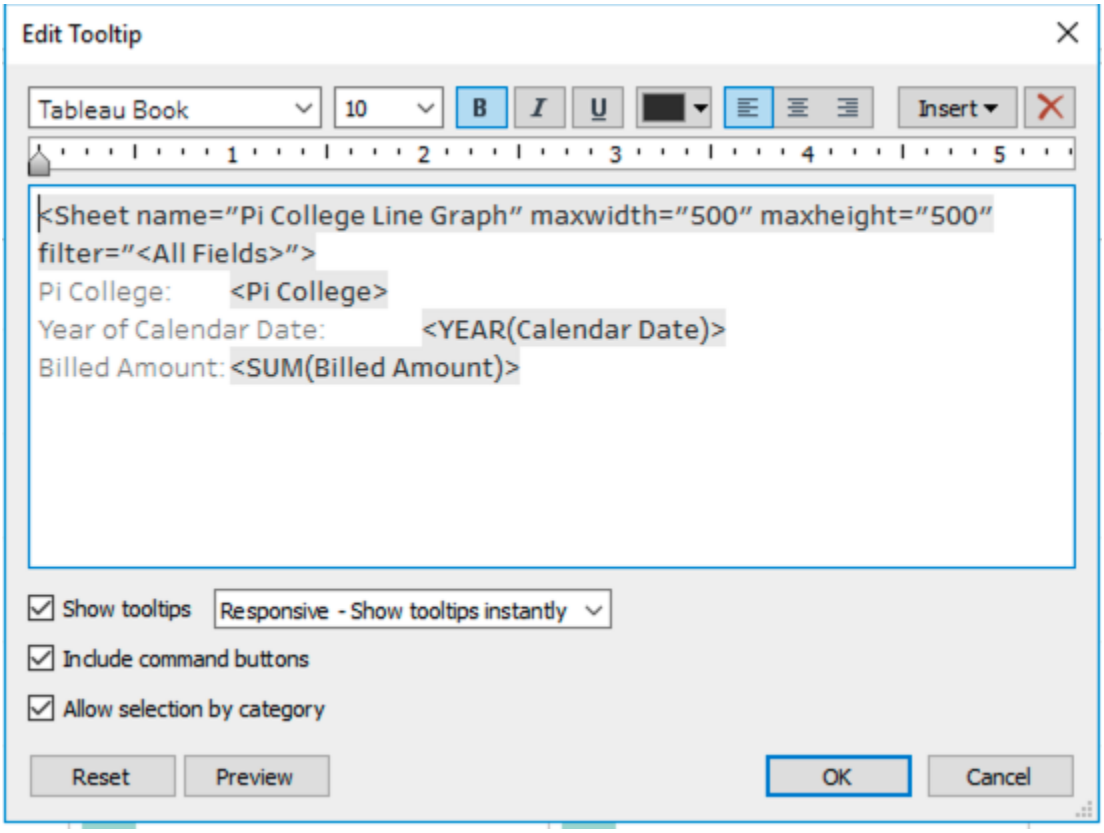
Once the popup appears, select “Insert” in the top right corner, then “Sheets”, and then “**Pi College Line Graph**”.





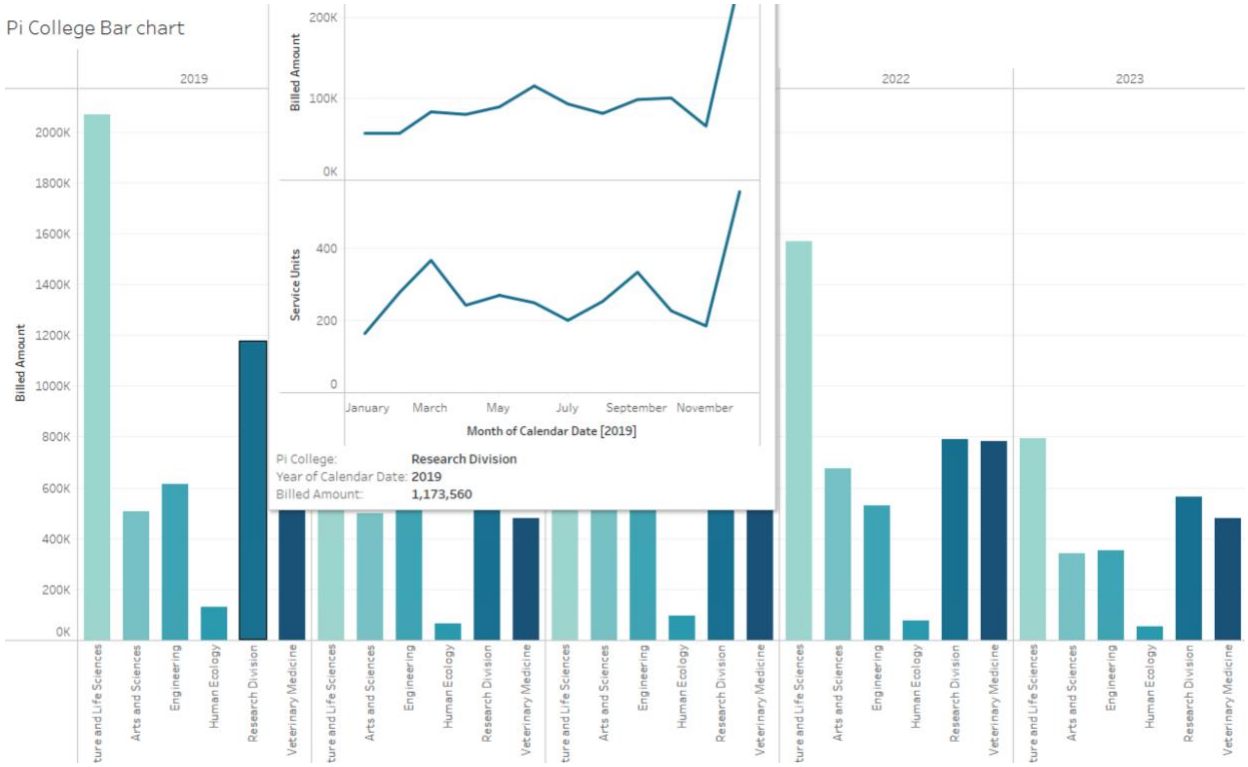
Assignment 6: Tableau Dashboard

Change the maxwidth and maxheight fields to 500.



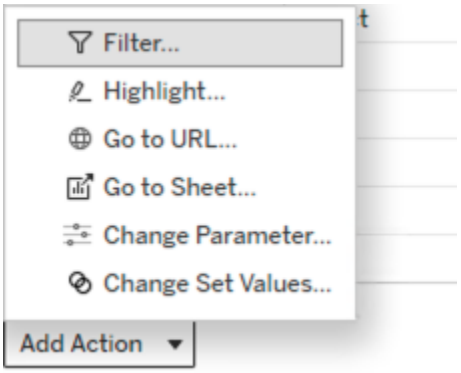
Click “OK”.

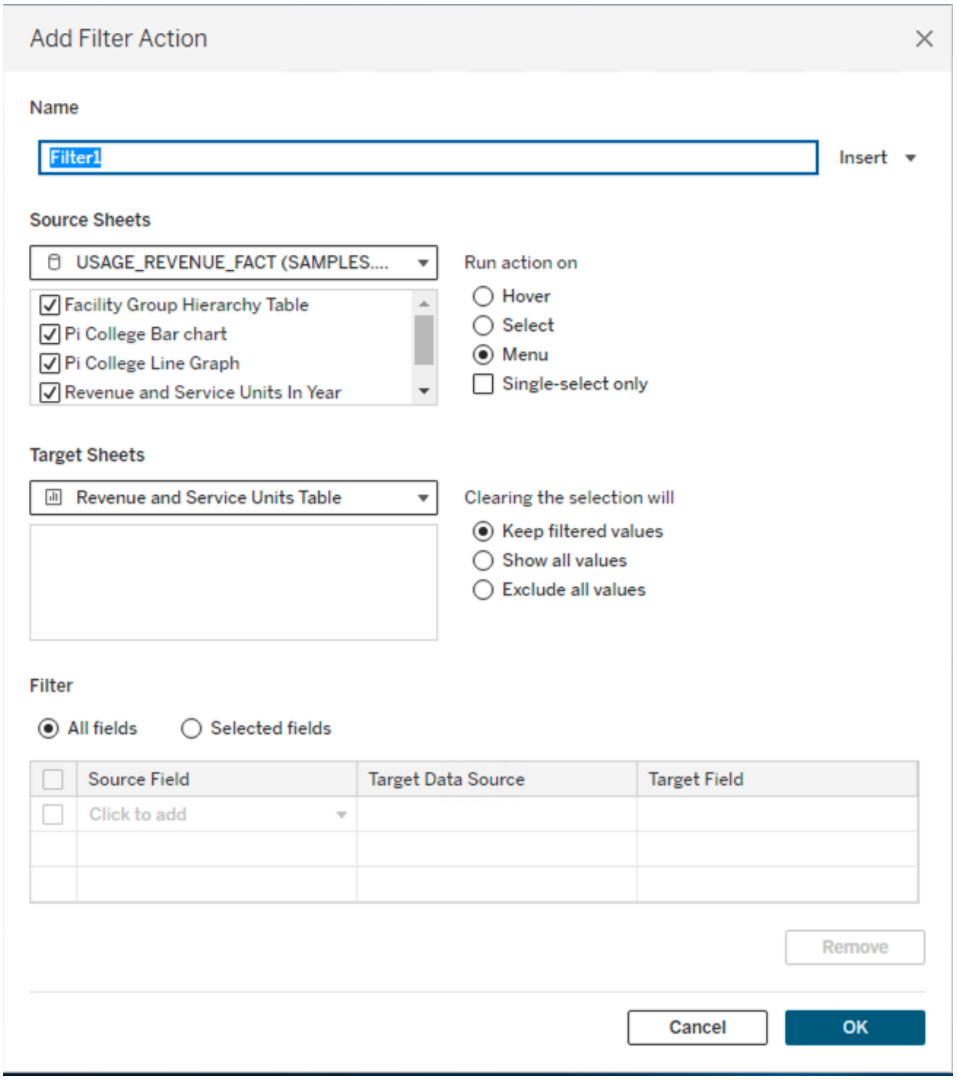
Now when you hover over bars on the chart, the line graph should appear in the tooltip. The line graph should also change based on the year and PI college that you are hovering over.



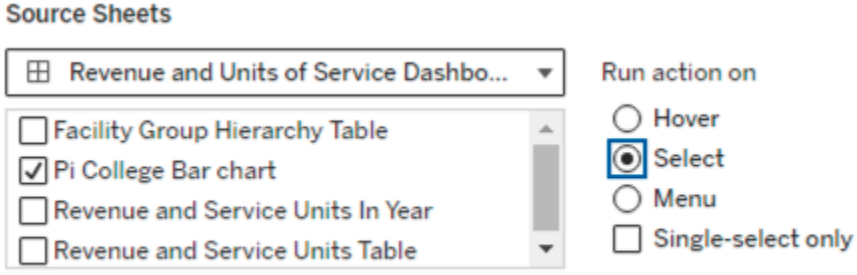
Pi Department Filter Action

Click Worksheet -> Actions -> Add Action -> Filter...



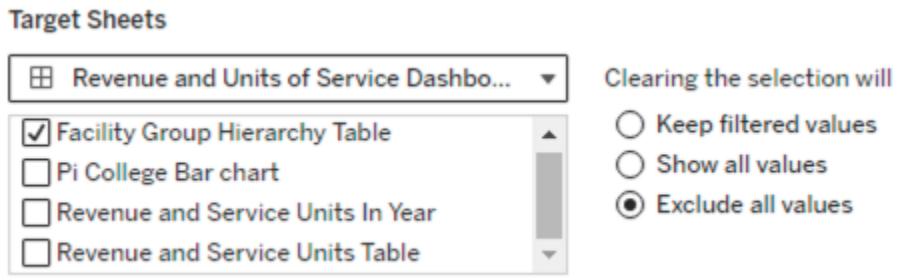


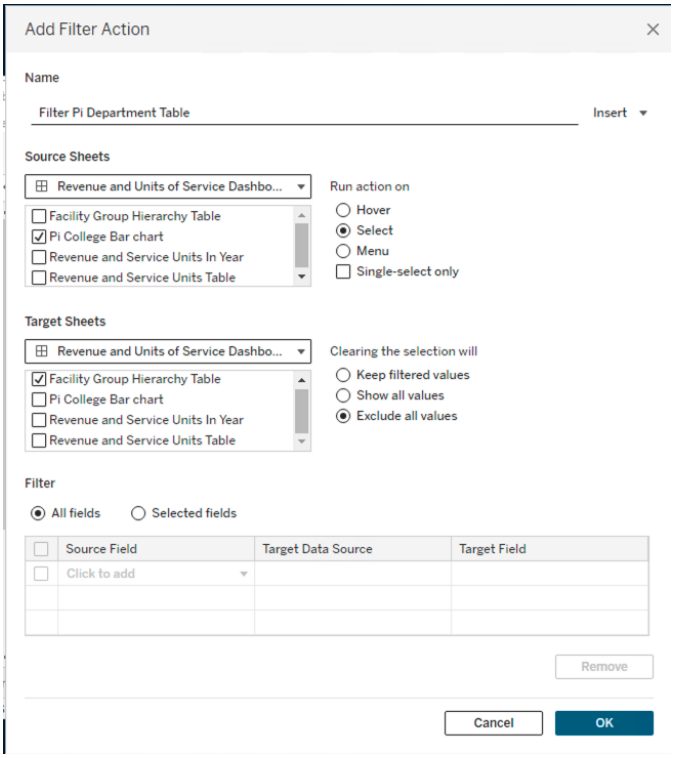
1. Rename the action to “Filter Pi Department Table”
2. Select “**Revenue and Units of Service Dashboard”** under “Source Sheets”
    a. Select “**Pi College Bar Chart**” from the checklist
    b. Click “**Select**” under “Run action on”



3. Select “**Revenue and Units of Service Dashboard**” under “Target Sheets”
    a. Select “**Facility Group Hierarchy Table**” from the checklist

b. Click "**Exclude all Values**" under "**Clearing the selection will**"





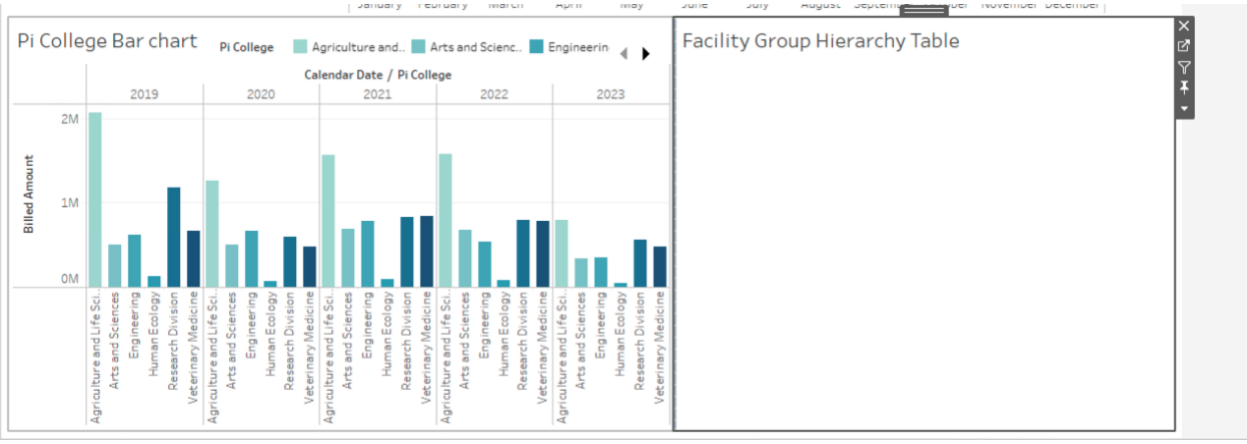
Click "OK".

At this point, the **Facility Group Hierarchy Table** should appear with the corresponding year/department information of a clicked on bar of the **Pi College Bar Chart**. It should disappear when none of the bars are selected. *Note it may be slow rendering the Facility Group Hierarchy Table and making it disappear when a bar not selected.*

Let's hide the title, so it does not appear. Go to your "Revenue and Units of Service" dashboard. Select the **Facility Group Hierarchy Table** view on the dashboard.

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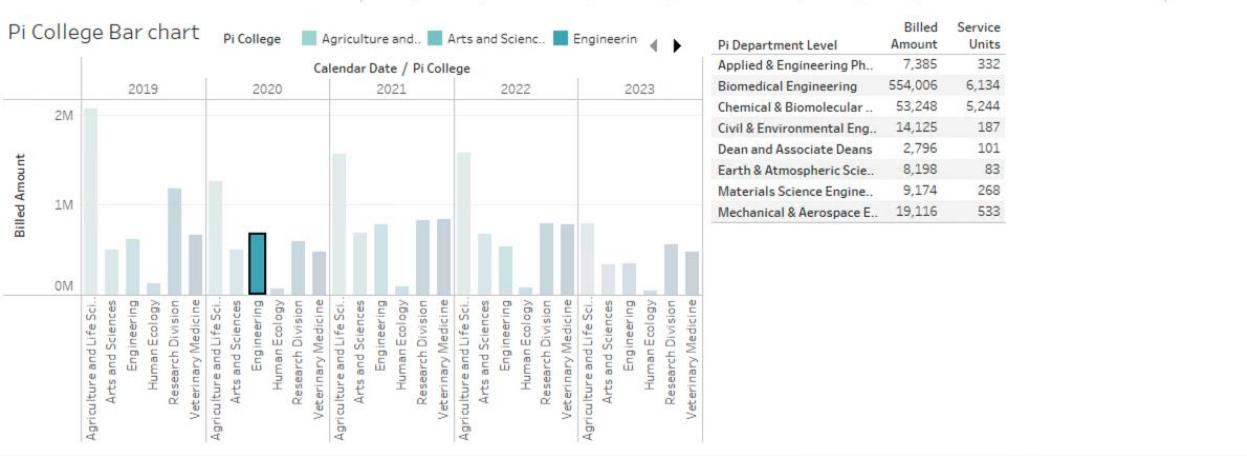
Assignment 6: Tableau Dashboard



Deselect the “Title”.



Here is an example of when a bar is selected. *Note that it may take a few minutes for the action to update. Also note that in the screenshot the hierarchy is at the high level, if you had drilled down prior, you will see the facility group level. Either are fine.*

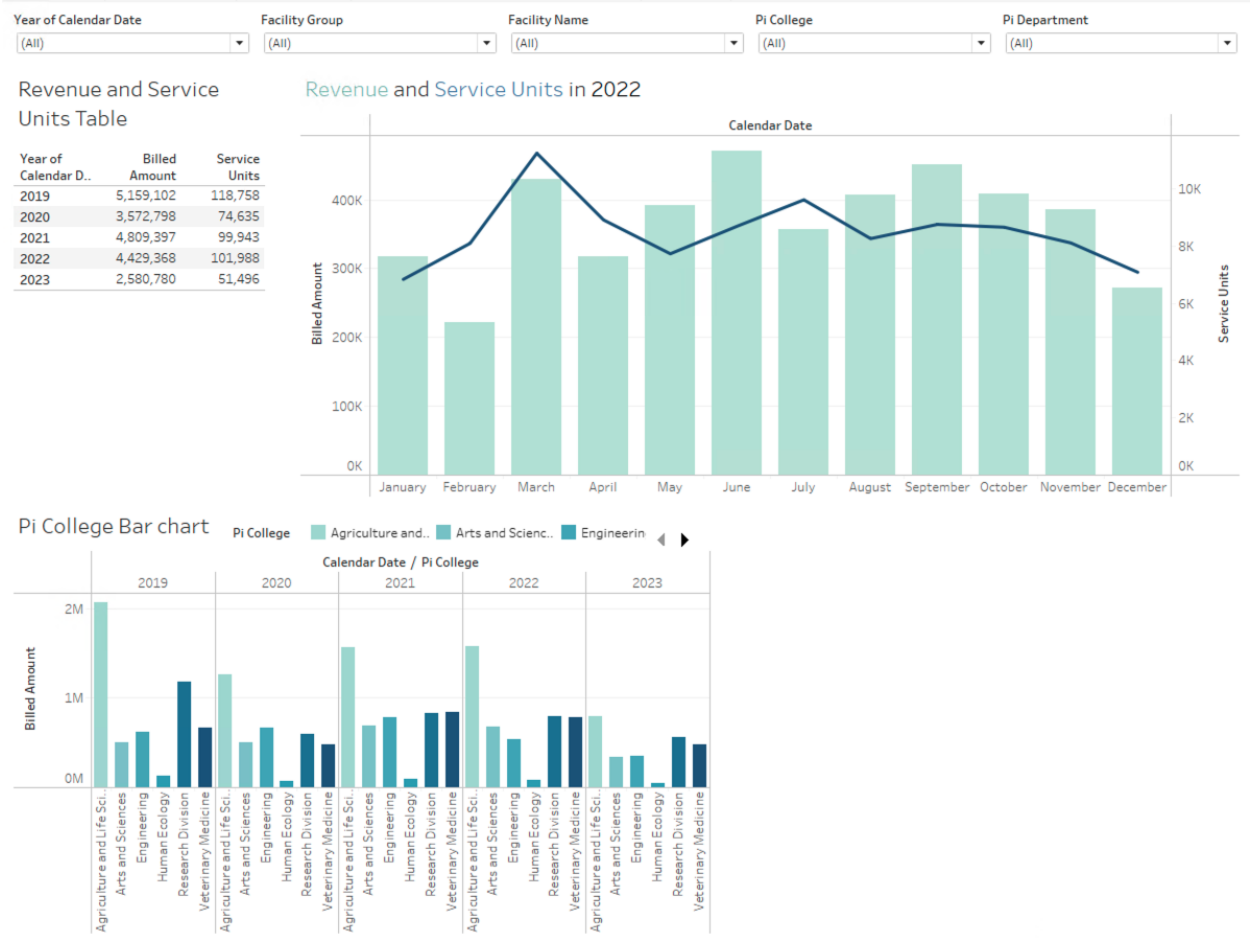


Pi Department Level	Billed Amount	Service Units
Applied & Engineering Ph..	7,385	332
Biomedical Engineering	554,006	6,134
Chemical & Biomolecular ..	53,248	5,244
Civil & Environmental Eng..	14,125	187
Dean and Associate Deans	2,796	101
Earth & Atmospheric Scie..	8,198	83
Materials Science Engine..	9,174	268
Mechanical & Aerospace E..	19,116	533

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Assignment 6: Tableau Dashboard

When nothing is selected, the table will not appear.



Additionally, notice how the chart can be expanded to include **Facility Group Level** due to the hierarchy we included earlier.

Pi Department Level	Facility Group Level	Billed A..	Service ..
CALS Animal Science	Bioinformatics	698	3
	Flow	836	13
	Genomics Facilities	5,427	1,009
	Imaging	4,887	180
	Preoteomics-Metabolomi..	1,671	147
CALS Biological & Environmental Engr	Bioinformatics	345	12
	Flow	3,631	79
	Genomics Facilities	3,555	484
	Imaging	17,705	267
	Preoteomics-Metabolomi..	4,865	72
CALS Biology Stats & Com..	Bioinformatics	1,704	1
CALS Computational Biology	Bioinformatics	6,093	72
	Genomics Facilities	10,104	378
CALS Earth and Atmosphe..	Bioinformatics	1,075	1
CALS Ecology & Evolutionary Biology	Bioinformatics	24,244	85
	Genomics Facilities	95,656	429
	Imaging	4,063	37
	Unknown	2,085	20
CALS Entomology	Bioinformatics	16,899	184

Congratulations! You have created an interactive Tableau dashboard. Save your file, remember the format is [netid]_financial.twb, ex. Jrc42_financial.twb. Grading will be based on your Tableau file containing all of the features covered by this tutorial. Upload your tableau file to Canvas by the stated deadline.